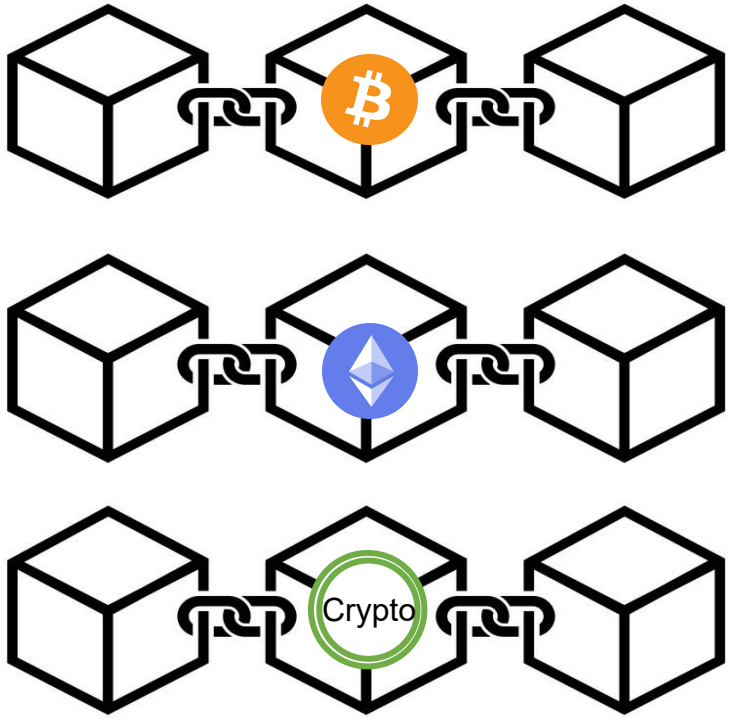
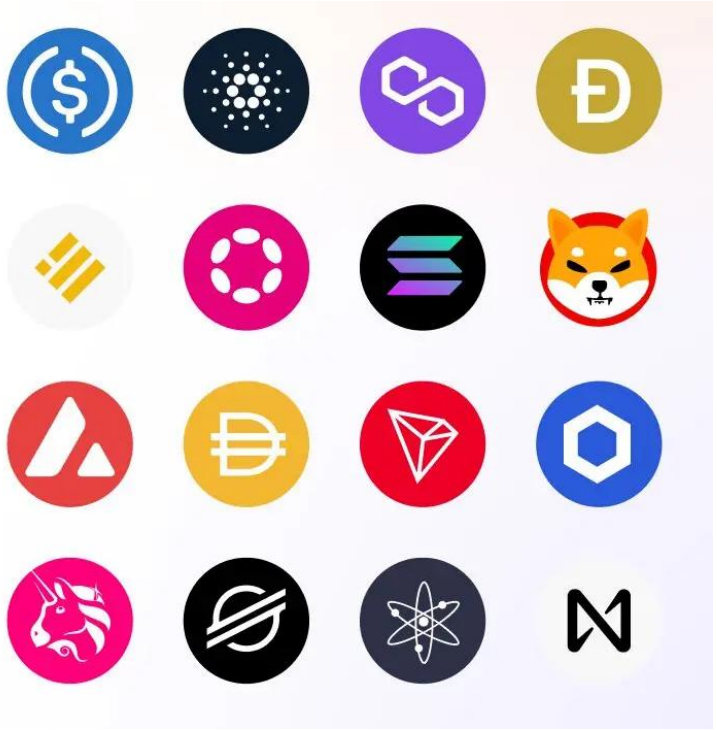
Lecture 05 Cryptocurrency

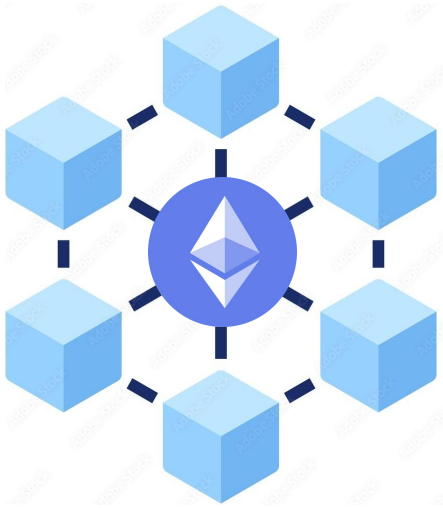
Blockchain vs Cryptocurrency



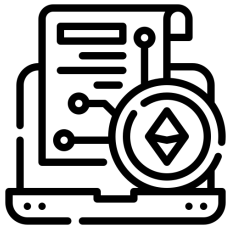


Cryptocurrency

Native Token



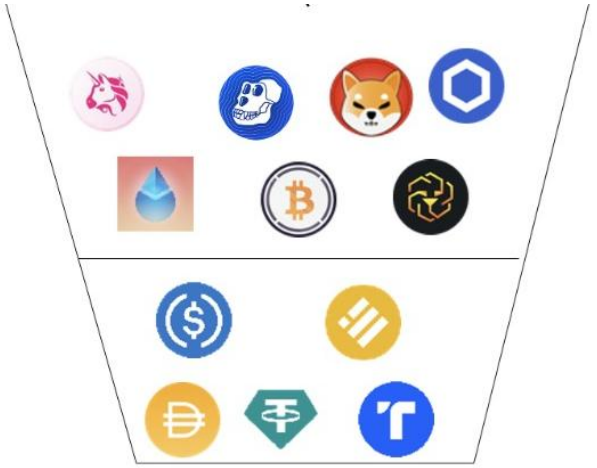
Build-in logic in protocol of a blockchain



Smart Contracts



Non-native Token

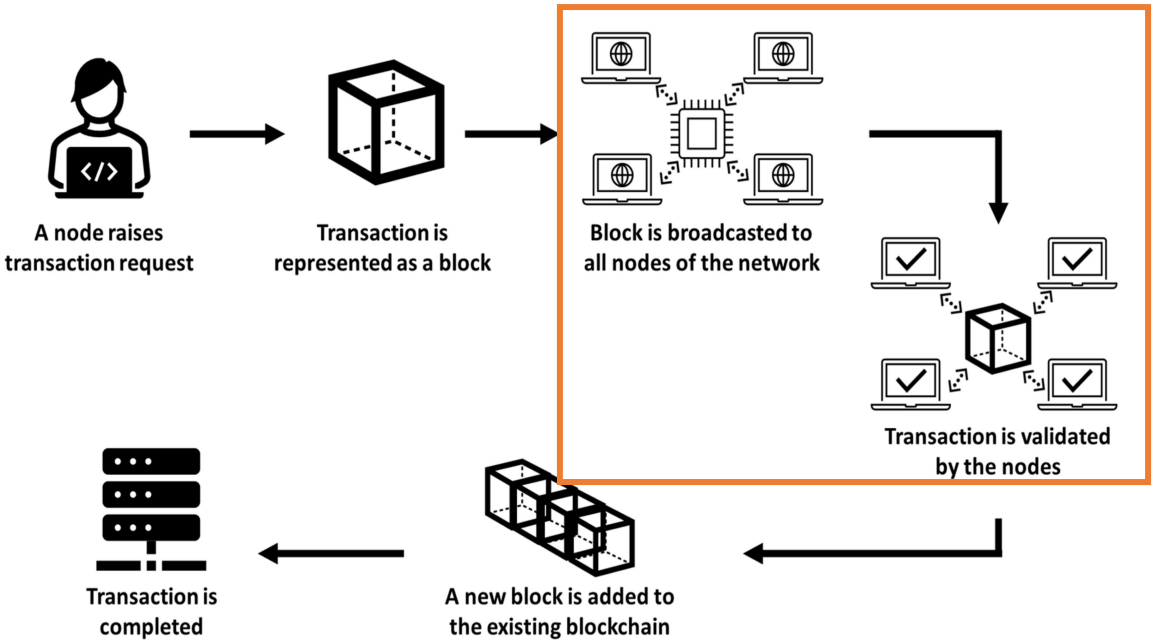


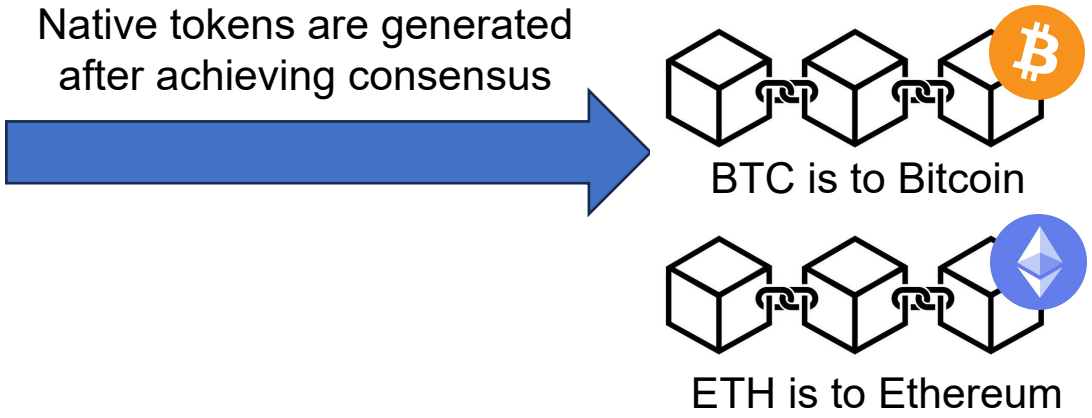
Programmable logic in smart contract deployed to a blockchain

Native Token

Native tokens are digital assets that are native to their own blockchain. They are independent and operate on their own network. Bitcoin (BTC) and Ether (ETH) are examples of native tokens.

UTXO (Unspent Transaction Output) and Account-Based are two different blockchain transaction models, each with distinct ways of managing manage native tokens.

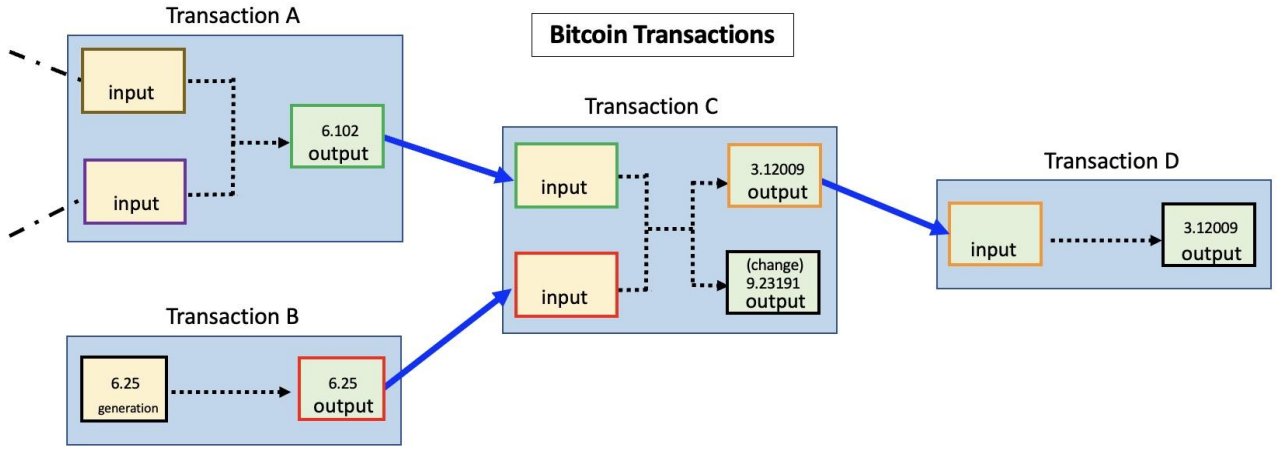


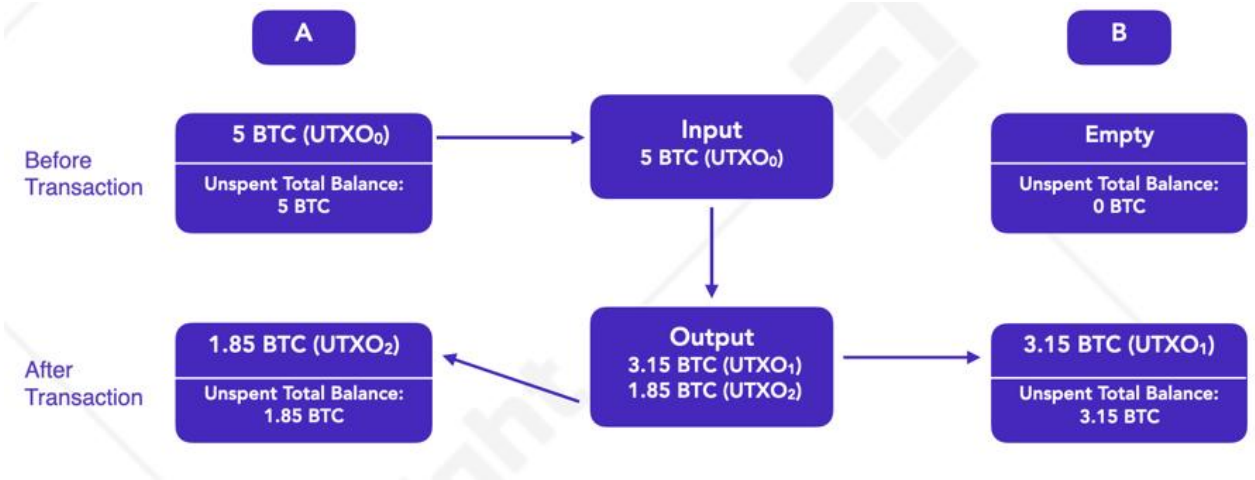


Recall Transaction Models: The UTXO Vs Account Model

Unspent Transaction Output (UTXO) Model

- An Unspent Transaction Output (UTXO) is an unused token.
- Every time a transaction is executed, the input UTXOs are deleted and the output UTXOs are generated. Any output of the transaction as a new UTXO that can be later spent in a new transaction.
- The main differences between the cash and the UTXO model is that bitcoin isn't bound by set amounts, i.e. $5, $10, $20, etc. Any amount of bitcoin (up to eight decimal places) can be an UTXO.
- Double Spent is impossible in this model as every UTXO will have the state whether it is spent already in the same global ledger.

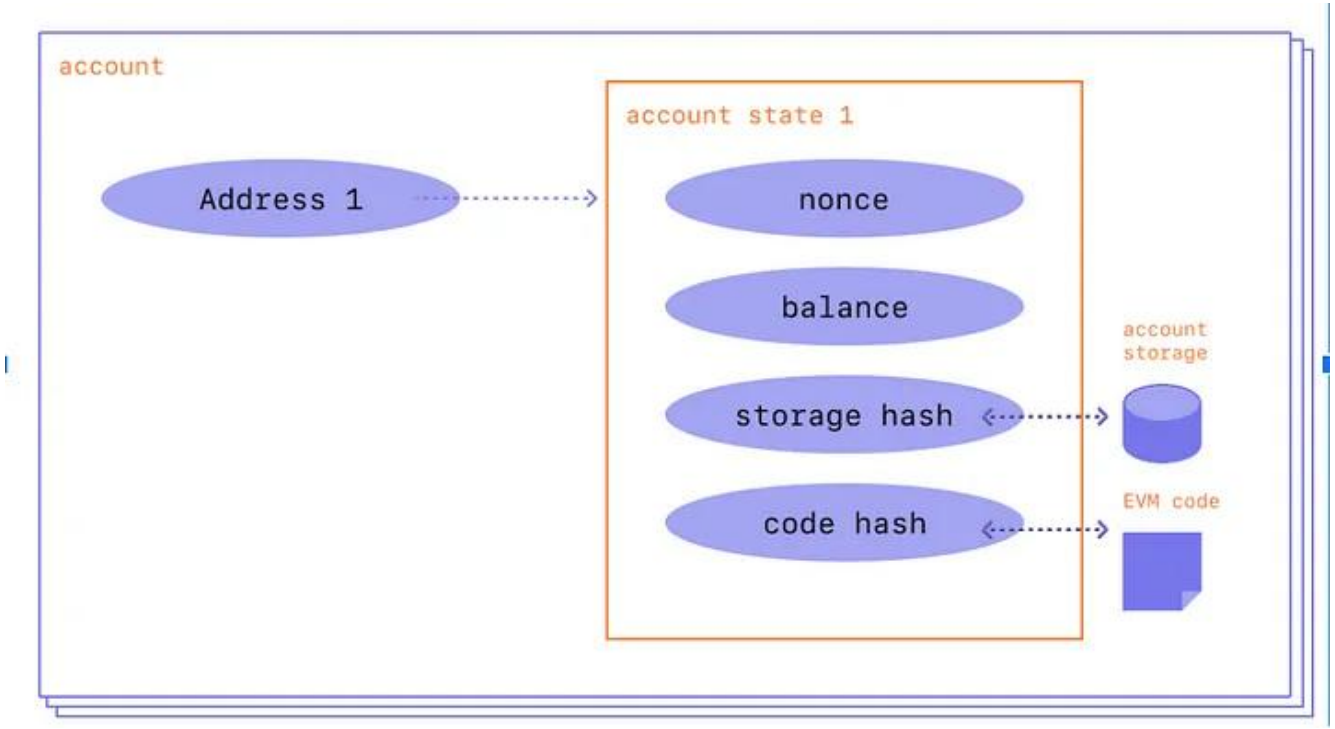


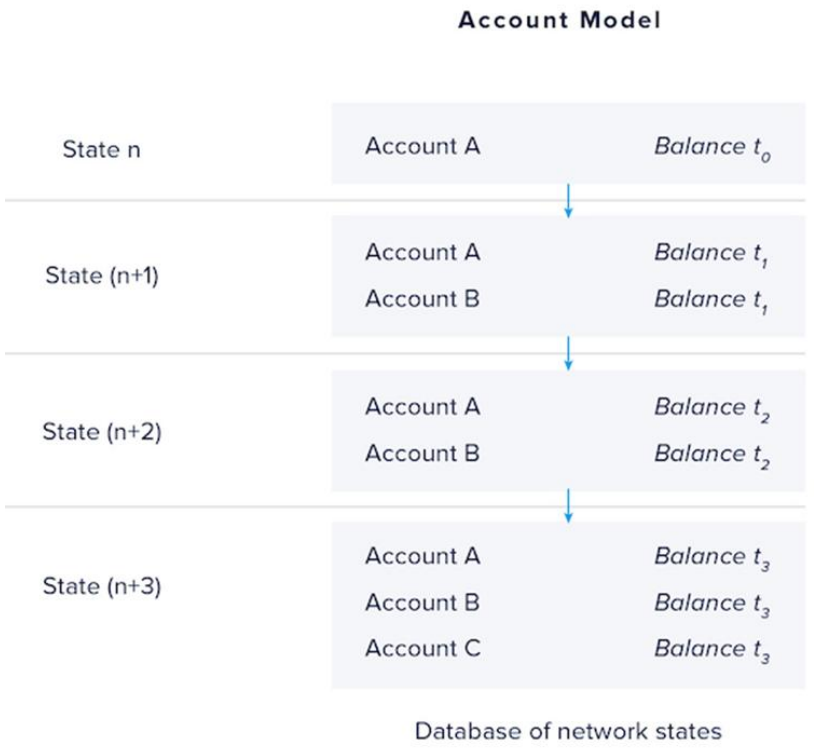


Recall Transaction Models: The UTXO Vs Account Model

Account-based Model——Like Bank Accounts

- **Definition:** The account model maintains the state of accounts directly on the blockchain.
- This is enabled by global state tracking - a database of all accounts, keys, contract code, and balances across the network.

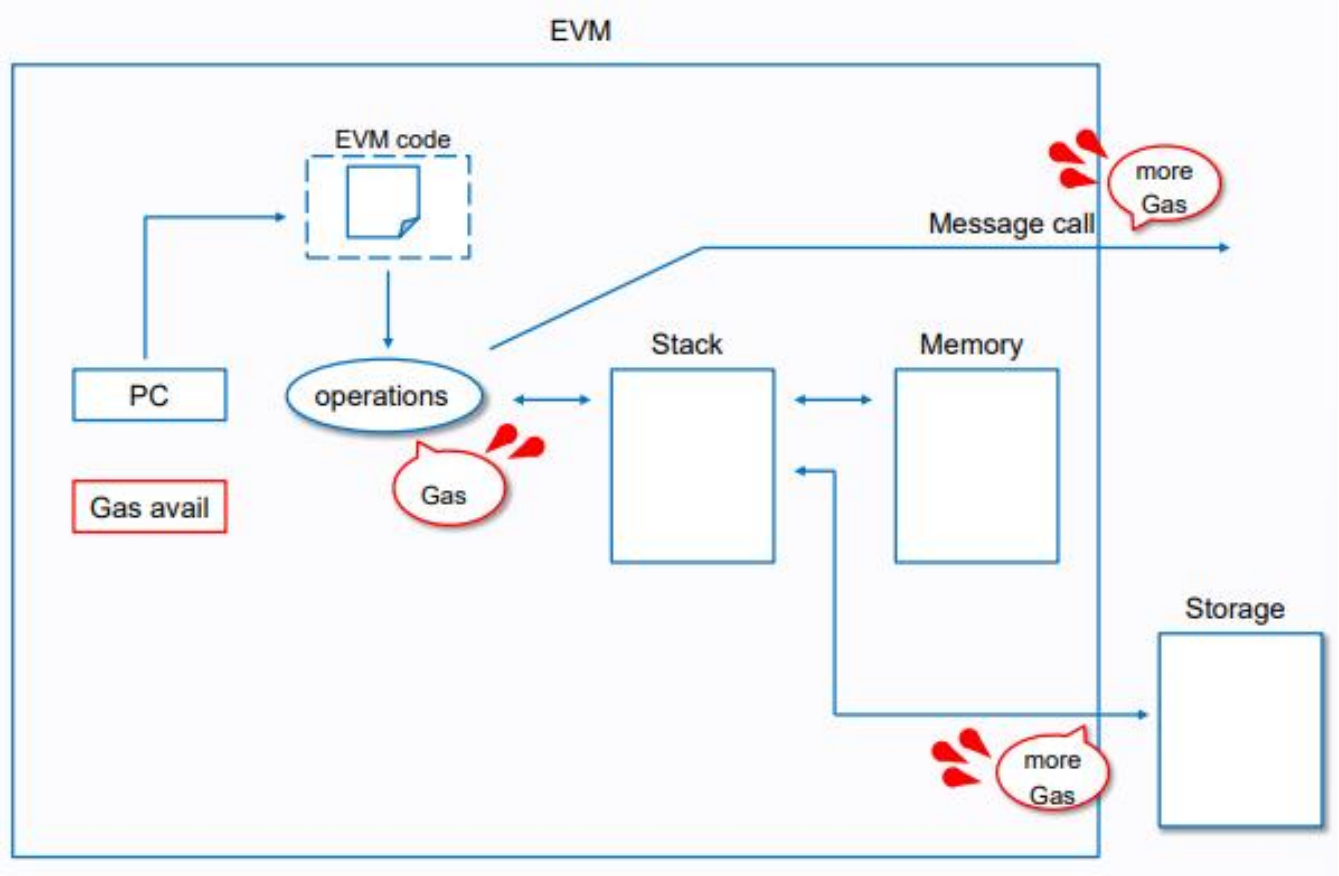




Applications of Native Tokens

■ Transaction Fees

All transactions and smart contract executions on blockchain networks require the consumption of native tokens (such as ETH on Ethereum) as Gas fees to prioritize and secure network operations.



Applications of Native Tokens

■ Network Incentive Mechanism

ETH is distributed as block rewards to validators (under Proof of Stake consensus), ensuring decentralized consensus and safeguarding network security.

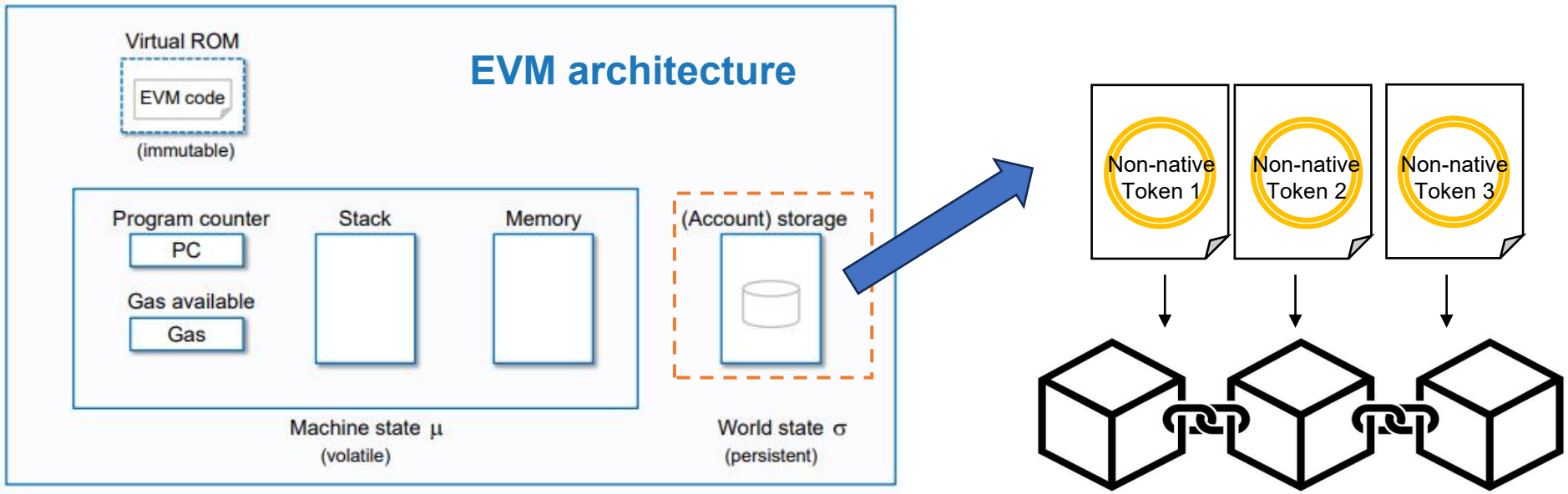
Recall Proof-of-Stake (PoS)

- Proof-of-Stake (PoS) is a consensus mechanism in blockchain technology where the creation of new blocks and the validation of transactions are determined by the stake (i.e., the amount of cryptocurrency) held by a network participant.
- Validators in PoS systems are required to stake the network's native token to participate in block validation and creation.
  - For example, in the Ethereum network, validators need to stake 32 ETH to become a validator.
  - The more tokens staked, the higher the probability of validating new blocks.

Non-native Token

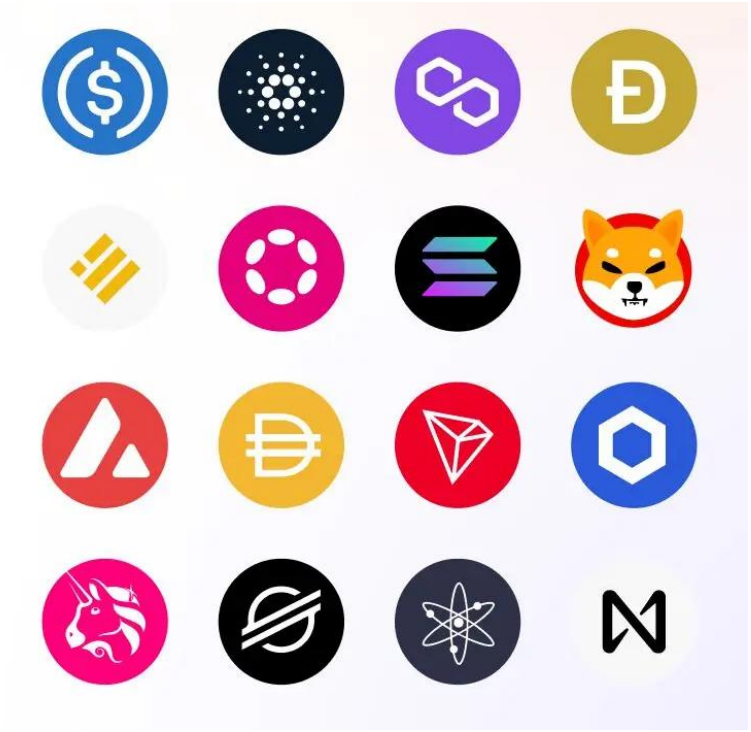
Non-native tokens are deployed and managed by smart contracts and balances are stored in a data-structure in the storage space of that contract.

Non-native tokens operate on existing blockchain networks (such as Binance Smart Chain, Ethereum, etc.) rather than on their own independent blockchains.



Non-native Token

Tokens can be developed for a wide range of use cases other than being a form of money. There are some popular types of tokens.

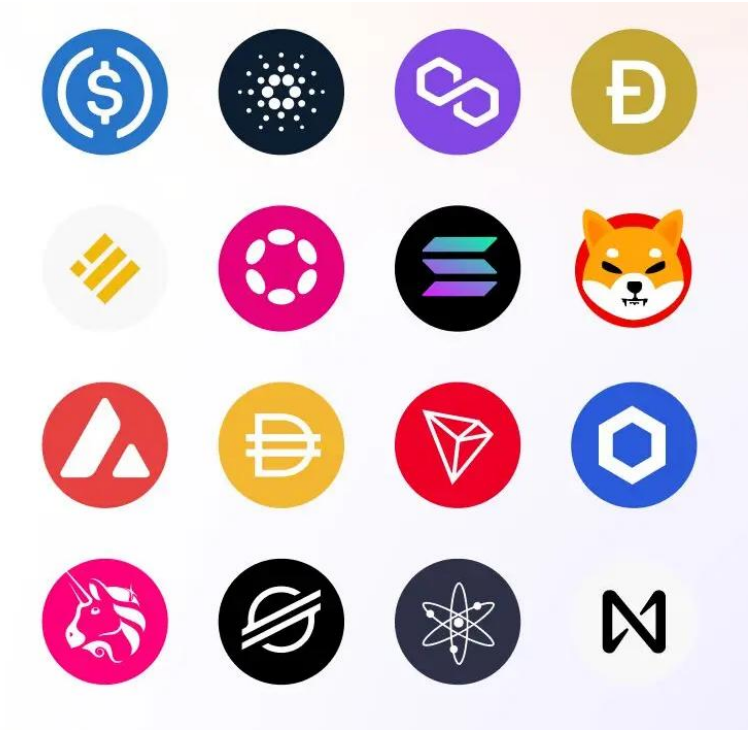


How to ensure that tokens on a blockchain are consistent, interoperable, and can be used seamlessly across various applications.



Non-native Token

Tokens can be developed for a wide range of use cases other than being a form of money. There are some popular types of tokens.



How to ensure that tokens on a blockchain are consistent, interoperable, and can be used seamlessly across various applications.

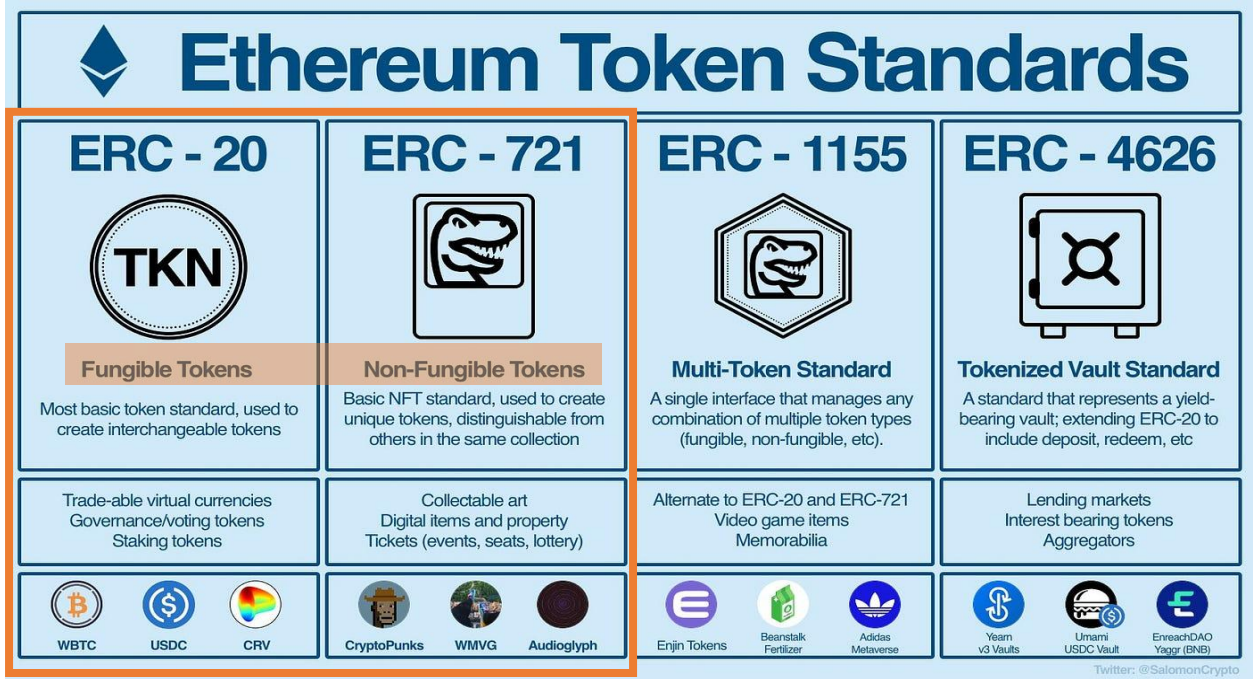


Build standard!

Ethereum Token Standard

Token standards are a subset of smart contract standards. For blockchains that support smart contracts, token standards represent a guide for the creation, issuance, and deployment of new tokens on them.

Ethereum Request for Comments (ERC) is essentially a set of technical documents containing guidelines on developing a smart contract.



The ERC-20 token standard is a blueprint for creating fungible tokens on the Ethereum network.

In contrast with ERC-20, ERC-721 is a token standard for non-fungible tokens (NFTs).

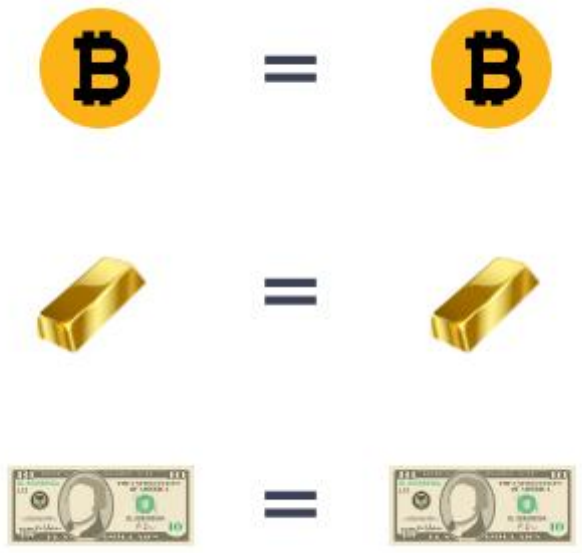
Fungible and Nonfungible Token

Fungible tokens are divisible and non-unique.

Nonfungible tokens are unique and non-divisible.

They should be considered as a type of deed or title of ownership of a unique, non-replicable item.

Fungible



Non-fungible



ERC-20 Token Standard

ERC-20 is a technical standard for fungible tokens on Ethereum, ensuring uniform behavior across all compliant tokens.

- It defines a standard interface for tokens, enabling seamless interoperability with wallets, dApps, and decentralized exchanges (DEXs).
- The standardized rules simplify token integration for developers and ensure compatibility across the Ethereum ecosystem.
- Any smart contract implementing the required methods and events is ERC-20 compliant. Once deployed, it manages token balances, transfers, and approvals.

ERC-20 Token Standard

ERC-20 is a standard API for fungible tokens.

A standard interface allows other contracts to interact with every ERC-20 token. No need for special logic for each token.

Official link: https://eips.ethereum.org/EIPS/eip-20

Core components of the ERC-20 standard:

- Basic information → **3 optional functions**
- Market supply and token balance
- Transfer and approval function

  } **6 mandatory functions**
- Event

ERC-20 Token Standard

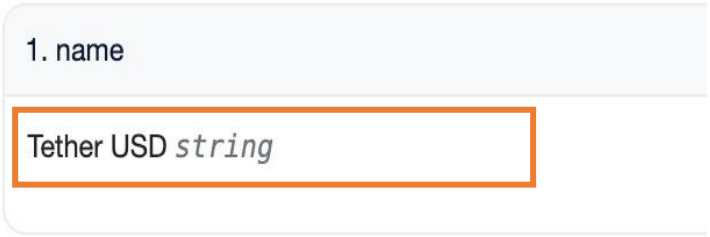
Function: basic information of the token

OPTIONAL - This method can be used to improve usability, but interfaces and other contracts MUST NOT expect these values to be present.

- ***Name***

function name() public view returns (string)

Returns the name of the token.



ERC-20 Token Standard

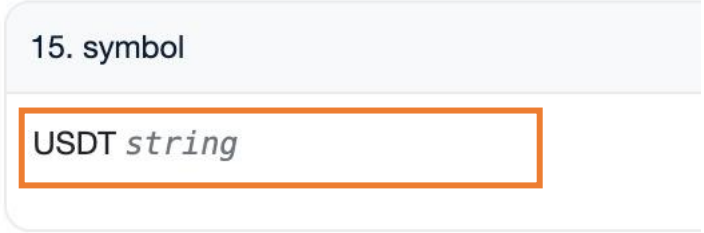
Function: basic information of the token

OPTIONAL - This method can be used to improve usability, but interfaces and other contracts MUST NOT expect these values to be present.

- ***Symbol***

function symbol() public view returns (string)

Returns the symbol of the token.



ERC-20 Token Standard

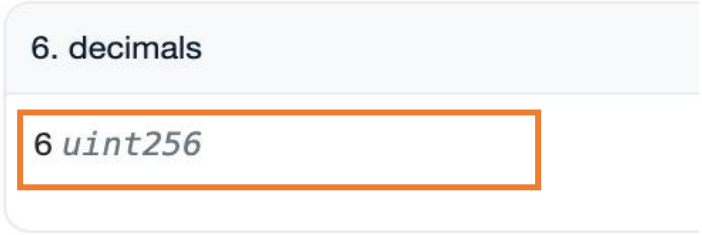
Function: basic information of the token

OPTIONAL - This method can be used to improve usability, but interfaces and other contracts MUST NOT expect these values to be present.

- ***Decimals***

function decimals() public view returns (uint8)

Returns the number of decimals the token uses - e.g. 6, means to divide the token amount by 1000000 to get its user representation.



ERC-20 Token Standard

The functions a token must have: total supply of the token available on the network.

- **TotalSupply:** The total number of tokens that will ever be issued

function totalSupply() public view returns (uint256)

Returns the total token supply.



ERC-20 Token Standard

The functions a token must have: token balance of an account.

- **BalanceOf:** The account balance of a token owner's account

function balanceOf(address _owner) public view returns (uint256 balance)

Returns the account balance of address _owner.



ERC-20 Token Standard

The functions a token must have: transfer tokens from one account to another.

- **Transfer:** Automatically executes transfers of a specified number of tokens to a specified address for transactions using the token

function transfer(address _to, uint256 _value) public returns (bool success)

Transfers [_value] amount of tokens to address [_to], and MUST fire the Transfer event. The function SHOULD [throw] if the message caller's account balance does not have enough tokens to spend.

Note Transfers of 0 values MUST be treated as normal transfers and fire the [Transfer] event.

ERC-20 Token Standard

The functions a token must have: transfer tokens from one account to another.

- **TransferFrom:** Automatically executes transfers of a specified number of tokens from a specified address using the token

function transferFrom(address _from, address _to, uint256 _value) public returns (bool success)

Transfers [_value] amount of tokens from address [_from] to address [_to], and MUST fire the [Transfer] event.

The [transferFrom] method is used for a withdraw workflow, allowing contracts to transfer tokens on your behalf. This can be used for example to allow a contract to transfer tokens on your behalf and/or to charge fees in sub-currencies. The function SHOULD [throw] unless the [_from] account has deliberately authorized the sender of the message via some mechanism.

Note Transfers of 0 values MUST be treated as normal transfers and fire the Transfer event.

ERC-20 Token Standard

The functions a token must have: transfer tokens from one account to another.

- *function* **transfer**(*address _to, uint256 _value) public returns (bool success)*
- *function* **transferFrom**(*address _from, address _to, uint256 _value) public returns (bool success)*

Transaction Hash	Type	Method	Date Time (UTC)	From	OR	To	Amount	Asset
0x5a5a971485c...	ERC-20	Transfer	2024-01-26 13:13:35	0xE99131...8A1588Ae	→	0x8E5D91...Ac3CA1d2	50	Tether USD (USDT)
0x4b3d94445c4...	ERC-20	Transfer	2024-01-26 13:13:35	0x484464...e014E3B8	→	0xe74081...AEc7Bd48	100	Tether USD (USDT)
0x5e8de1122c6...	ERC-20	Transfer	2024-01-26 13:13:35	Binance 18	→	0x5673d2...d5D53169	211	Tether USD (USDT)
0x1494d0e3cd8...	ERC-20	Transfer	2024-01-26 13:13:35	0x3eD24E...A5CDc844	→	0xf4C203...6D70e2B8	27,162.748933	Tether USD (USDT)
0x3521d5188ef8...	ERC-20	Transfer	2024-01-26 13:13:35	0x7F47A2...47c81d3C	→	0xbCC10e...Fc115c16	336.154975	Tether USD (USDT)

ERC-20 Token Standard

The functions a token must have: approve whether an amount of token from an account can be spent by a third-party account.

- **Approve:** Allows a spender to withdraw a set number of tokens from a specified account, up to a specific amount

function approve(address _spender, uint256 _value) public returns (bool success)

Allows [_spender] to withdraw from your account multiple times, up to the [_value] amount. If this function is called again it overwrites the current allowance with [_value].

ERC-20 Token Standard

The functions a token must have: approve whether an amount of token from an account can be spent by a third-party account.

- **Allowance:** Returns a set number of tokens from a spender to the owner

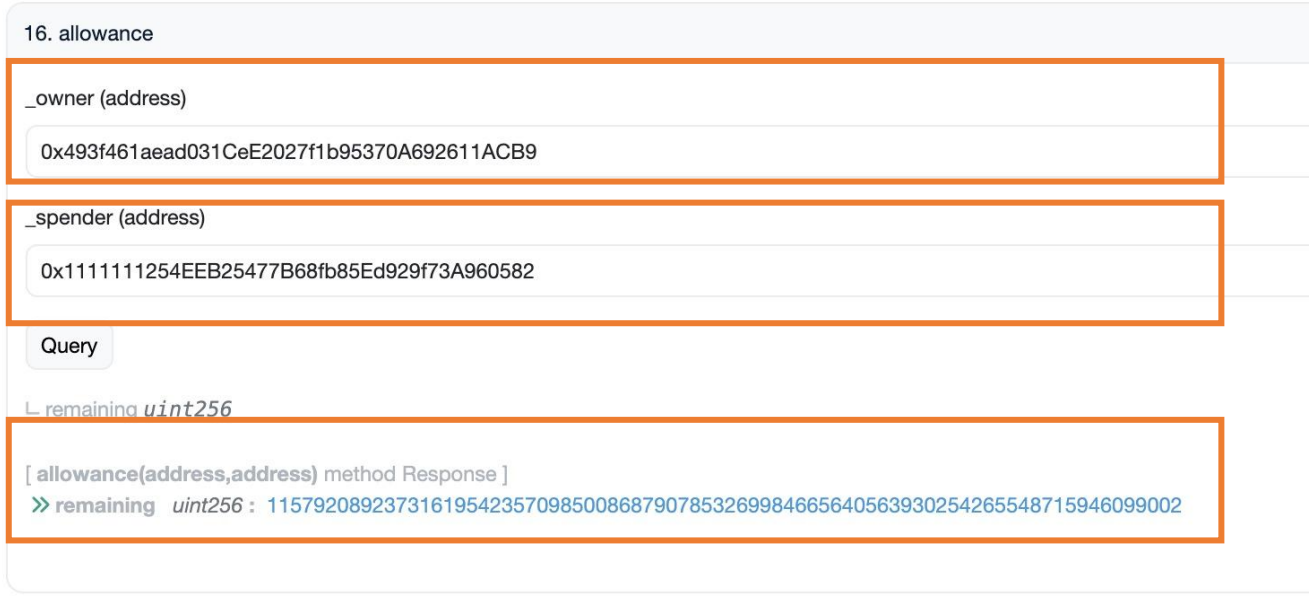
function allowance(address _owner, address _spender) public view returns (uint256 remaining)

Returns the amount which _spender is still allowed to withdraw from _owner.

ERC-20 Token Standard

The functions a token must have: approve whether an amount of token from an account can be spent by a third-party account.

- **function approve(address _spender, uint256 _value) public returns (bool success)**
- ***function allowance(address _owner, address _spender) public view returns (uint256 remaining)***



ERC-20 Token Standard

The events that must be included in the token:

- **Transfer**

event Transfer (address indexed _from, address indexed _to, uint256 _value)

Must trigger when tokens are transferred, including zero value transfers.

A token contract which creates new tokens SHOULD trigger a Transfer event with the [_from] address set to [0x0] when tokens are created.

ERC-20 Token Standard

The events that must be included in the token:

- **Approval**

event Approval *(address indexed _owner, address indexed _spender, uint256 _value)*

Must trigger on any successful call to [approve(address _spender, uint256 _value)].

Smart contracts emit event logs when certain predefined actions are completed. The emitted event logs are stored on the blockchain and are publicly accessible. Event logs are an important tool for smart contract developers to communicate with the outside world, as well as for data analysts to keep track of what happens inside of a smart contract.

ERC-20 Token Standard

Event logs

40

Address Wall Street Memes: WSM Token

Name `Transfer (index_topic_1 address from, index_topic_2 address to, uint256 value)` View Source

Topics

0 `0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef`

1: from Dec → `0x493f461aead031CeE2027f1b95370A692611ACB9`

2: to Dec → `0xdB0D7E28E7e1b9972dF333bE72a320cf387B9132`

Data

value : 1756838567567567567567568

41

Address Wall Street Memes: WSM Token

Name `Approval (index_topic_1 address owner, index_topic_2 address spender, uint256 value)` View Source

Topics

0 `0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925`

1: owner Dec → `0x493f461aead031CeE2027f1b95370A692611ACB9`

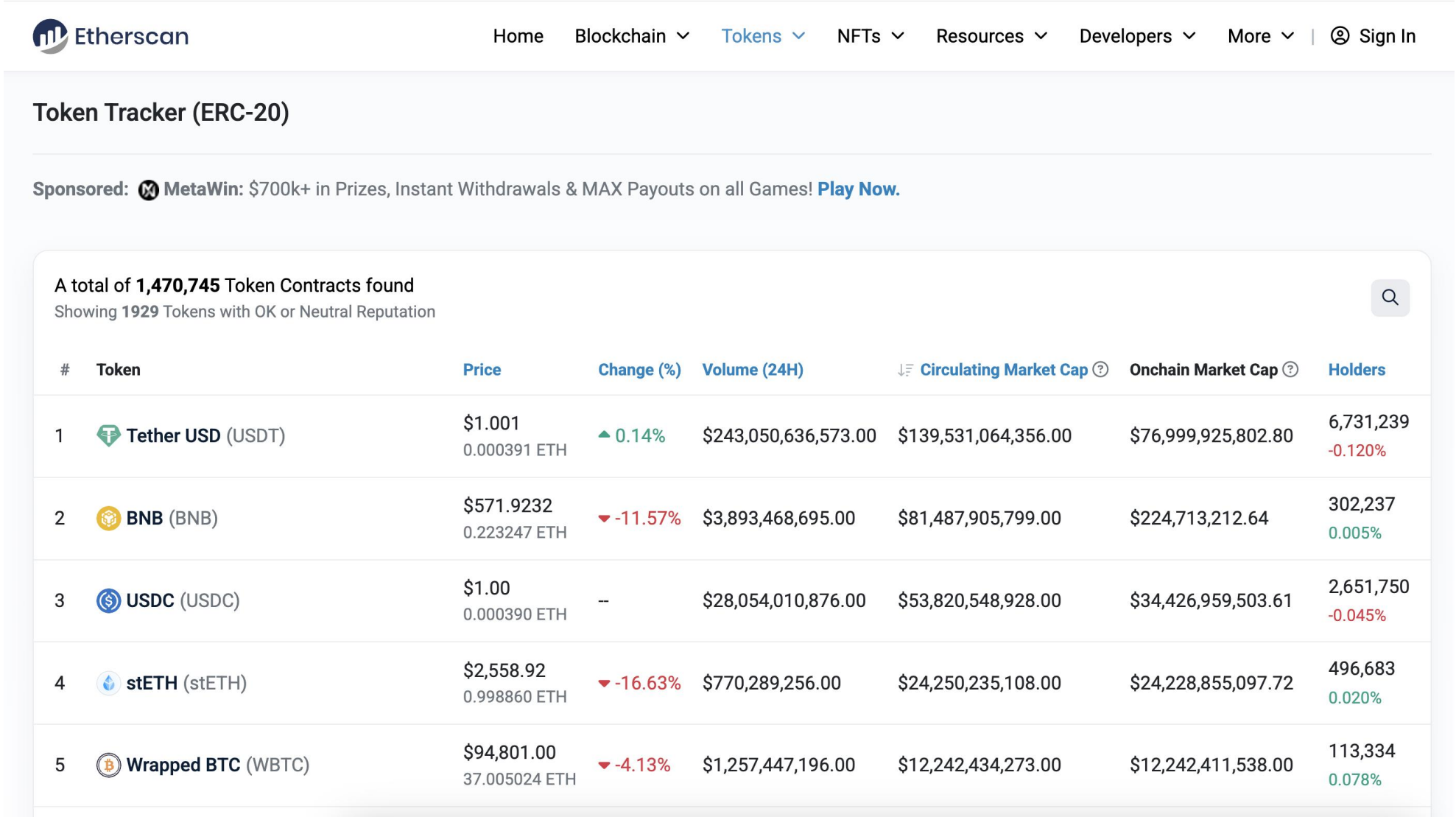
2: spender Dec → `0x1111111254EEB25477B68fb85Ed929f73A960582`

Data

value : 115792089237316195423570985008687907853269984665640545389987133759779942520732

ERC-20 Tokens

https://etherscan.io/tokens

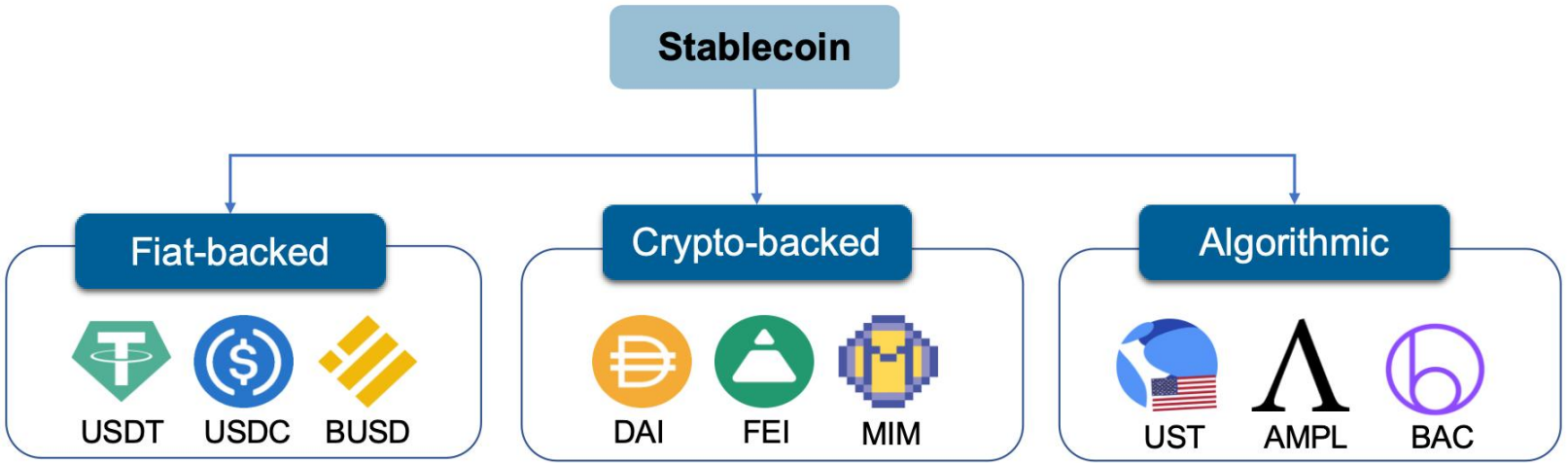


Uses and Value of ERC-20 Tokens

■ Stablecoins

Due to the extreme price fluctuations of cryptocurrencies, they fall short of fulfilling the essential functions of a currency, which has consequently limited their applicability in various use cases. Stablecoins are tokens designed to stay at a fixed value.

- ■ Examples: 1 coin = 1 USD, 1 coin = 1 EUR
- ■ Goals:
  - Integrate real-world currencies into on-chain applications.
  - Enable people without easy access to USD, to hold and trade a USD-equivalent asset.



Uses and Value of ERC-20 Tokens

■ Wrapped Tokens

A wrapped token is a cryptocurrency that represents another cryptocurrency on a different blockchain, enabling interoperability and functionality across blockchain networks. In order to enable tokens to function in an environment or blockchain other than their native one, they are encapsulated.

Example: Wrapped Bitcoin (WBTC) is a token that represents Bitcoin (BTC) on the Ethereum blockchain. It is the first ERC20 token backed 1:1 by Bitcoin. WBTC delivers the power of Bitcoin with the flexibility of an ERC20 token.



WBTC can be used within the Ethereum ecosystem, enabling Bitcoin holders to access decentralized applications (dApps), decentralized exchanges (DEXs), and other Ethereum-based services.

Uses and Value of ERC-20 Tokens

- **Decentralized Autonomous Organization (DAO) Voting/Governance**

Example: UNI, the governance token of Uniswap (a decentralized exchange), is a non-native ERC-20 token on Ethereum. Holding UNI grants voting rights to propose or decide on protocol upgrades (e.g., fee structures), even though Uniswap operates independently of Ethereum's native governance.

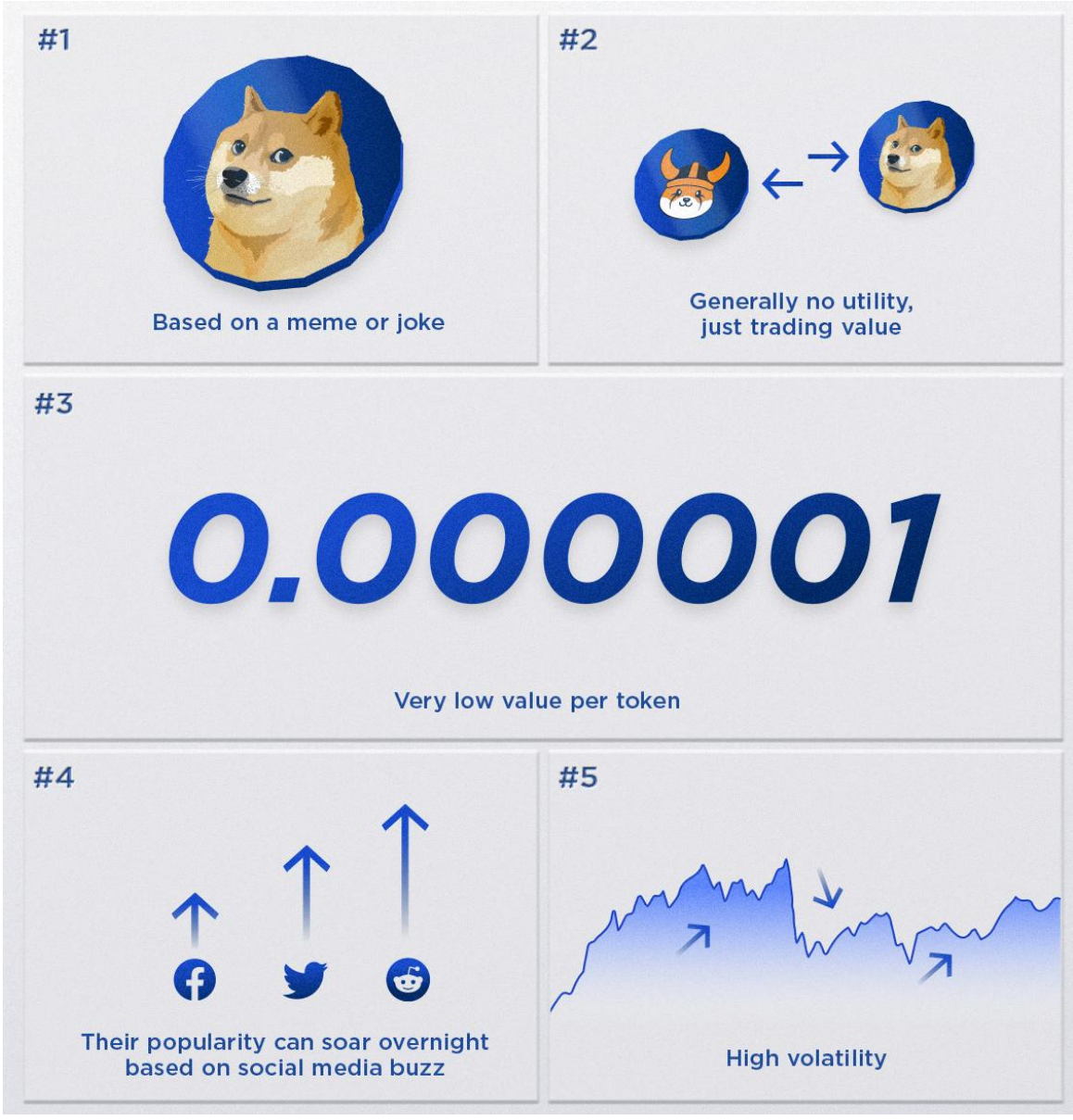


Uses and Value of ERC-20 Tokens

■ Meme coins

Example: Meme coins can be seen as digital assets that represent a community or a cultural phenomenon. For example, Dogecoin started as a joke based on the "Doge" meme but has since gained significant value and a strong community following.



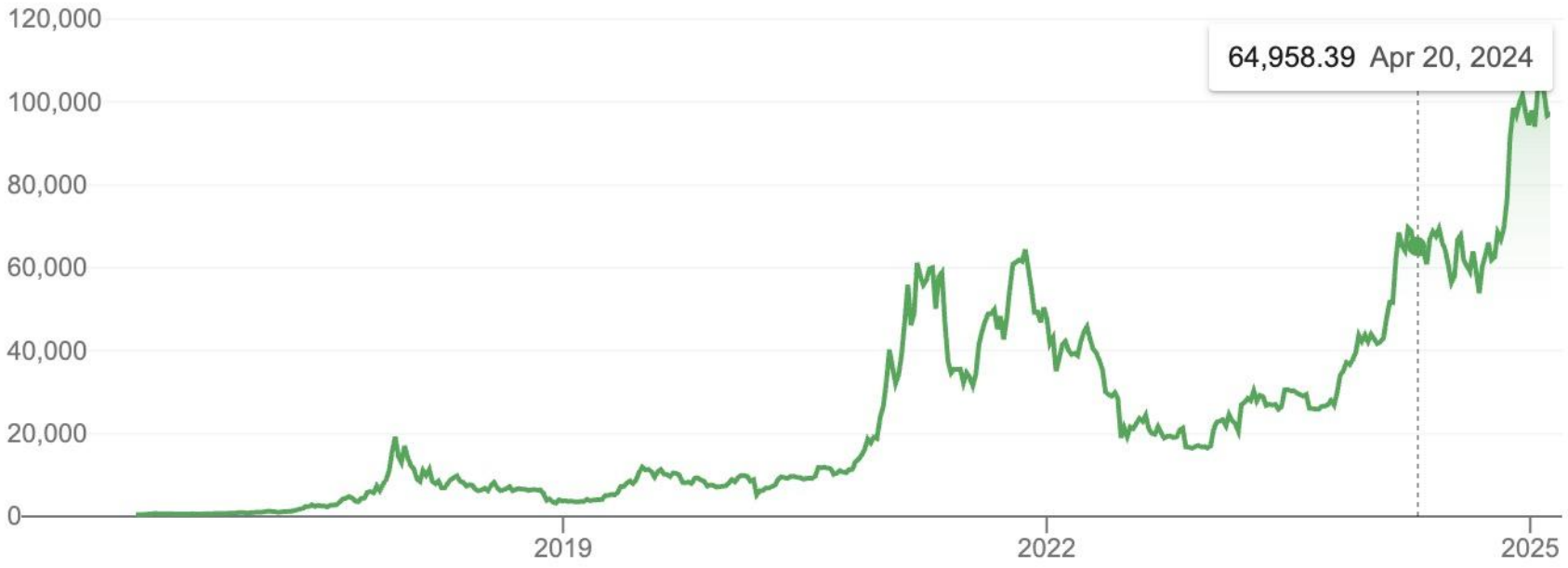


Why stablecoins?



Why stablecoins?

■ **Due to the extreme price fluctuations of cryptocurrencies, they fall short of fulfilling the essential functions of a currency, which has consequently limited their applicability in various use cases.**



The price of Bitcoin

Stablecoin

Stablecoins are tokens designed to stay at a fixed value.

- Due to the extreme price fluctuations of cryptocurrencies, they fall short of fulfilling the essential functions of a currency, which has consequently limited their applicability in various use cases.



Stablecoin

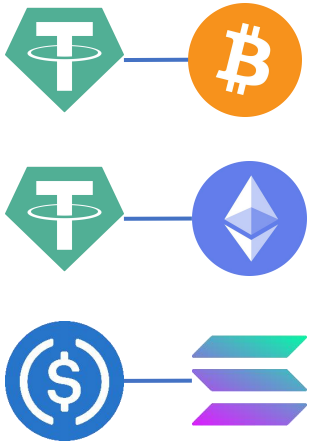
	#	Name	Price	1h %	24h %	7d %	Market Cap	Volume(24h)
☆	1	Bitcoin BTC	$34,062.85	▼0.40%	▼3.09%	▲19.78%	$665,014,405,961	$39,820,143,130 1,166,840 BTC
☆	2	Ethereum ETH	$1,790.36	▼0.45%	▼1.72%	▲14.13%	$215,313,563,078	$14,505,061,423 8,083,709 ETH
☆	3	Tether USDt USDT	$1.00	▲0.02%	▲0.05%	▼0.03%	$84,266,050,687	$54,669,128,002 54,664,674,036 USDT
☆	4	BNB BNB	$226.80	▼0.63%	▼3.08%	▲7.17%	$34,407,127,237	$595,949,679 2,621,020 BNB
☆	5	XRP XRP	$0.5564	▼0.30%	▲0.44%	▲13.32%	$29,732,661,714	$2,584,604,024 4,630,386,515 XRP
☆	6	USDC USDC	$1.00	▲0.01%	▲0.06%	▲0.01%	$25,081,632,234	$7,597,239,736 7,596,768,006 USDC

Source: CoinMarketCap

CYBER AND CRYPTO LAB

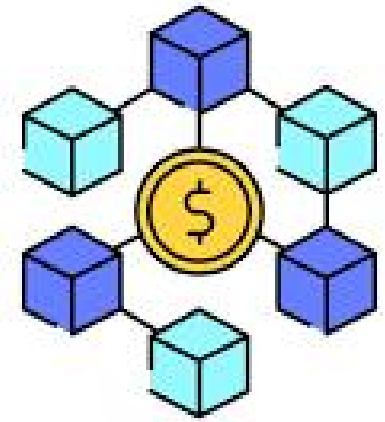
Use cases

Trading



...

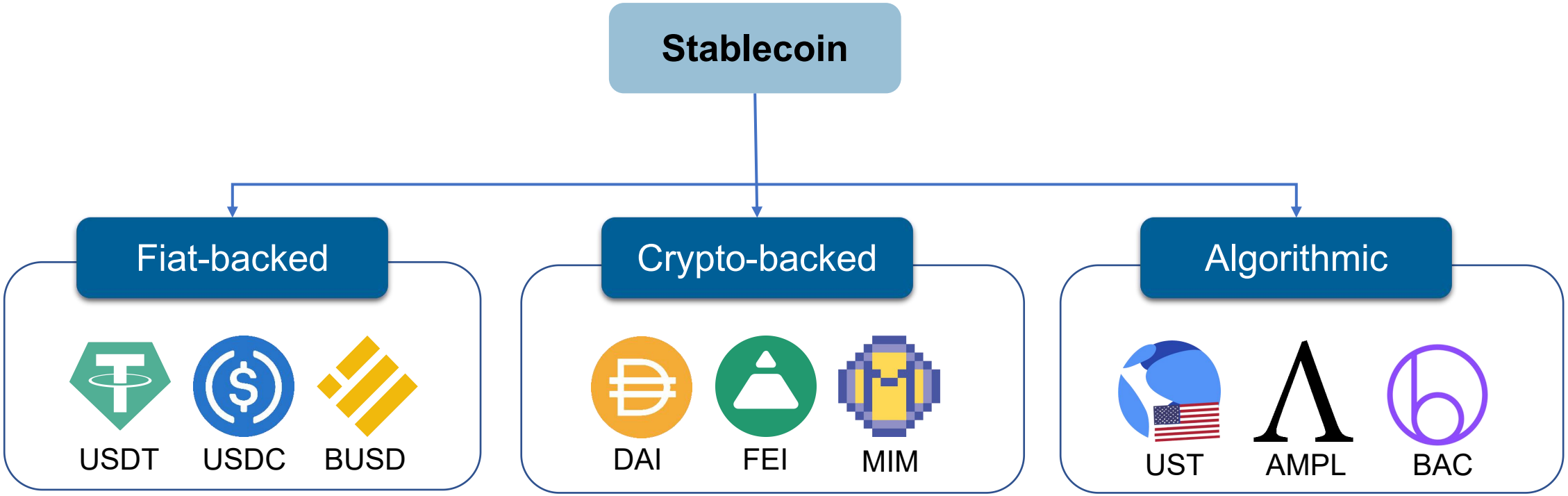
Financial applications and composability



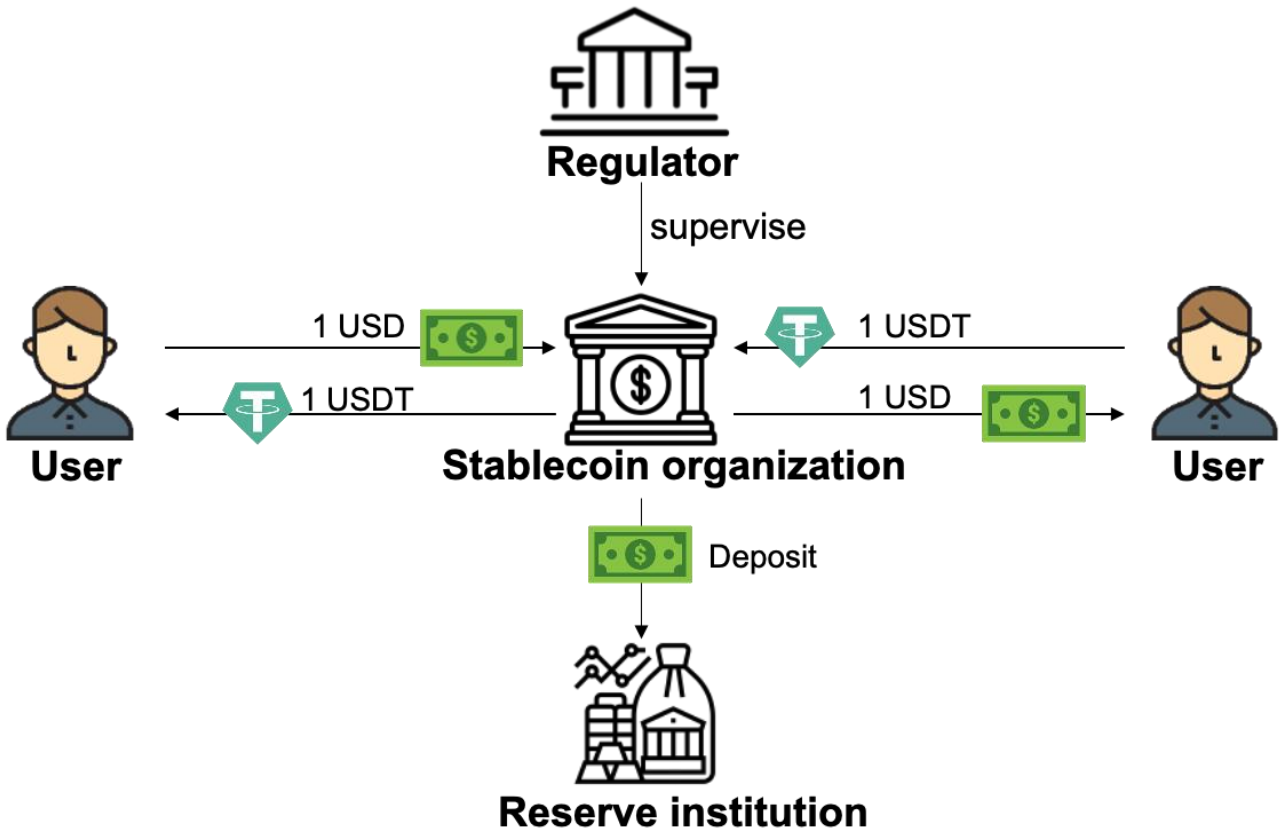
Store of wealth



Types of stablecoins

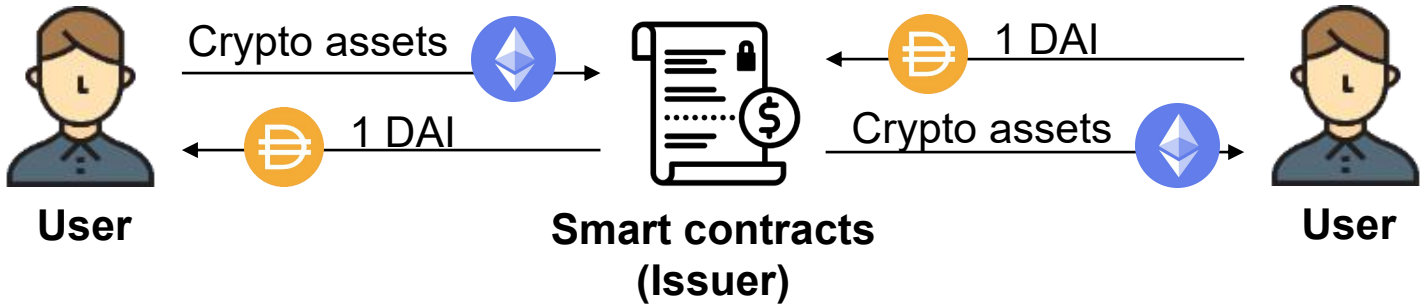


Fiat-backed stablecoin



- Fiat-backed stablecoins are cryptocurrencies pegged to the value of real-world currencies, such as the U.S. dollar or euro, and backed by reserves in that currency.

Crypto-backed stablecoin

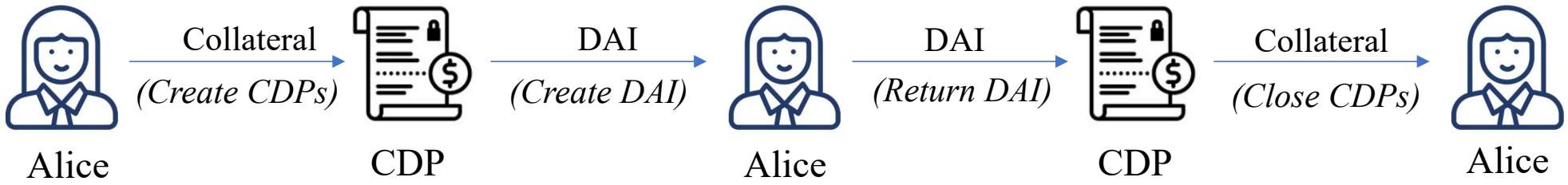


- Crypto-backed stablecoins are backed by other cryptocurrencies. Because the reserve cryptocurrency may also be prone to high volatility, such stablecoins are **overcollateralized—that is, the value of cryptocurrency held in reserves exceeds the value of the stablecoins issued**.

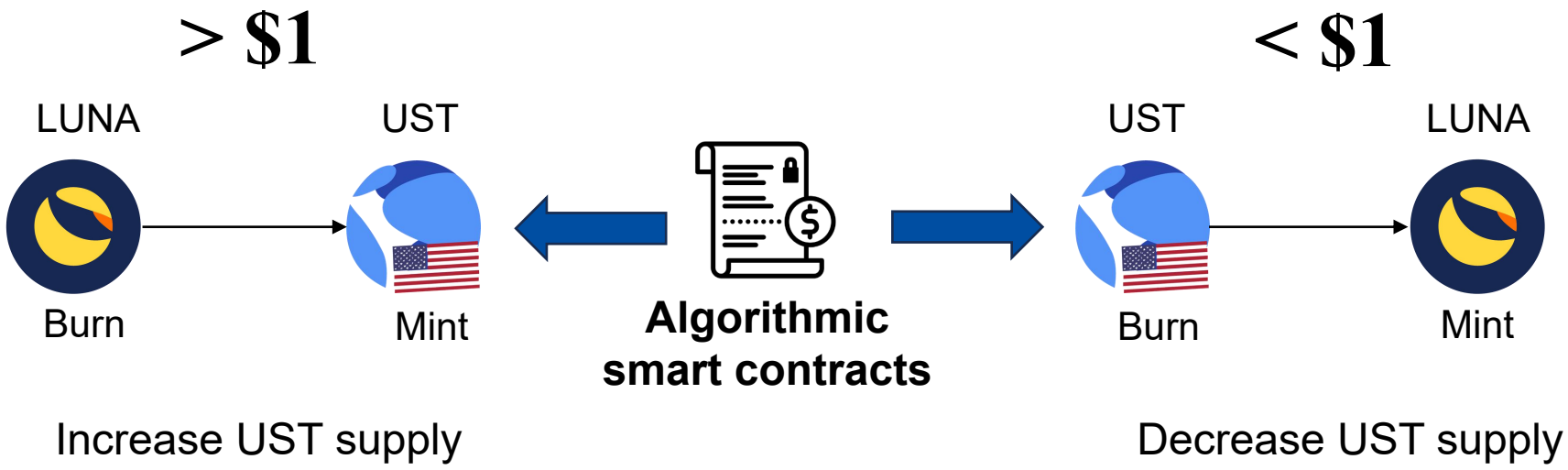
Take DAI as an example

Collateralized Debt Positions (CDPs):

- To generate DAI via the MakerDAO platform, users need to create CDPs, which is a smart contract on the MakerDAO where users lock up their ETH as collateral and generate DAI based on the value of that collateral.
- Users are required to maintain a certain collateralization ratio to ensure the stability of the system. If the value of the locked ETH falls below a certain threshold, the CDP may be liquidated, and the collateral auctioned to cover the debt.



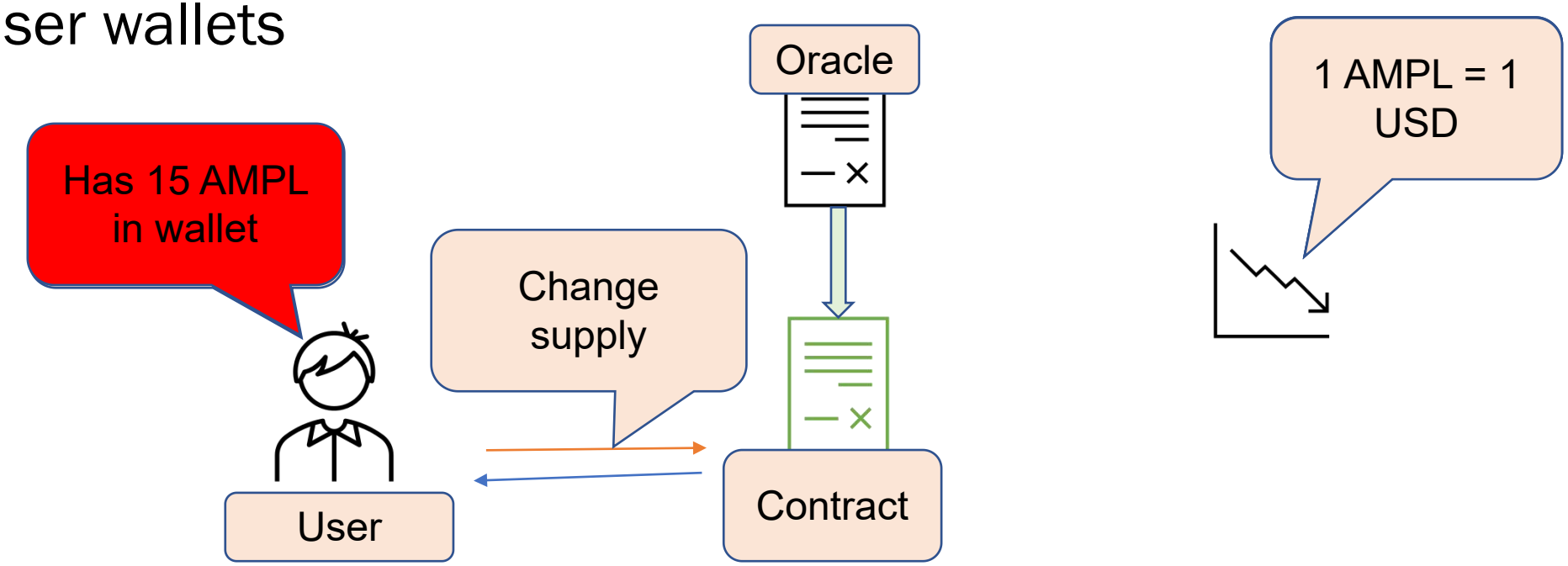
Algorithmic stablecoin



- Algorithmic stablecoins do not hold reserve assets. Their primary distinction is the strategy of keeping the stablecoin's value stable by controlling its supply through an algorithm, essentially a computer program running a preset formula.
- UST is an algorithmic stablecoin. It is anchored to $1 by burning (destroying) LUNA tokens in order to mint (create) new UST tokens.

Algorithmic stablecoins: Rebasing

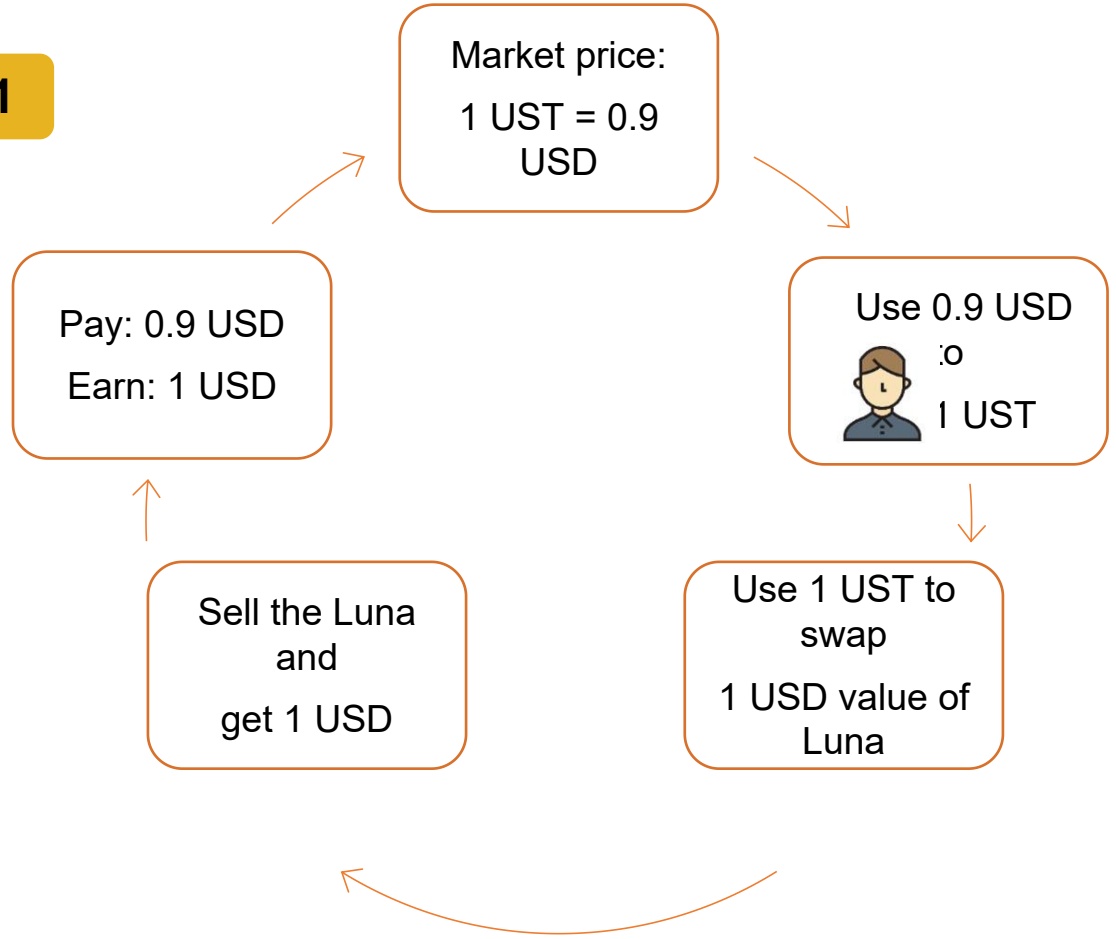
- Follows the simplest way to achieve price stability
- If oracle says price above peg, directly increase token supply in user wallets
- If oracle says price below peg, directly decrease token supply in user wallets



Take UST as an example

In the Terra ecosystem, users can swap the LUNA token for UST at a guaranteed price of $1.

The price of UST < $1





Burn UST





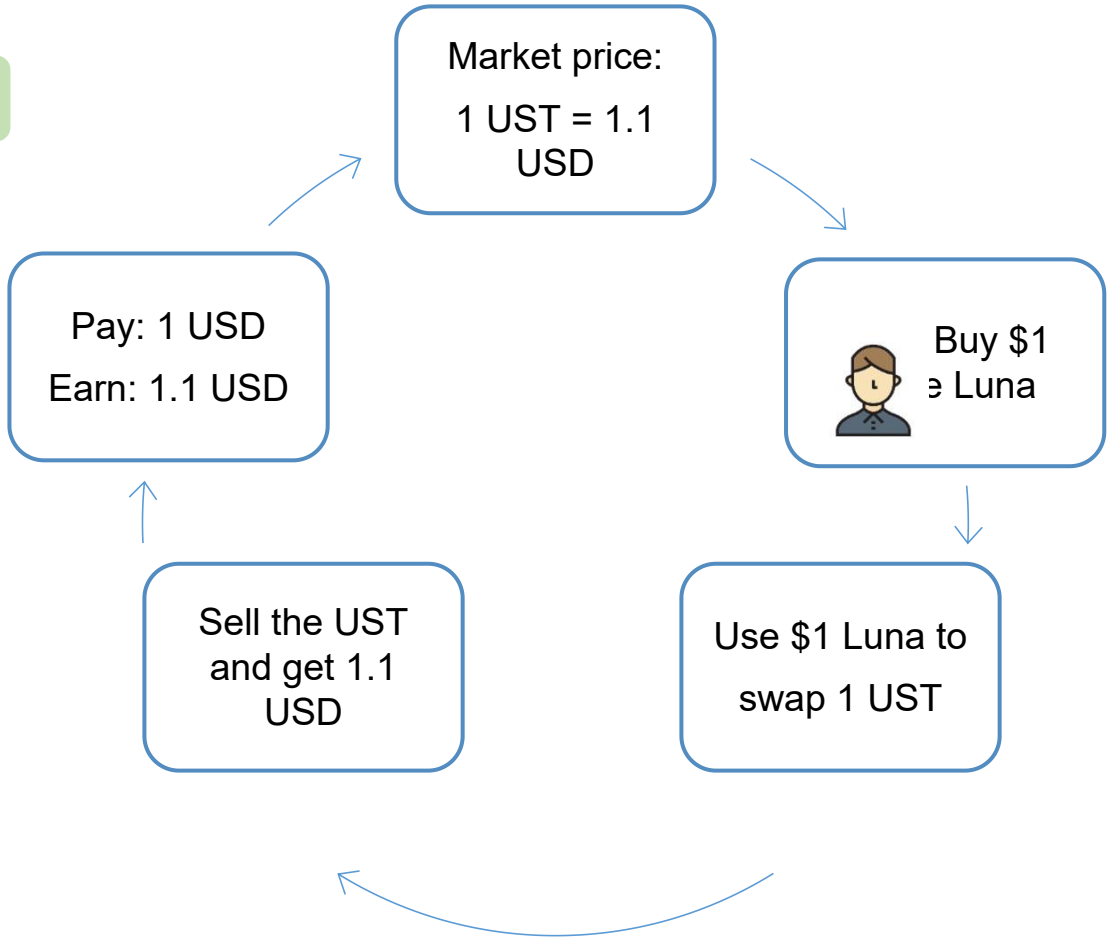
Mint Luna

Decrease supply

Take UST as an example

In the Terra ecosystem, users can swap the LUNA token for UST at a guaranteed price of $1.

The price of UST > $1





Burn Luna

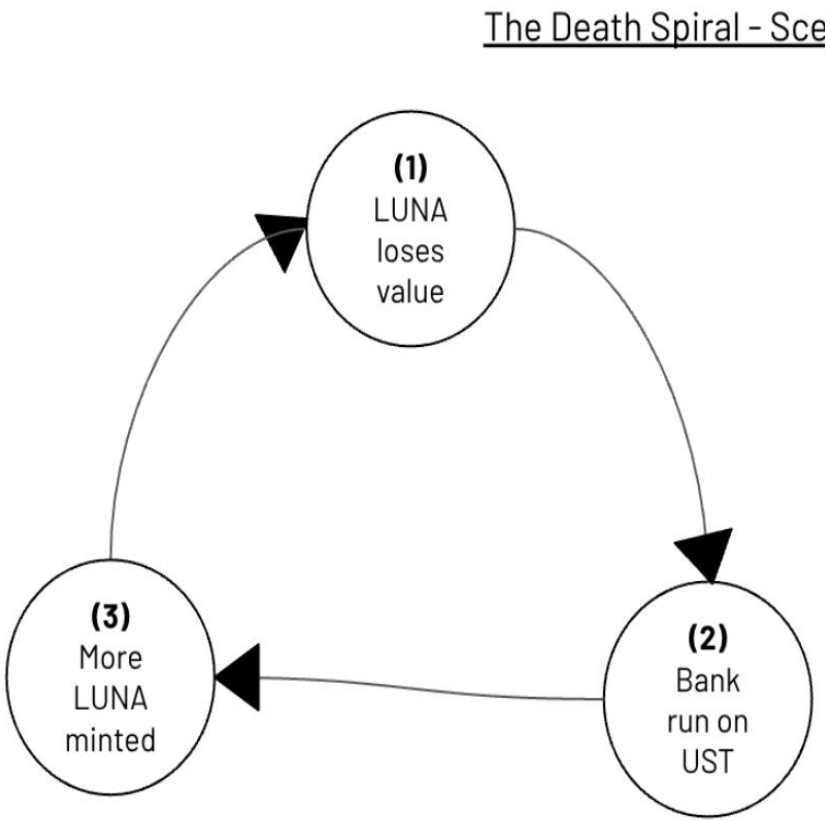




Mint UST

Increase supply

The “Death Spiral”

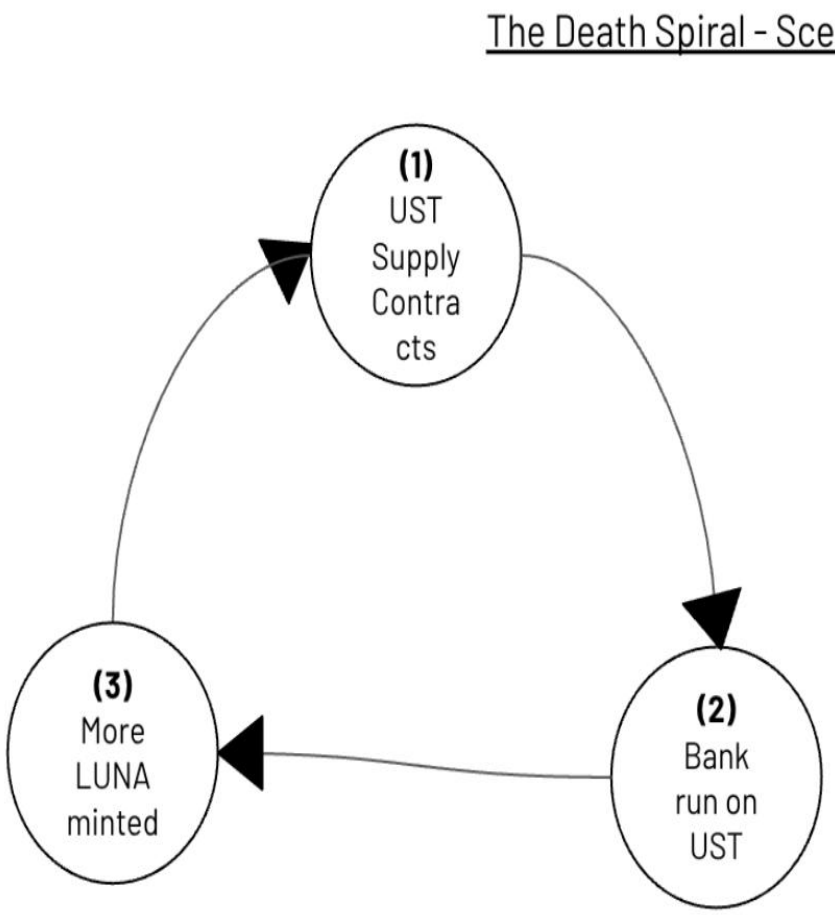


(1) A broader bear market causes the share token ($LUNA) to fall in value

(2) $UST holders fear a peg break and sell their UST

(3) More $LUNA is minted, leading to further price declines

The “Death Spiral”



(1) A contraction in UST supply leads to minting $LUNA

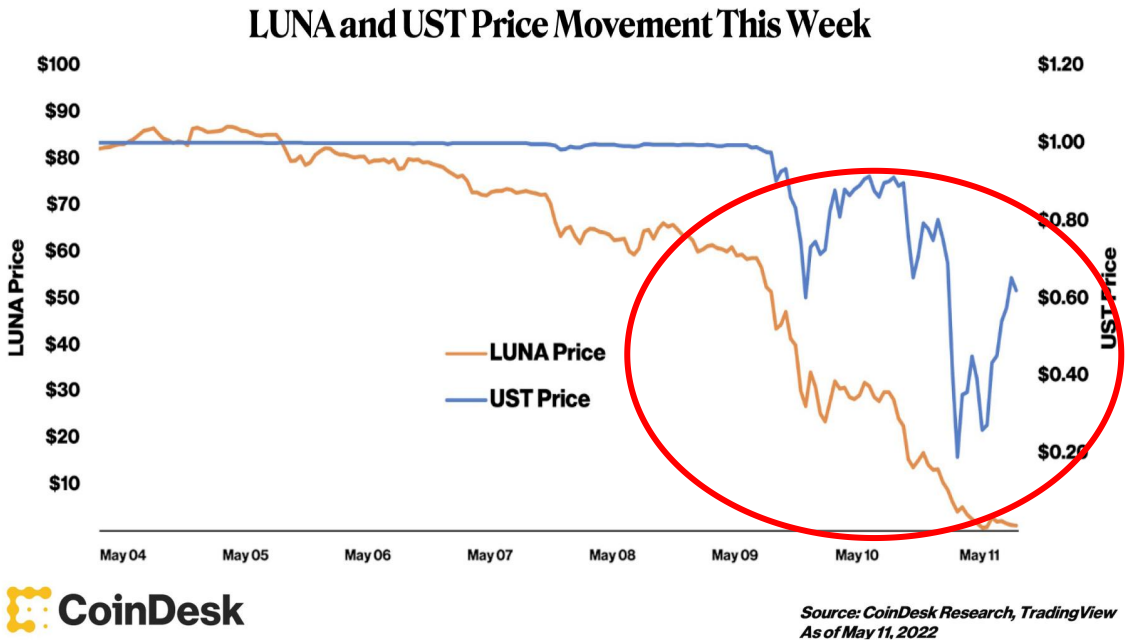
(2) $LUNA price declines. A sufficiently large decline may create a crisis of confidence in $UST as holders rush to redeem it for $LUNA

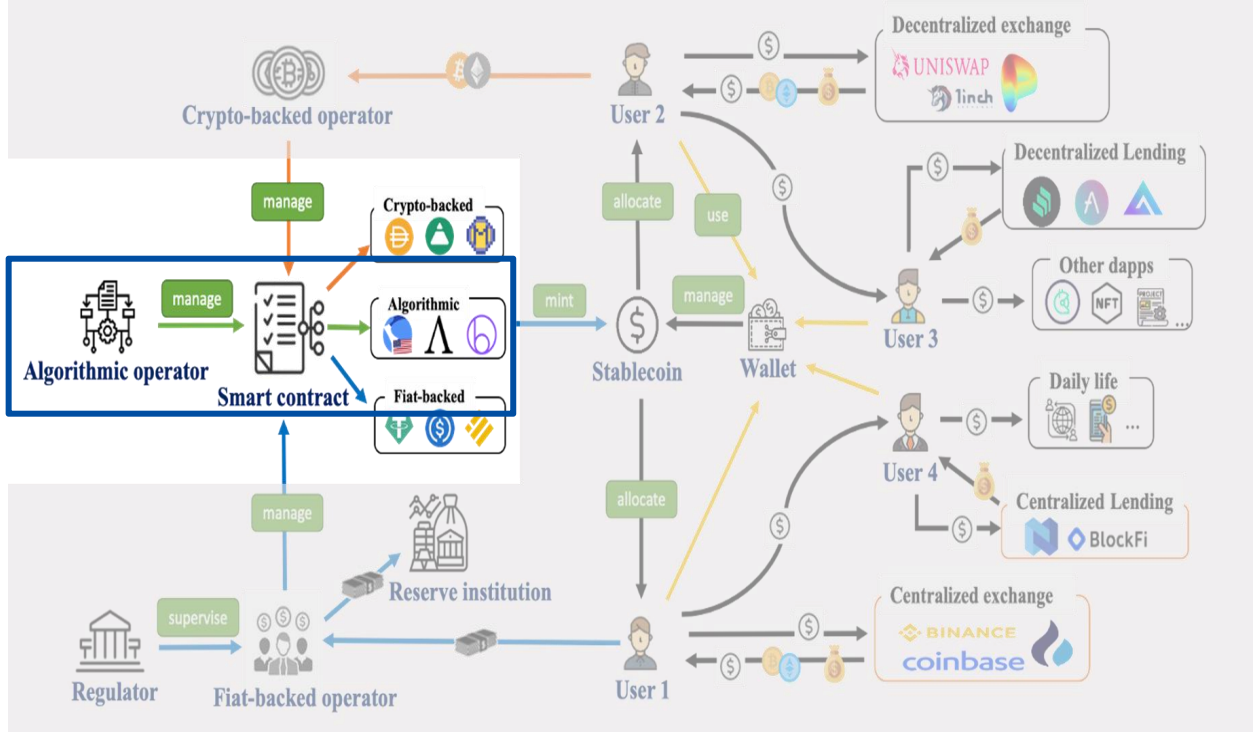
(3) More $LUNA is minted, leading to further price declines

Stablecoins are not “stable”

- Operator risk
- Smart contract risk
- Third-party risk
- Operational risk
- Collateral risk
- Regulatory risk

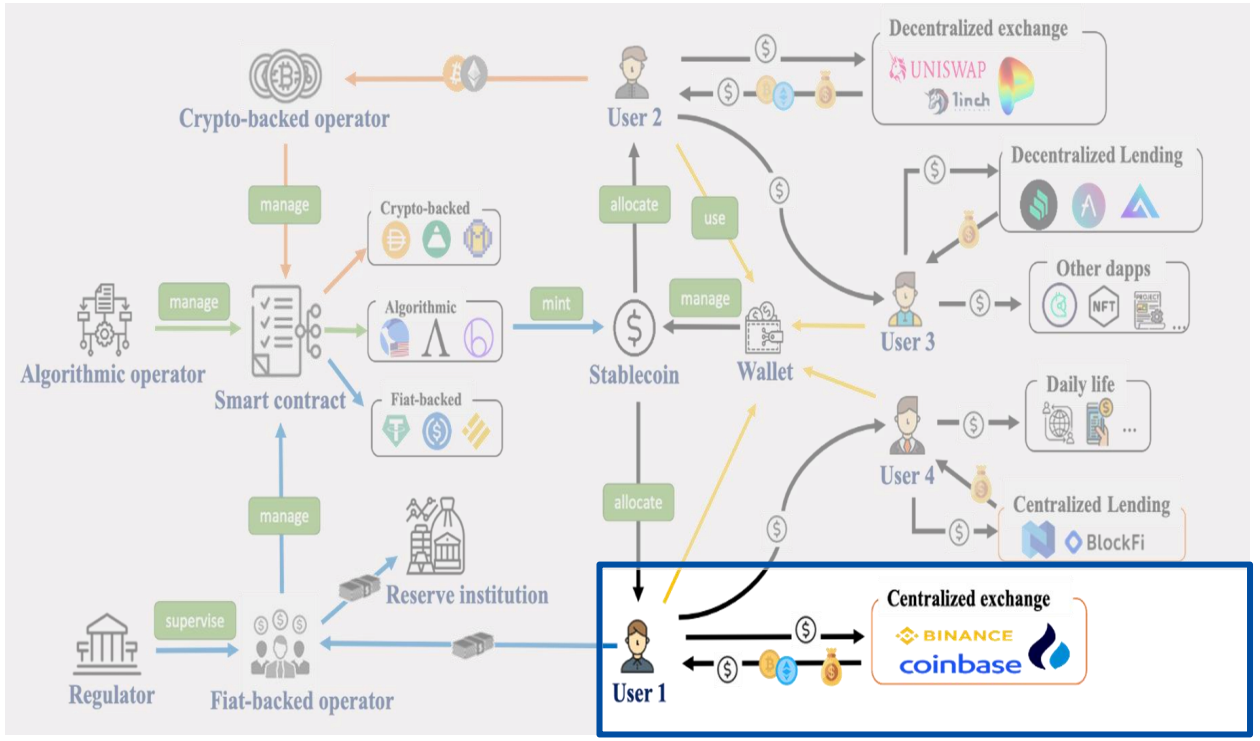
Terra-Luna Crash





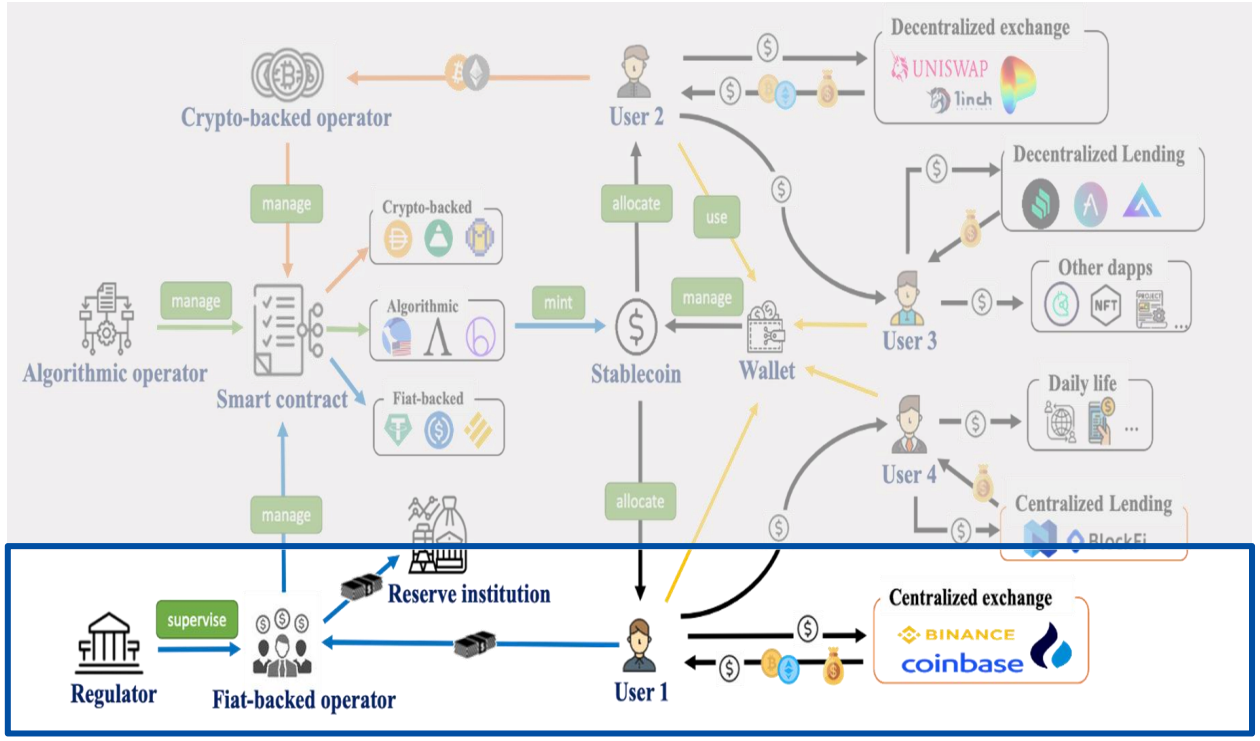
FTX Failure


BANKRUPTCY
FTX


Crypto-backed operator
User 2
Decentralized exchange
UNISWAP
1inch
manage
Crypto-backed
allocate
use
Decentralized Lending
Algorithmic operator
manage
Smart contract
Algorithmic
mint
Stablecoin
manage
Wallet
User 3
Other dapps
NFT
Fiat-backed
Daily life
User 4
manage
allocate
Centralized Lending
BlockFi
Reserve institution
Centralized exchange
BINANCE
coinbase
supervise
Regulator
Fiat-backed operator
User 1

Suspension of BUSD Issuance





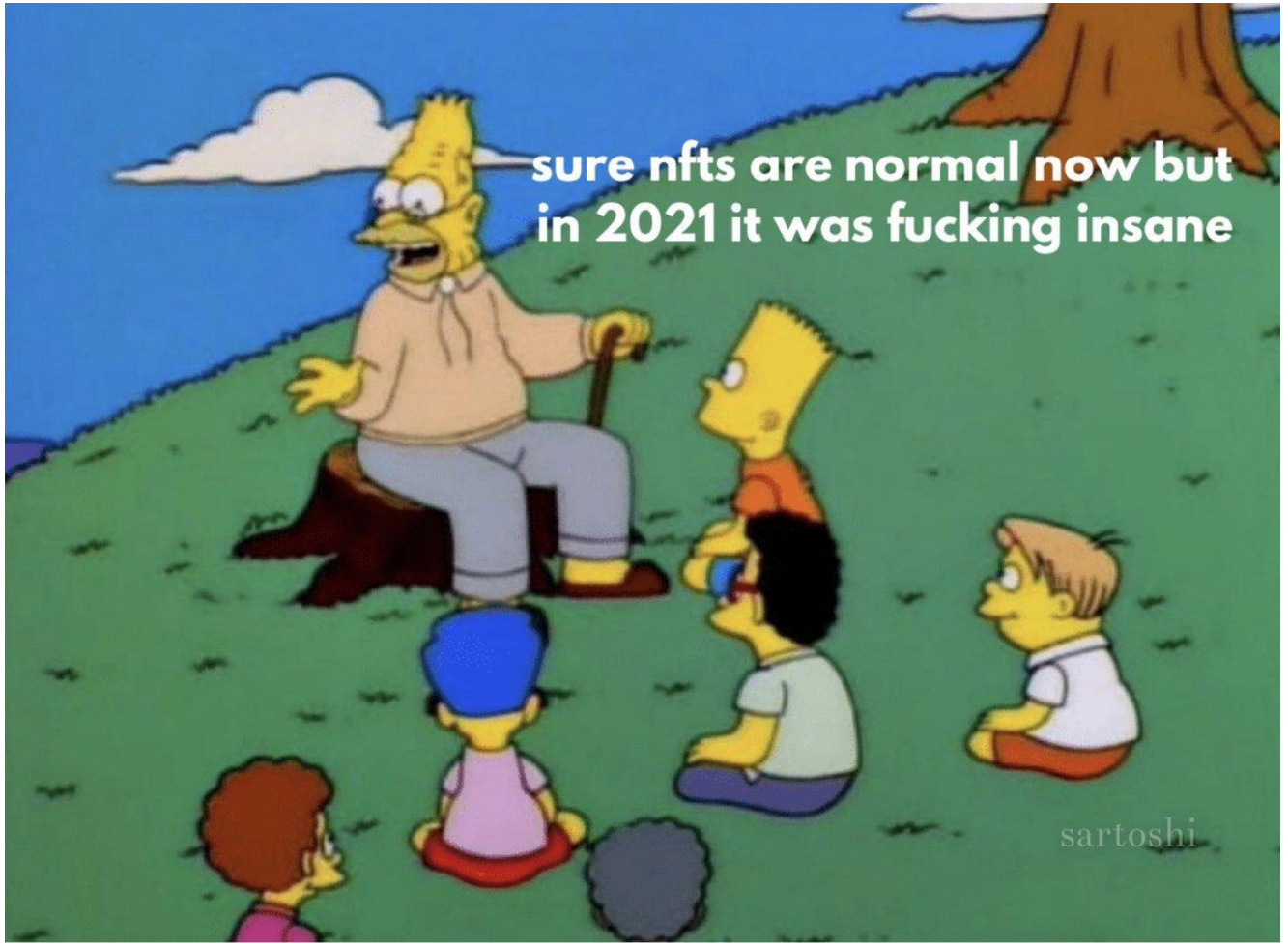
Fungible Token

- Divisible: $1 = $0.1*10, 1$BTC = 0.1$BTC*10
- Non-Unique: $1 = $1, 1$BTC = 1$BTC

- ERC-20 Standard
  - Token Transfer
  - Tokens Approval
  - ...

- Real-World Purpose: Payment System, Store of Value

Something New since 2015

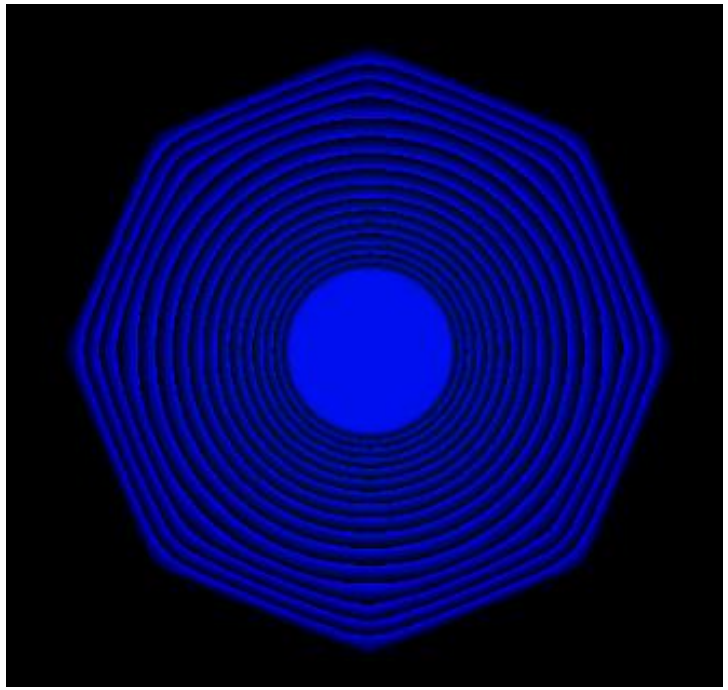


Non-Fungible Tokens (NFT)

- Unique
- Non-Divisible
- Non-Replicable
- Associate with a Title of Ownership: Usually can be transferred
- ERC-721 Standard
- Real-World Purpose: IP, Stock, Artwork, Gaming, Academic Title ...

Quantum: The First NFT

- Animation Art
- by Kevin McCoy in May 3, 2014
- On the NameCoin blockchain
- first to conceive of putting unique digital works of art on the blockchain
- sold for $1.47m in June 2021 during Sotheby's "Natively Digital" auction

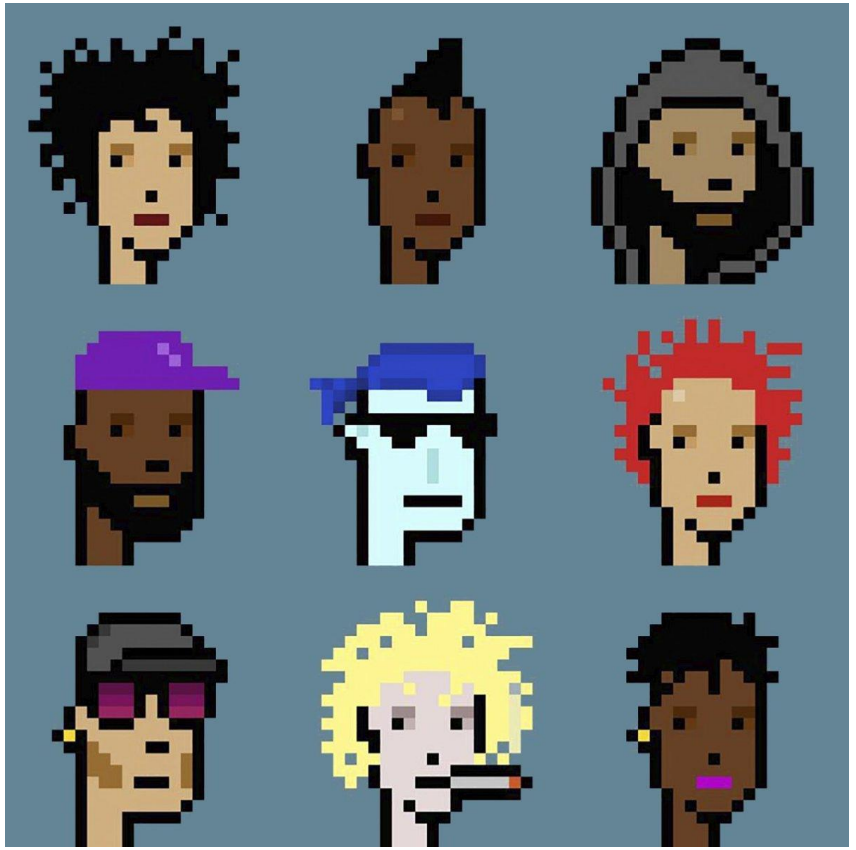


Ξtheria: The First NFT Project

- a virtual world in which players can own tiles, farm and build
  - the entire state of the world and all player actions are on-chain
- launched and demonstrated in 2017
  - at Ethereum's 1st developer conference
- 457 purchasable and tradable hexagonal tiles
  - each hardcoded to 1 ETH (US$0.43 at the time of launch)
  - went unsold for more than 5 years until March 13, 2021
  - sold within 24 hours for a total of US$1.4 million

CryptoPunks

- Collection Information
  - 10K Randomly Generated Collectibles
  - Release: June 2017
  - Developer(s): Larva Labs
  - Designer(s): J. Watkinson & M. Hall
  - Platform(s): Ethereum
- featured in New York Times, Christie's of London, Art|Basel Miami…
- inspired the ERC-721 standard



The Boom: CryptoKitties

- Unlimited Quantity Digital Cats
- by Dapper Labs
- Release: November 2017
- Genre: Virtual Pet
- Defined as a blockchain game
- Players purchase, breed and trade virtual cats that have different visual features of varying levels of rarity.



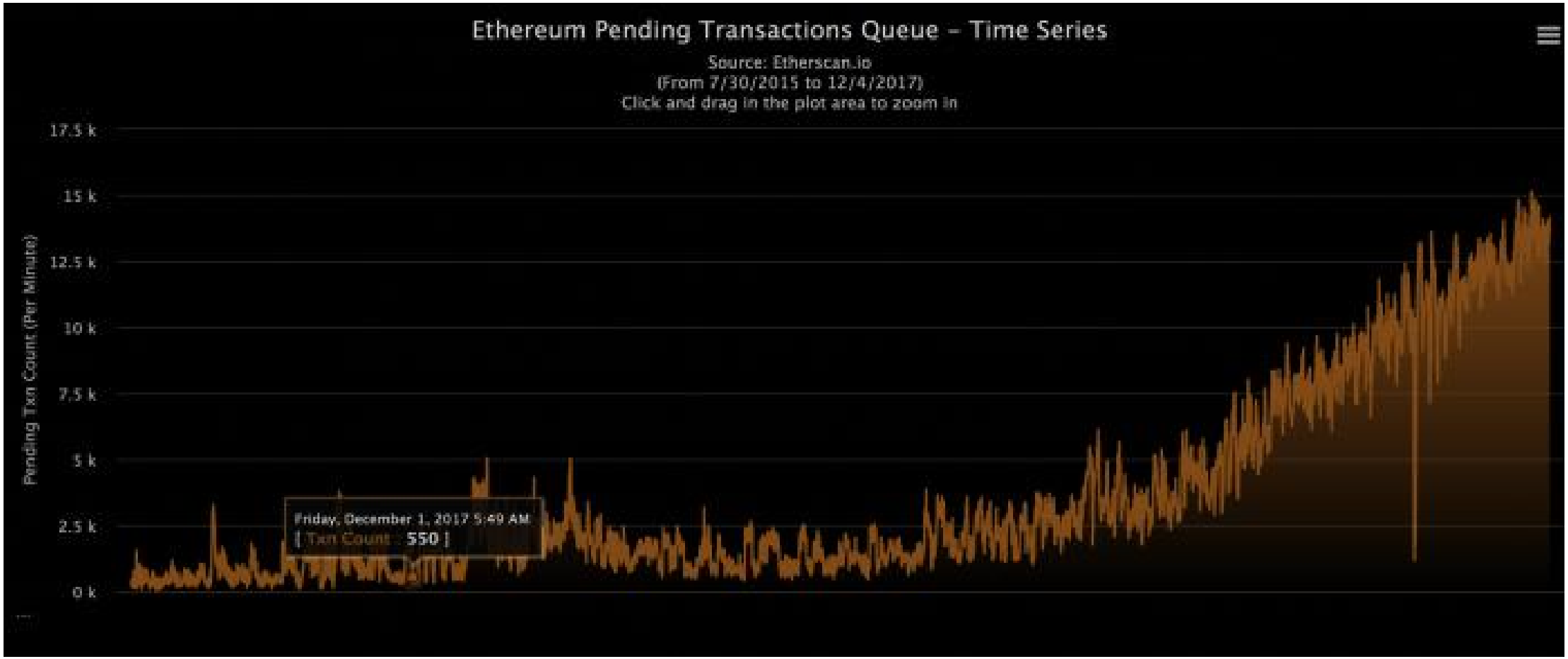


The Boom: CryptoKitties



CryptoKitties

Traffic Congestion Caused by CryptoKitties



The NFT Market

- Rapid growth during 2020
  - value tripling to $250 million
  - $200+ million spent on NFTs in first 3 months of 2021
- Trademark Applications to US Patent and Trademark Office
  - 3 in 2020, 1200 in 2021, 450 in January 2022
  - Including the NYSE, Star Trek, Walmart, Ticketmaster, Yahoo ...
- Collapsing since May 2022
  - Daily sales of NFT tokens had declined 92% from Sep 2021
  - Number of active wallets in the NFT market fell 88% from Nov 2021

What do you get as a NFT owner?

- On-chain Proof of Ownership,
- No Inherent Legal Meaning (Uncertain)
  - Copyright
  - Intellectual Property Rights
  - Other Legal Rights
- Does not restrict the sharing or copying of its associated digital file
- Does not prevent the creation of NFTs that reference identical files

Full Commercial Rights to the NFT Owner?

Do the rights transfer if NFT token sold on secondary?

Create a derivative and then later sell the original NFT?

Recommend Reading:
https://cryptoconsigliere.substack.com/p/the-nft-linked-trust

ERC-721: Non-Fungible Token Standard

- A free, open standard Interface proposed by Dappers Lab
  - based on EIP-721 on Ethereum
  - Provide basic functionality to track and transfer NFTs
  - Allow wallet/broker/auction applications to work with any NFT
- NFT Identifiers
  - every NFT is identified by a unique uint256 tokenId
  - the pair (contract address, uint256 tokenId) is globally unique

ERC-721 Interface

totalSupply()	• Total amount of emitted tokens.
balanceOf(_owner)	• Amount of tokens in specific _owner's wallet.
ownerOf(_tokenId)	• Returns wallet address of the specific tokens owner.
transfer(_to, _tokenId)	• Transfers token with _tokenId from senders wallet to specific wallet.
takeOwnership(_tokenId)	• Claims the ownership of a given token ID
approve(_to, _tokenId)	• Approves another address to claim for the ownership of the given token ID
Transfer & Approval	• Two Events for NFT transfer

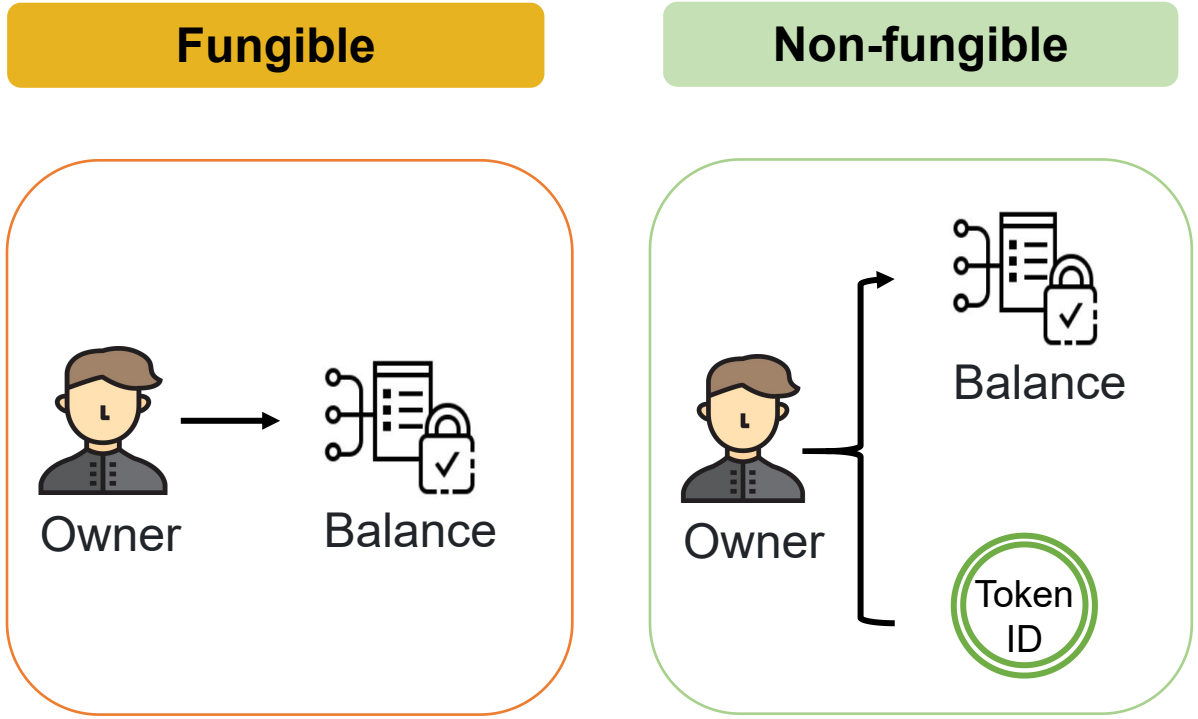
ERC-721 Token Standard

The ERC-721 introduces a standard for NFT, in other words, this type of Token is unique and can have different value than another Token from the same Smart Contract, maybe due to its age, rarity or even something else like its visual.

Official Link: http://eips.ethereum.org/EIPS/eip-721

Core components of the ERC-721 standard:

- Basic information
- Owner of the token
- Market supply and token balance
- Transfer and approval function
- Event



ERC-721 Token Standard

Function: basic information of the token

The metadata extension is OPTIONAL for ERC-721 smart contracts. This allows your smart contract to be interrogated for its name and for details about the assets which your NFTs represent.

- **function name() external view returns (string _name);**
- **function symbol() external view returns (string _symbol);**
- **function tokenURI(uint256 _tokenId) external view returns (string);**





ERC-721 Token Standard

Function: owner of the token

- **OwnerOf**

function ownerOf(uint256 _tokenId) **external view returns** (**address**);

Returns the owner of the tokenId token.



12. ownerOf

tokenId (uint256)

4990

Query

address

[**ownerOf(uint256)** method Response]

» *address* : 0x29469395eAf6f95920E59F858042f0e28D98a20B

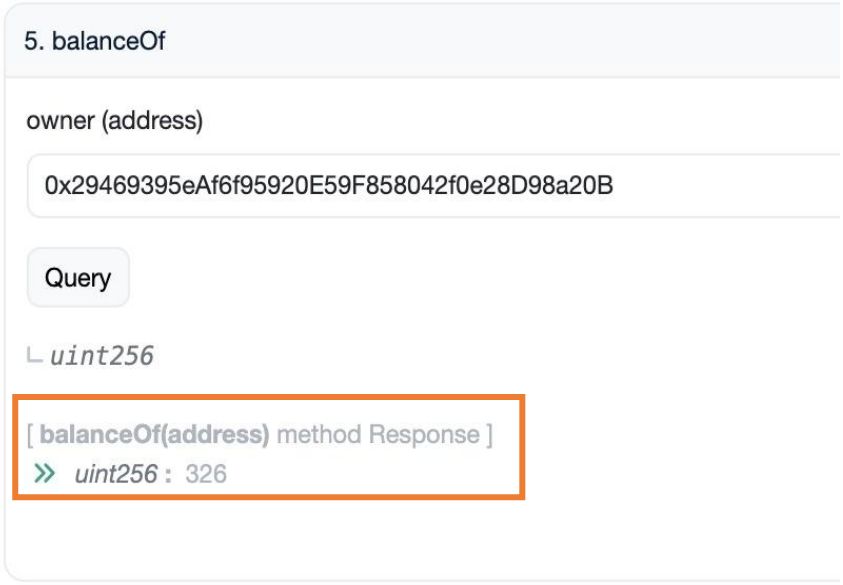
ERC-721 Token Standard

Function: owner of the token

- **BalanceOf**

function balanceOf(address _owner) **external view returns** (**uint256**);

Returns the number of tokens in owner's account.



ERC-721 Token Standard

Function: transfer tokens from one account to another

If a function is payable, it means you can send token when calling it.

- **function transferFrom(address _from, address _to, uint256 _tokenId) external payable;**

Transfers the [_tokenId] token from the [_from] address to the [_to] address.

Key Notes:

No safety checks for contract recipients.

Risk of locking tokens if [_to] is an incompatible contract.

Use only when [_to] is known to be an externally owned account (EOA).

ERC-721 Token Standard

Function: transfer tokens from one account to another

If a function is payable, it means you can send token when calling it.

- **function safeTransferFrom(address _from, address _to, uint256 _tokenId) external payable;**

Safely transfers the token without additional data.

Safety Mechanism:

Checks if [_to] is a contract:

- *If yes, verifies that [_to] implements the onERC721Received method and returns a valid value.*
- *Reverts the transfer if the check fails.*

Prevents accidental transfers to contracts that cannot handle tokens.

ERC-721 Token Standard

Function: approve whether an amount of token from an account can be spent by a third-party account

- **function getApproved**(**uint256** _tokenId) **external view returns** (**address**);

  Gets the approved address for a tokenId.

- **function approve**(**address** _approved, **uint256** _tokenId) **external payable**;

  Sets the [_approved] address as the authorized operator for the [_tokenId].

- **function isApprovedForAll**(**address** _owner, **address** _operator) **external view returns (bool**);

  Gets whether an operator is approved by a given owner.

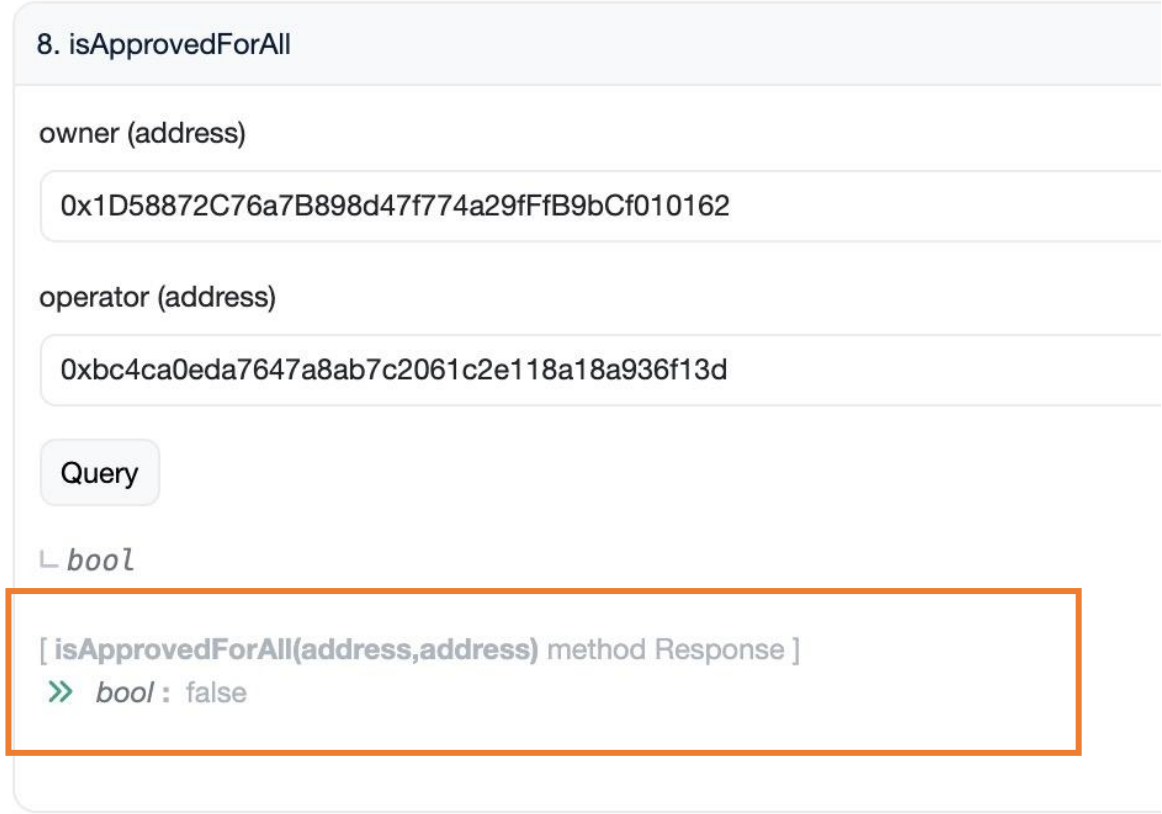
- **function setApprovalForAll**(**address** _operator, **bool** _approved) **external;**

  Allows the owner to authorize (or revoke) another address (operator) to manage all his token.

ERC-721 Token Standard

Function: approve whether an amount of token from an account can be spent by a third-party account





ERC-721 Token Standard

Event

- **event Transfer(address indexed _from, address indexed _to, uint256 _tokenId);**

  Record the transfer of token ownership.

  This emits when “tokenId” token is transferred from address [_from] to address [_to].

- **event Approval(address indexed _owner, address indexed _approved, uint256 _tokenId);**

  Record the authorization operations for a single token.

  This emits when the operator is enabled for an owner to manage the tokenId of the owner.

- **event ApprovalForAll(address indexed _owner, address indexed _operator, bool _approved);**

  Record the enabling or disabling of global authorization (for all tokens).

  This emits when the operator is enabled or disabled for an owner to manage all tokenIds of the owner.

Event Log

Transaction Receipt Event Logs

151

Address Bored Ape Yacht Club: BAYC Token

Name `Approval (index_topic_1 address owner, index_topic_2 address approved, index_topic_3 uint256 tokenId)` View Source

Topics

0 `0x8c5be1e5ebec7d5bd14f71427d1e84f3dd0314c0f7b2291e5b200ac8c7c3b925`

1: owner Dec → `0x8bfe553Ab794615B6009fbA5F077509A453DCBFd`

2: approved Dec → `0x0000000000000000000000000000000000000000`

3: tokenId Dec → 2549

Data `0x`

152

Address Bored Ape Yacht Club: BAYC Token

Name `Transfer (index_topic_1 address from, index_topic_2 address to, index_topic_3 uint256 tokenId)` View Source

Topics

0 `0xddf252ad1be2c89b69c2b068fc378daa952ba7f163c4a11628f55a4df523b3ef`

1: from Dec → `0x8bfe553Ab794615B6009fbA5F077509A453DCBFd`

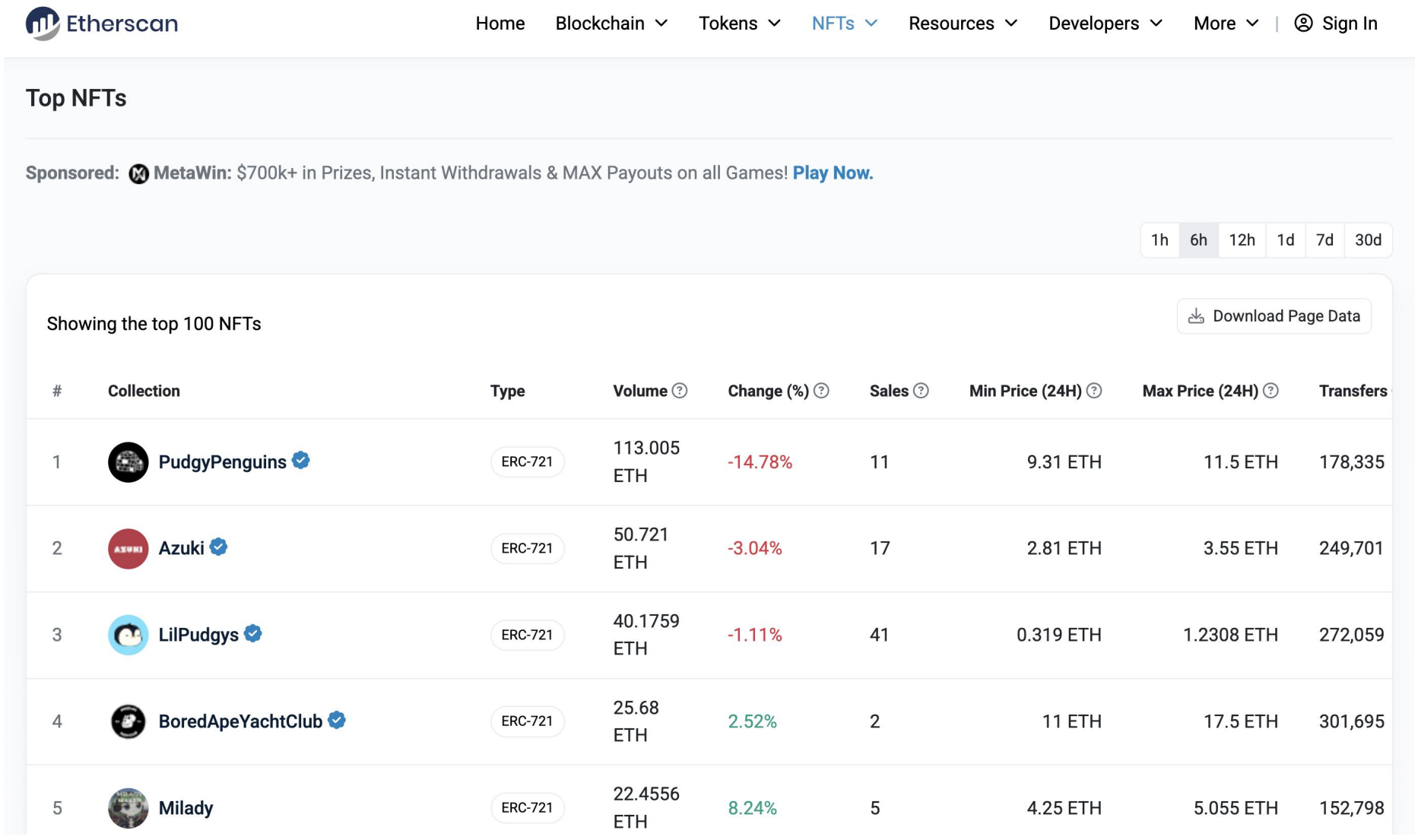
2: to Dec → `0x1D58872C76a7B898d47f774a29fFfB9bCf010162`

3: tokenId Dec → 2549

Data `0x`

ERC-721 Tokens

https://etherscan.io/nft-top-contracts



Zeppelin ERC721 Implementation

- Zeppelin Implementation
  - provides some additional useful functions:
    - *tokensOf(_owner)* - Returns list of token ID's of specific _owner
    - *approvedFor(_tokenId)* - Returns the approved address to take a given token ID
    - https://github.com/OpenZeppelin/openzeppelin-contracts/blob/master/contracts/token/ERC721
- Reference
  - ERC-721
    - http://erc721.org/
  - Coding a Crypto Collectible in 15 minutes - Ethereum ERC721
    - https://www.youtube.com/watch?v=7TiXsOLiIrc

NFT Metadata: Motivation

- NFTs usually represent digital assets (JPEG file, GIF, MP4, etc.)
  - Technical and Financial catastrophe
  - 1 GB of data on Ethereum costs about 17,500 ETH
- NFT metadata is essentially a workaround
- NFT metadata specifies
  - what that data is
  - the visual or auditory asset
  - other information like transactional history

NFT Metadata

- Each ERC-721 contains a “metadata” string in its definition
  - defines what the NFT actually is (e.g., .JPEG)
  - references the file that exists online: HTTP URL or an IPFS hash

- ERC-721 specifies metadata in a JSON format
  - ERC-721 stores the metadata as a URI inside the contract

```
function tokenURI(uint256 _tokenId) public view returns (string)
```

the URI string points to a location stores the token’s JSON description

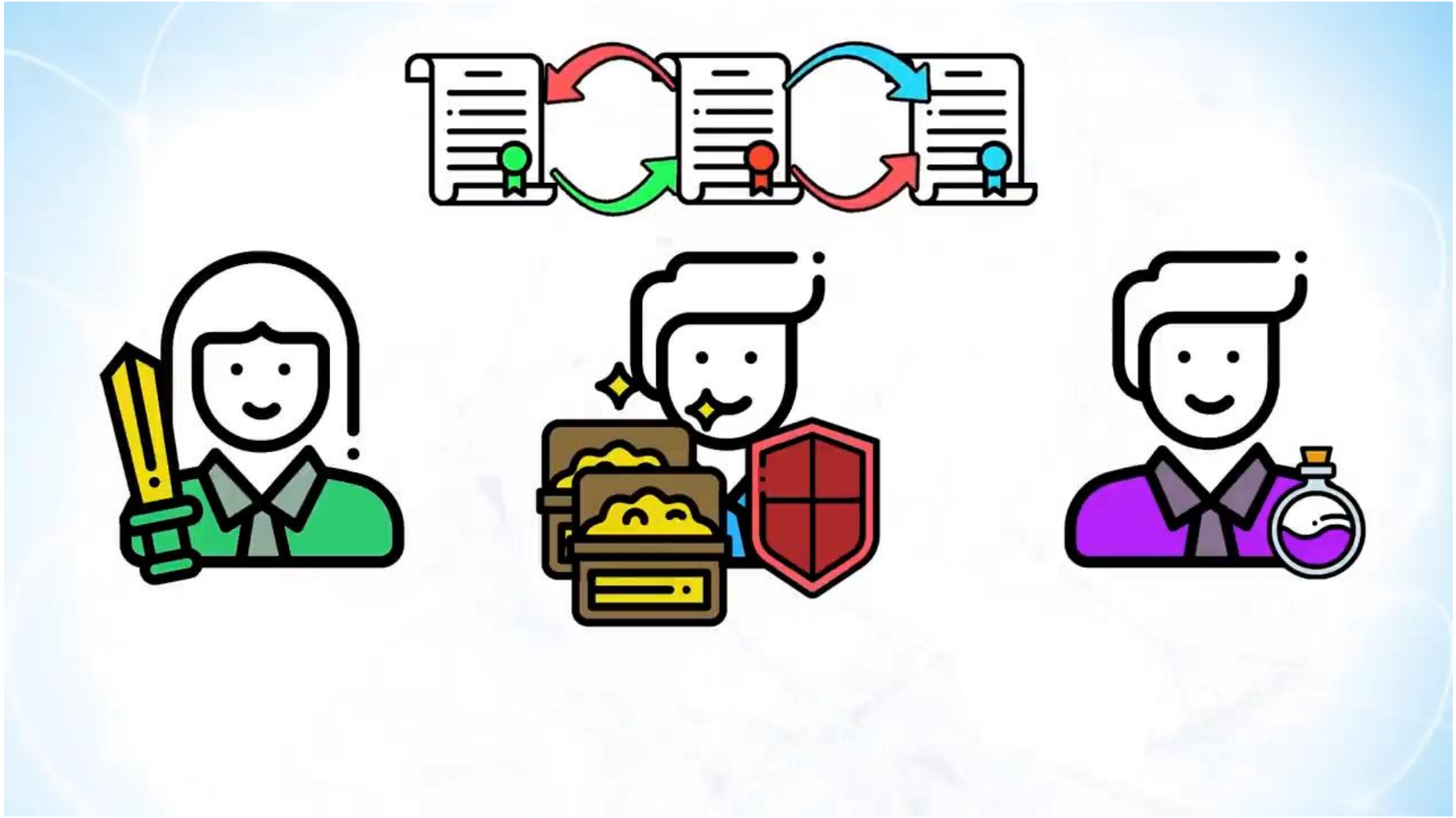
NFT Metadata: An Example

```
{
  "title": "Asset Metadata",
  "type": "object",
  "properties": {
    "name": {
      "type": "string",
      "description": "Identifies the asset to which this NFT represents",
    },
    "description": {
      "type": "string",
      "description": "Describes the asset to which this NFT represents",
    },
    "image": {
      "type": "string",
      "description": "A URI pointing to a resource with mime type image/* representing the asset to which this NFT represents. Consider making any images at a width between 320 and 1080 pixels and aspect ratio between 1.91:1 and 4:5 inclusive.",
    }
  }
}
```

ERC-721 Variance

- ERC-721A
  - by Azuki
  - minting multiple tokens for close to the cost of one
- ERC-721R
  - by CryptoFighters
  - allows minters to return the minted NFT for a refund

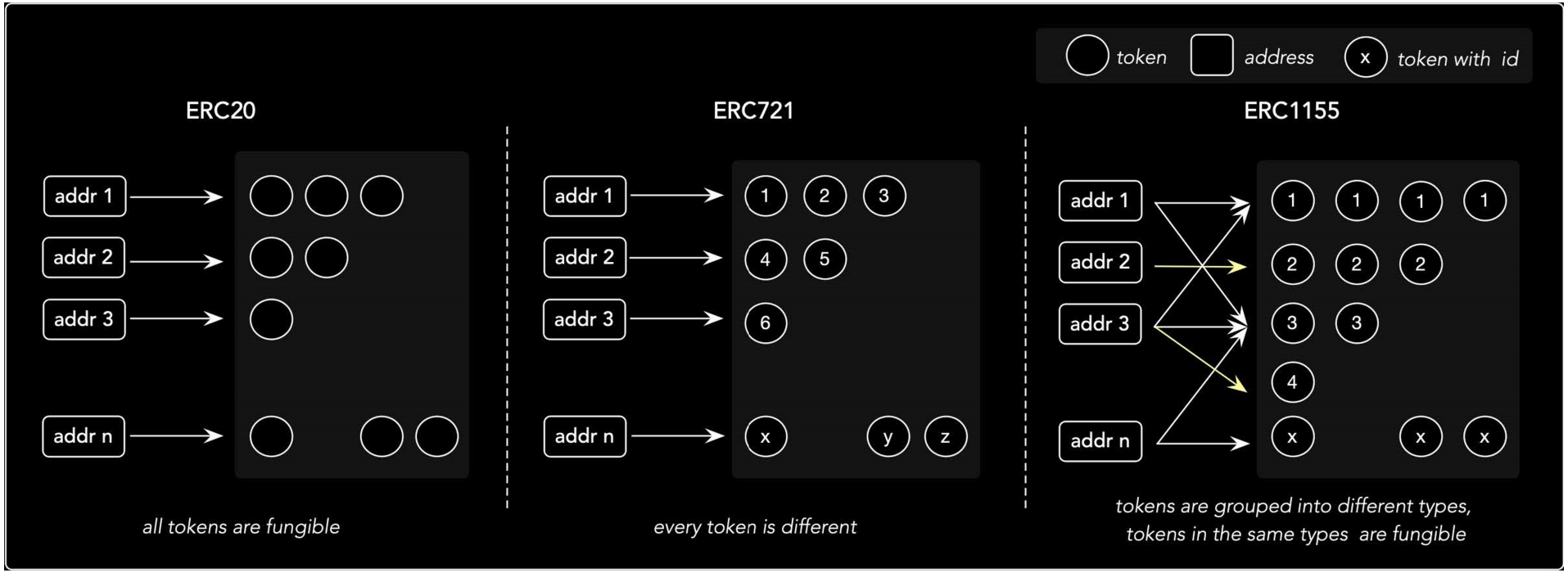
ERC-1155: Multi-Token Standard



ERC-1155 Explained

ERC-1155: Multi-Token Standard

- Can represent any number of fungible and non-fungible token types
- Each tokenId can represent different configurable token types.



ERC-1155: Enjin Coin



Other NFT related ERC: Improves 721

- ERC-2309: ERC-721 Consecutive Transfer Extension
  - Create, transfer, and burn a massive amount of tokens
- ERC-4341: Ordered NFT Batch Standard
  - Create unicoded NFTs to represent logographic characters
- ERC-4494: Permit for ERC-721 NFTs
  - An approve architecture with signed message-based approval flow
- ERC-4521: 721/20-compatible transfer
  - Add a backwards-compatible transfer() in ERC-20

Other NFT related ERC: Improves 721

- ERC-4885: Subscription Token Standard for NFTs and Multi Tokens
  - Subscribe to non-fungible and multi tokens
- ERC-4944: Contract with Exactly One Non-fungible Token
  - An ERC-721 compatible contract with a total supply of one
- ERC-5007: TimeNFT, ERC-721 Time Extension
  - Add start time and end time to ERC-721 tokens
- ERC-5008: ERC-721 Nonce Extension
  - Add a nonce function: nonce changed when a token is transferred

Other NFT related ERC: Composition

- ERC-998: ERC-998 Composable Non-Fungible Token Standard
  - Enable ERC721 tokens to own other ERC721 and ERC20 tokens
- ERC-3525: Semi-Fungible Token Standard
  - ERC-721 extension/compatible token standard
  - <ID, SLOT, VALUE> model that represents the token structure
  - Be suited for financial instruments
- ERC-4675: Multi-Fractional Non-Fungible Tokens
  - Fractionalize NFT

Other NFT related ERC: Metadata

- ERC-4906: EIP-721 Metadata Update Extension
  - Add a standard MetadataUpdate event
  - for third-party platforms to timely update the metadata of NFTs
- ERC-5185: NFT Updatable Metadata Extension
  - Allow controlled NFTs' metadata updates along predefined formulas
- ERC-4955: Vendor Specific Metadata Extension for NFT
  - Add a new field namespaces to the JSON schema for NFTs
- ERC-3569: Sealed NFT Metadata Standard
  - Immortalize NFT metadata

Other NFT related ERC: User and Rental

- ERC-4400: EIP-721 Consumable Extension
  - Add a consumer role for instances of EIP-721
- ERC-4799: NFT Ownership Designation Standard
  - Define arbitrarily complex ownership agreements on NFTs
  - For pre-existing NFTs without giving up the authenticity of the original
- ERC-4907: Rental NFT, ERC-721 User And Expires Extension
  - An additional role (user) which can be granted to addresses
  - A time where the role is automatically revoked (expires)
- ERC-5006: Rental NFT, NFT User Extension
  - An additional role (user) in ERC-1155

Other NFT related ERC: Right and License

- ERC-3754: A Vanilla Non-Fungible Token Standard
  - Represent concepts like rights, a form of abstract ownership
  - NFT options, oracle membership, virtual coupons, etc.
- ERC-4519: Non-Fungible Tokens Tied to Physical Assets
  - A standard interface for non-fungible tokens that represent physical assets
  - Manage smart assets (IoT devices) tied to NFTs in a secure and traceable way
- ERC-5218: NFT Rights Management
  - Create, transfer, and revoke licenses
  - Determine the current licensing state of an NFT

Other NFT related ERC: Fees

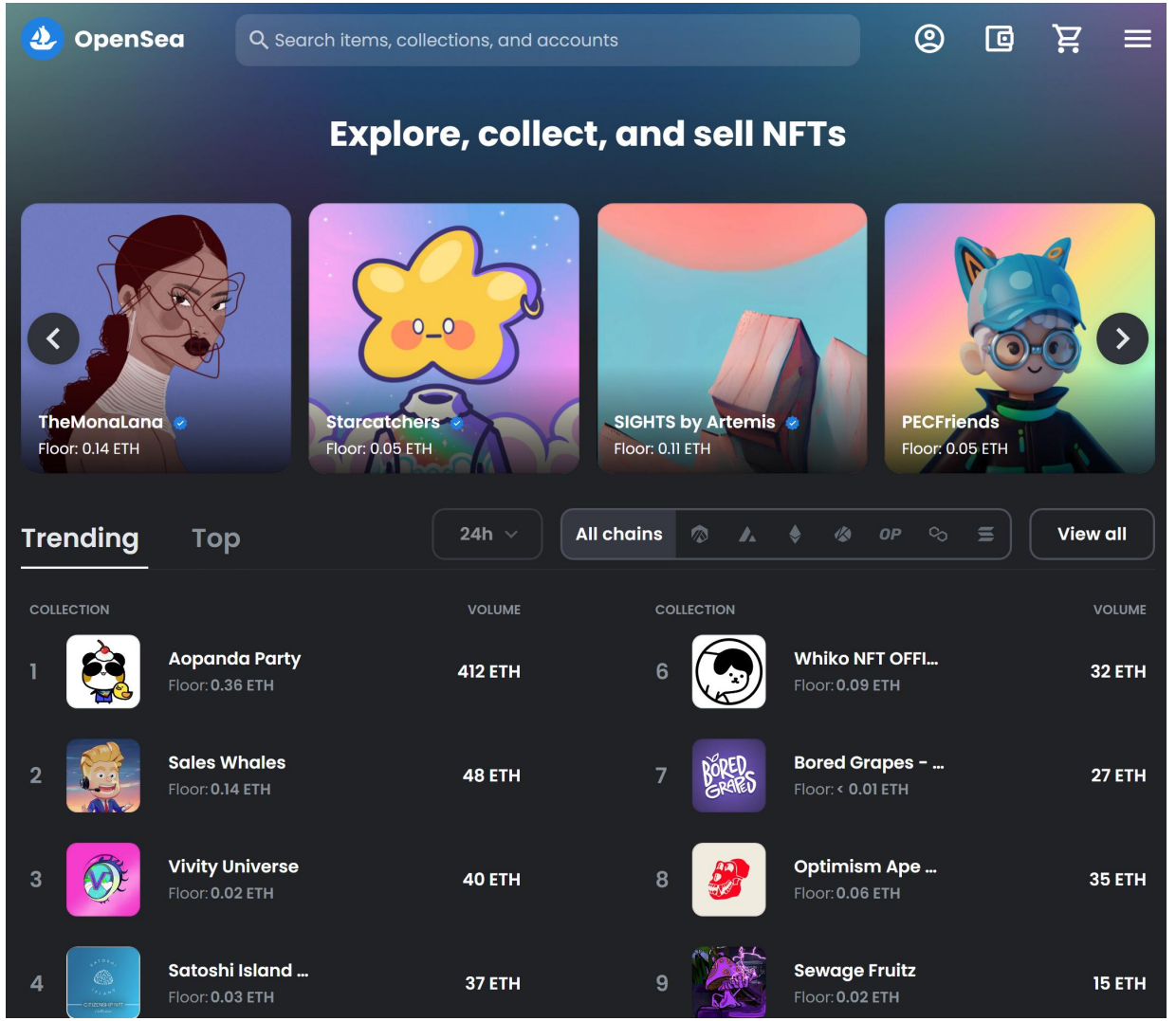
- ERC-2981: NFT Royalty Standard
  - Royalty payment information for NFTs
- ERC-4353: Interface for Staked Tokens in NFTs
  - Retrieve the staked token amount of an NFT
- ERC-4393: Micropayments Standard for NFTs and Multi Tokens
  - Enable tips to non-fungible and multi tokens
- ERC-5173: NFT Future Rewards (nFR)
  - Enable NFT owners to share future price increases after sold

NFT Marketplace

- Traditional Marketplaces
  - OpenSea
  - X2Y2
- Aggregators
  - blur.io
  - gem.xyz
  - et.xyz
- AMMs for NFT
  - Sudoswap
- Cross-Chain Marketplaces
  - TofuNFT
- For Artworks
  - SuperRare
  - Rarible
  - Async Art
  - Art Blocks
  - Nifty Gateway

OpenSea

- Founded in 2017
- Most Famous
- 90% of all NFT Trading
- Support Auction
- Easy NFT Creation
- $2.7 billion on May 1, 2022
- dropped by 99% in 4 months
- likely go public and IPO



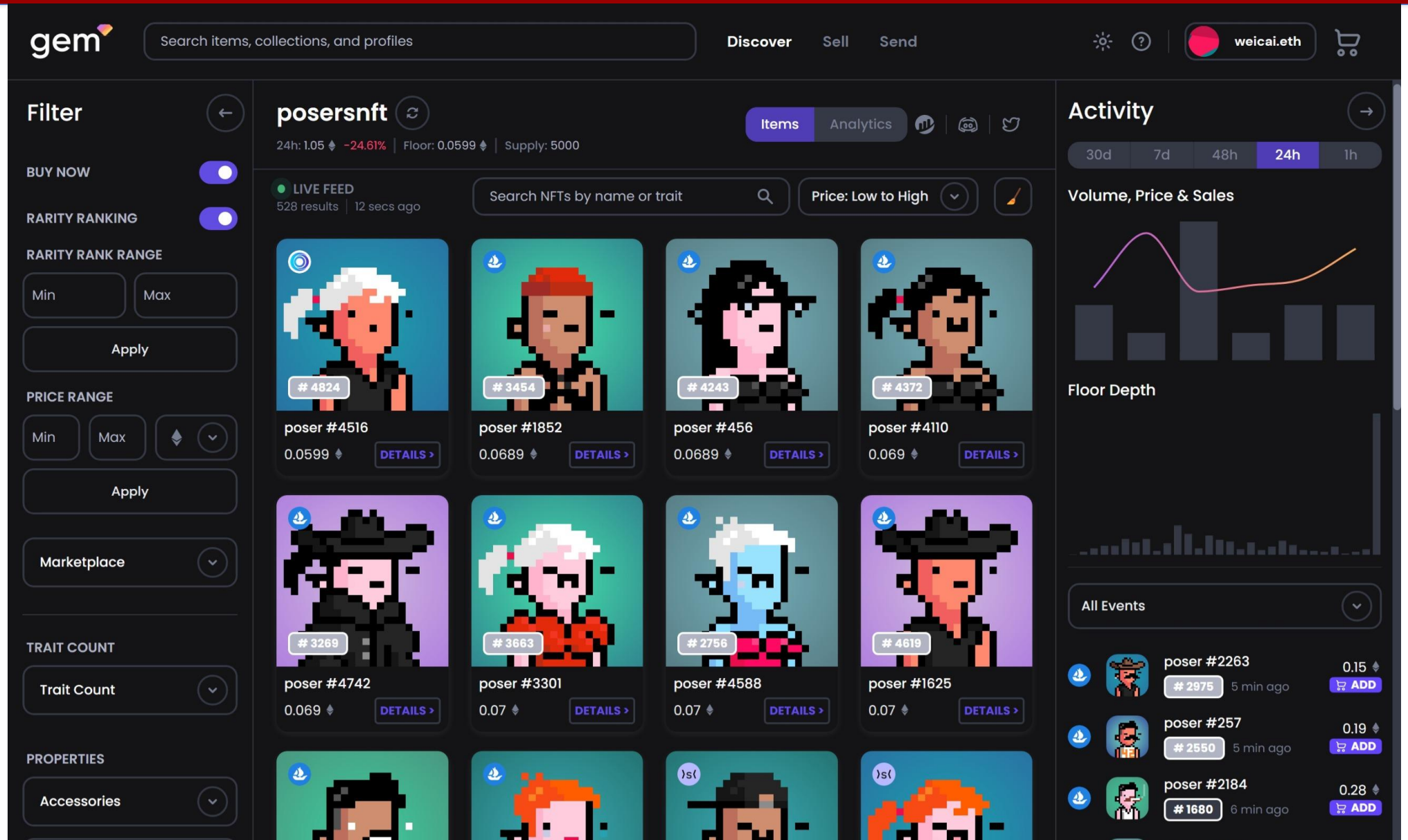
X2Y2

- Launched in February 2022
- To challenge the OpenSea Centralization in NFT trading
- Features
  - Profit-sharing: X2Y2 staking will receive a % of the NFT trading fees
  - Bulk listing: one transaction for many NFTs
  - Bulk purchasing: can purchase multiple NFTs with one transaction
  - Rarity tracking: new rarity tracker that can track the rarity of items
  - Real-time notifications: emails users the second there’s a bid or a purchase

NFT Marketplace Aggregator

- Goal: aggregate every NFT marketplace
- Locate the best deals across exchanges
- NFT analytics: sales volume, floor, and minting activity…
- Buy multiple NFTs in a single transaction
- Gas savings compared to buying from marketplaces directly

gem.xyz


gem
Search items, collections, and profiles
Discover
Sell
Send
weicai.eth
Filter
BUY NOW
RARITY RANKING
RARITY RANK RANGE
Min
Max
Apply
PRICE RANGE
Min
Max
Apply
Marketplace
TRAIT COUNT
Trait Count
PROPERTIES
Accessories
posersnft
24h: 1.05 -24.61% | Floor: 0.0599 | Supply: 5000
Items
Analytics
LIVE FEED
528 results | 12 secs ago
Search NFTs by name or trait
Price: Low to High
4824
poser #4516
0.0599
DETAILS
3454
poser #1852
0.0689
DETAILS
4243
poser #456
0.0689
DETAILS
4372
poser #4110
0.069
DETAILS
3269
poser #4742
0.069
DETAILS
3663
poser #3301
0.07
DETAILS
2756
poser #4588
0.07
DETAILS
4619
poser #1625
0.07
DETAILS
Activity
30d
7d
48h
24h
1h
Volume, Price & Sales
Floor Depth
All Events
poser #2263
2975
5 min ago
0.15
ADD
poser #257
2550
5 min ago
0.19
ADD
poser #2184
1680
6 min ago
0.28
ADD

blur.io

BLUR COLLECTIONS PORTFOLIO ACTIVITY AIRDROP

Collections, wallets, or ENS ⌘K

0 PTS | 0% LOYALTY

COLLECTIONS TRENDING FAVORITES POINTS

NAME	FLOOR PRICE	TOP BID	1D CHANGE	7D CHANGE	15M VOLUME	1D VOLUME ↓	7D VOLUME
BoredApeYachtClub ★	25.98	25.61	6.52%	2.37%	-	1456.02	9543.32
PudgyPenguins	5.08	5.01	1.16%	5.07%	-	566.26	2159.38
MutantApeYachtClub	4.94	4.87	3.13%	1.65%	-	381.12	3087.60
DeGods	3.32	3.29	3.79%	3.79%	3.60	332.28	1944.54
Azuki ★	4.38	4.34	4.41%	4.40%	-	182.31	887.92
Winds of Yawanawa by…	9.95	8.48	8.74%	44.62%	-	171.64	2383.49
Sugartown Oras	0.19	0.19	1.14%	-8.16%	1.48	168.10	276.81
CloneX	1.09	1.08	1.78%	0.20%	1.08	142.62	755.10
BoredApeKennelClub	1.53	1.52	0.75%	-0.39%	-	130.97	811.13
Potatoz	1.39	1.32	9.45%	30.06%	4.26	119.19	237.09
The Captainz	4.16	4.00	-0.45%	19.20%	-	86.65	503.33
yOOts	0.78	0.77	1.92%	-	-	69.81	338.52

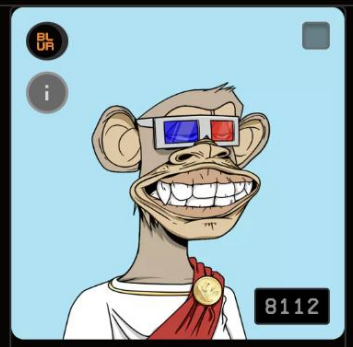
ITEMS BIDS LOANS HOLDERS Item ID

490 LISTED



#9670

PRICE 27.16 LAST SALE 29.00



#6974

PRICE 27.30 LAST SALE 26.30



#4827

PRICE 27.39 LAST SALE 23.06



#9522



#2841



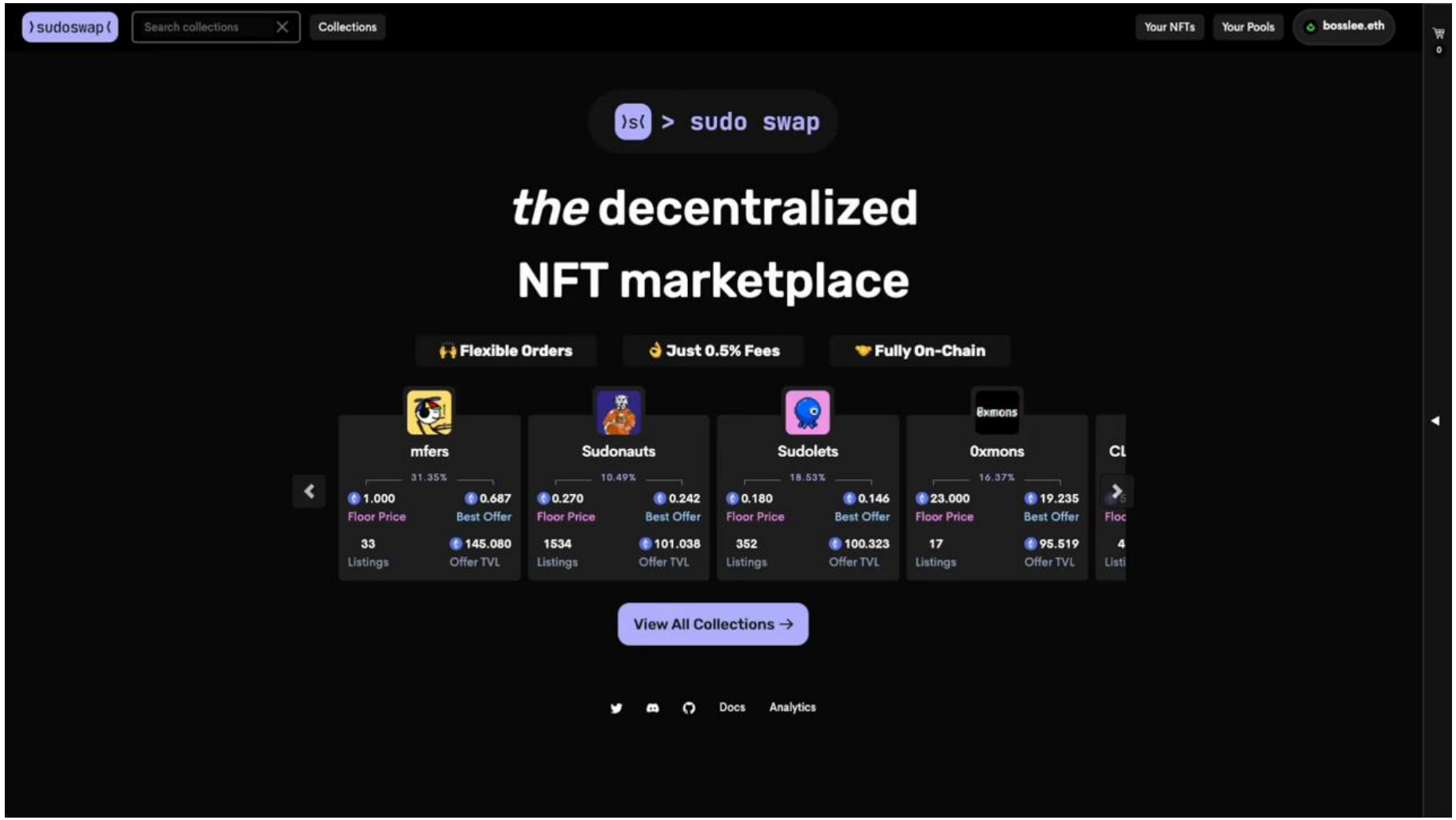
#5728

Sudoswap

Liquidity problem in NFT Trading!

Can we build decentralized AMM for NFT?

Sudoswap

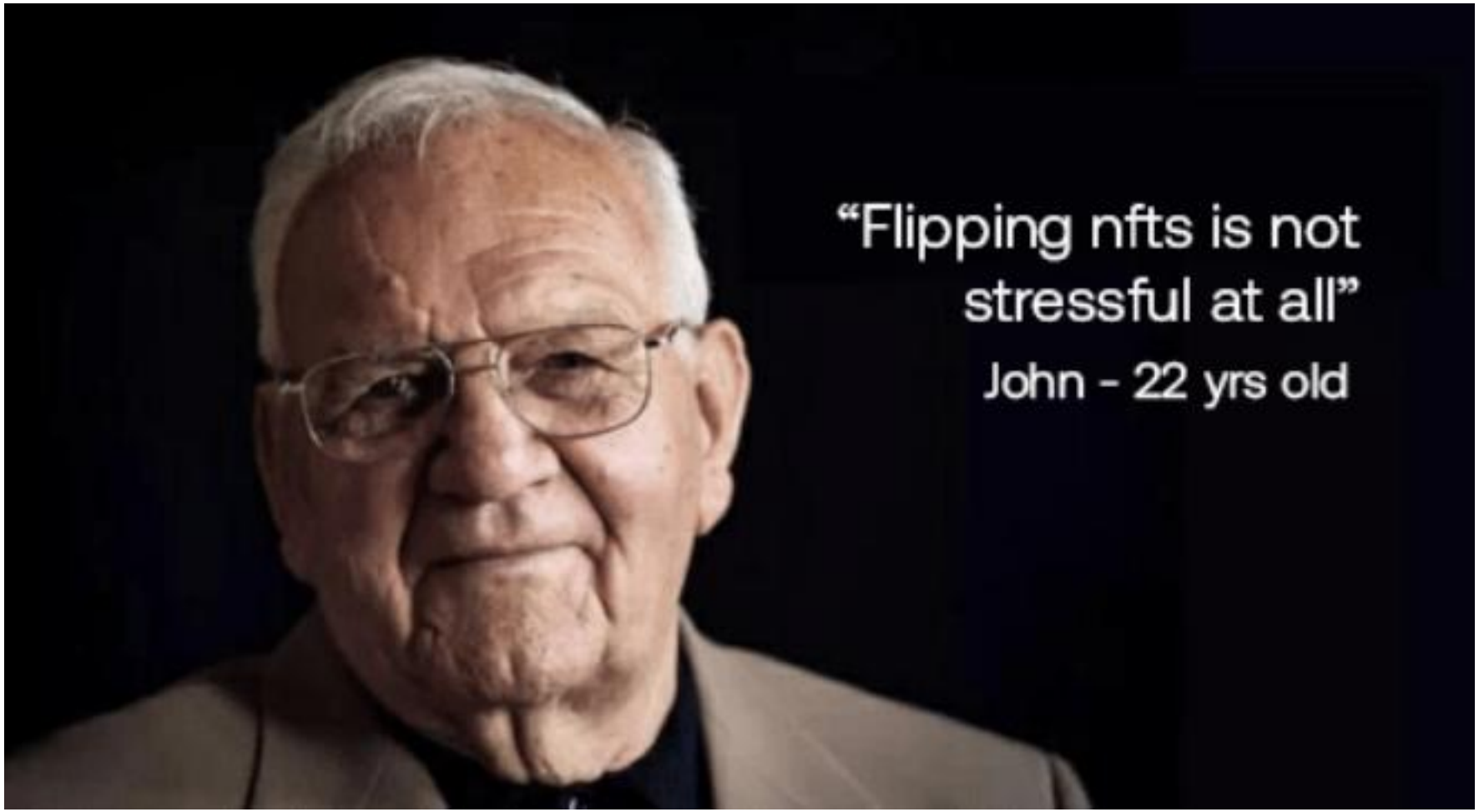


Sudoswap Explained

NFT Royalties

- Give creator a percentage of the each sale
  - for artists and creators to have their work yield substantial profit
- Executed by smart contracts automatically
- Creator usually can choose your royalty percentage
  - 5-10% is considered a standard royalty
- Discussed a lot recently

Trading NFTs





NFT traders be like...

Risks in NFT Trading

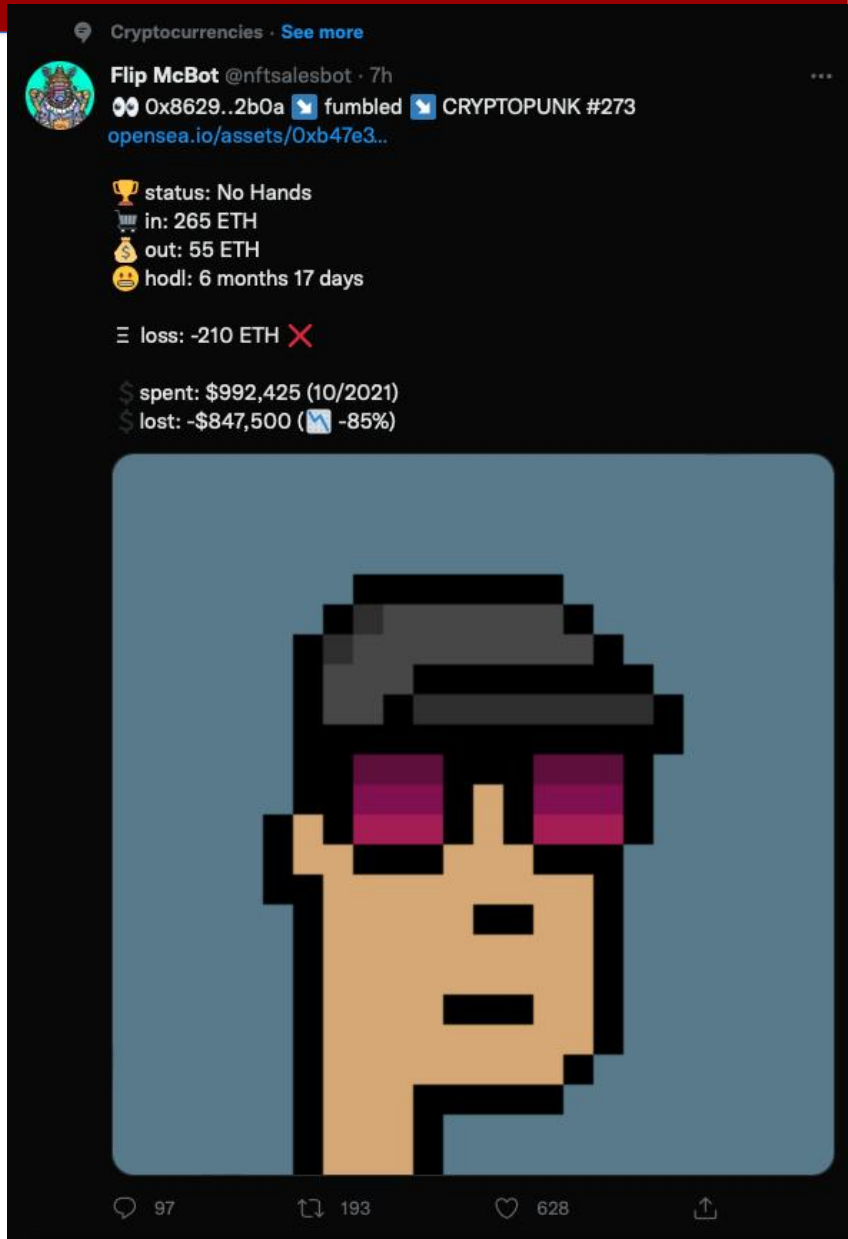
Item Activity

Filter

Sales ✕ Transfers ✕ Clear All

Event	Price	From	To	Date
Sale	55	86292F	punksotc.eth	8 hours ago
Sale	265	F18FCE	86292F	7 months ago
Transfer		yk111	F18FCE	9 months ago
Sale	26.5	comfygang	NullAddress	a year ago
Sale	20	Digital_Antiquities	comfygang	a year ago

https://opensea.io/assets/0xb47e3cd837ddf8e4c57f05d70ab865de6e193bbb/273



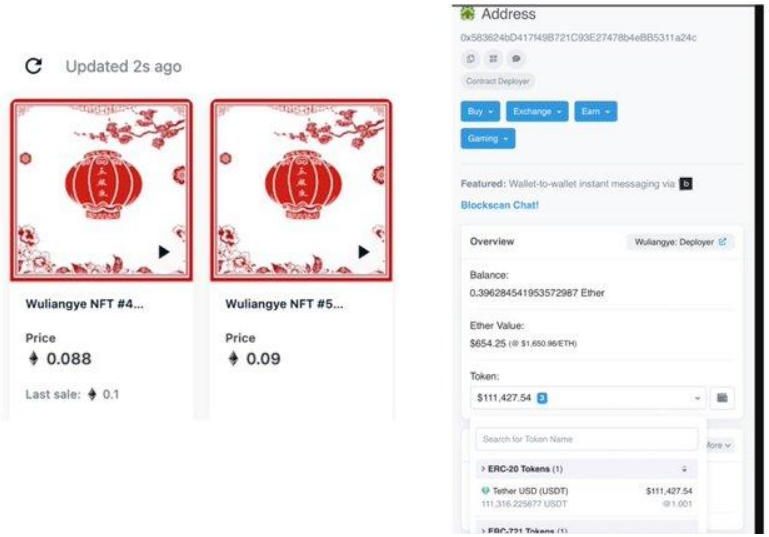
Rugs: Wuliangye Project

摆烂的项目： 五粮液

mint 两天+公售一天，数量高达 750 / 3000

mint 后不到24小时，地板0.08项目方直接摆烂，
一天成交量个位数，多个holder提点意见，直接被踢出频道

项目方不堪重负，果断换 U，疑似rug
国企背景？技术川大毕业应届生？ DC 运营外包？国产之光？笑哈哈！！



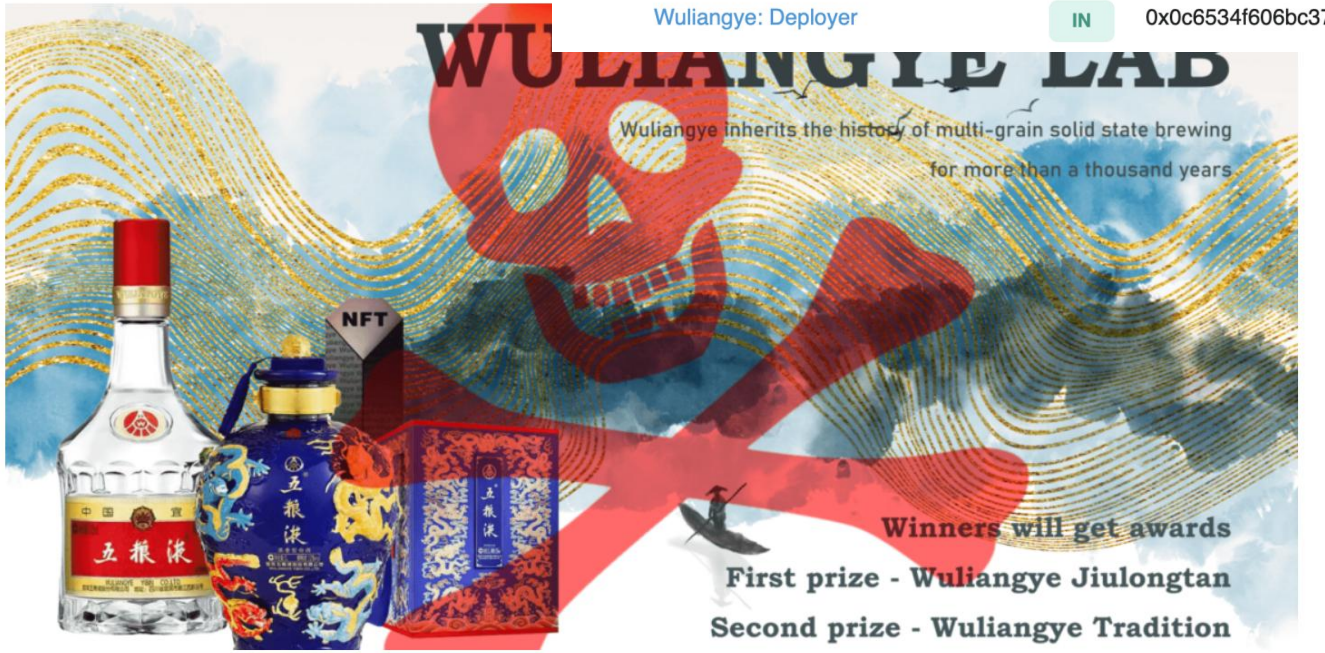
中國白酒兌換NFT《五糧液》傳Rug

by luc: — 2022-08-11 in nft, 即時新聞, 犯罪

項目方已捲走 111,316 枚USDT

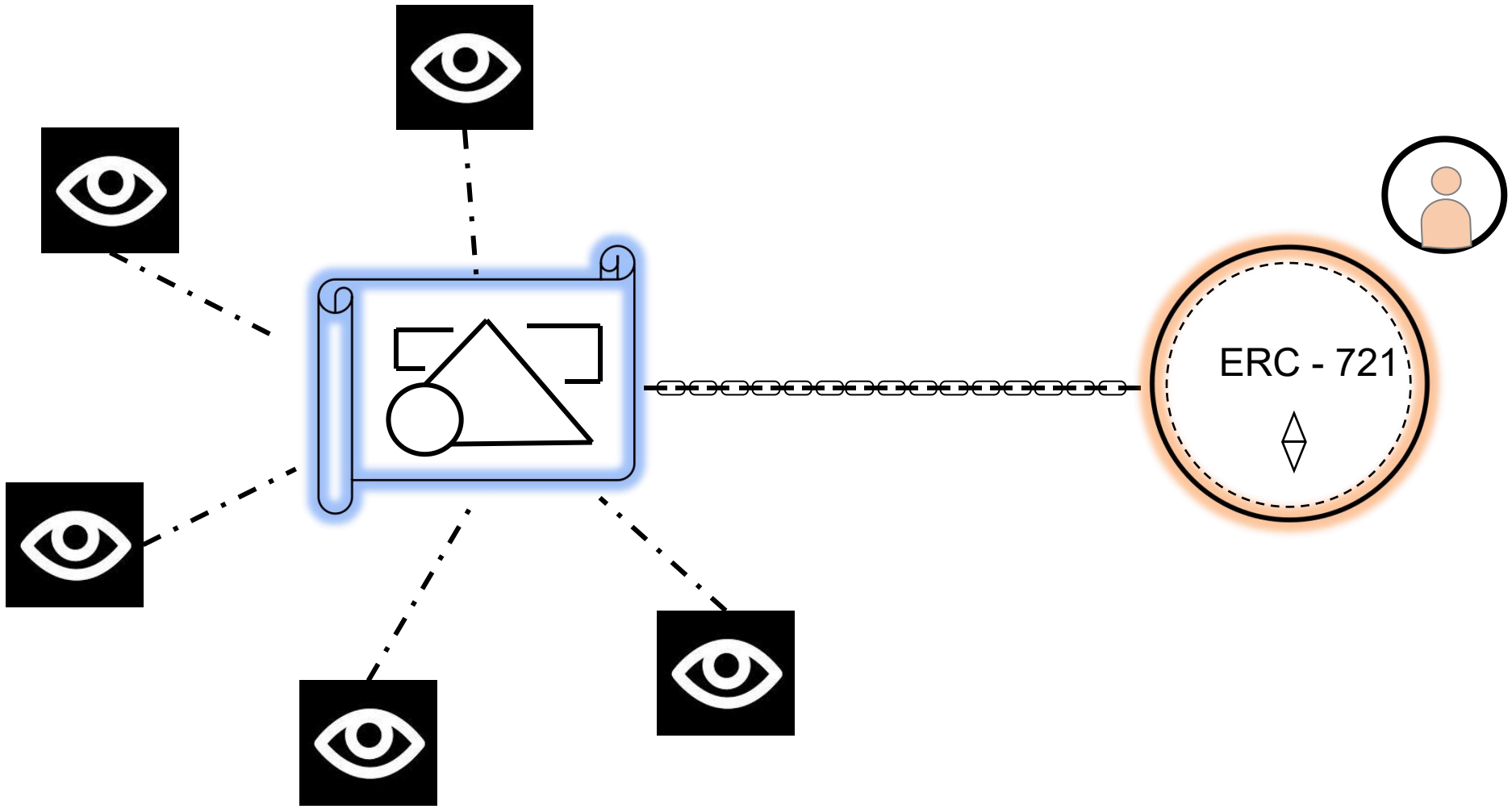
五糧液 NFT 部署地址在 3 日鑄造公售後，即將收入轉入 UniSwap 換成 USDT ，5 日時再將 111,316 枚USDT 轉至幣安交易所內。

From		To
0x0c6534f606bc37fac71...	OUT	Binance 14
Wuliangye: Deployer	IN	0x0c6534f606bc37fac71...
Wuliangye: Deployer	IN	0x0c6534f606bc37fac71...



圖源：Wuliangye NFT

What is NFT Artwork



The Value of NFT Artwork

- Usability in Virtual World
  - Decoration
  - With Usable Attributes

- Value as an Artwork
  - Uniqueness
  - Pioneering
  - Orderly Inheritance



Types of NFT Artwork

- Digital Artwork with NFT sales
  - Interactive
- Gamified Artwork
  - Incentive Mechanism Driven
  - Collaborative

Autoglyphs

- created in 2019 by Larva Labs
- the first “on-chain” generative art created by Ethereum code
- to anyone who was willing to donate the creation fee
  - 0.2Ξ (around $35 at the time) to chosen charity, 350.org
- the creator of each glyph became the first owner of that glyph
- 512 glyphs in total

Autoglyphs on OpenSea

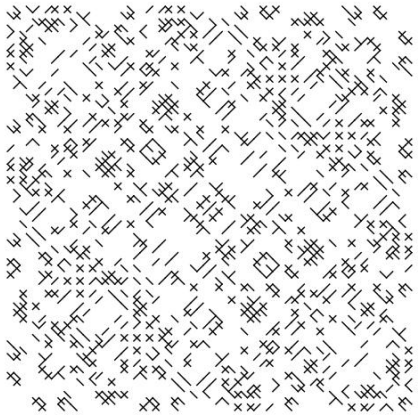


Autoglyphs

512	152	177	18.9K
items	owners	floor price	volume traded

Autoglyphs are the first “on-chain” generative art on the Ethereum blockchain. A completely self-contained mechanism for the creation and ownership of an artwork.

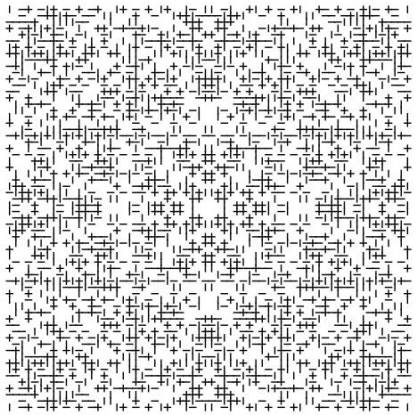
+ Add to



Autoglyphs
Autoglyph #192

Price
210
Last 5

116



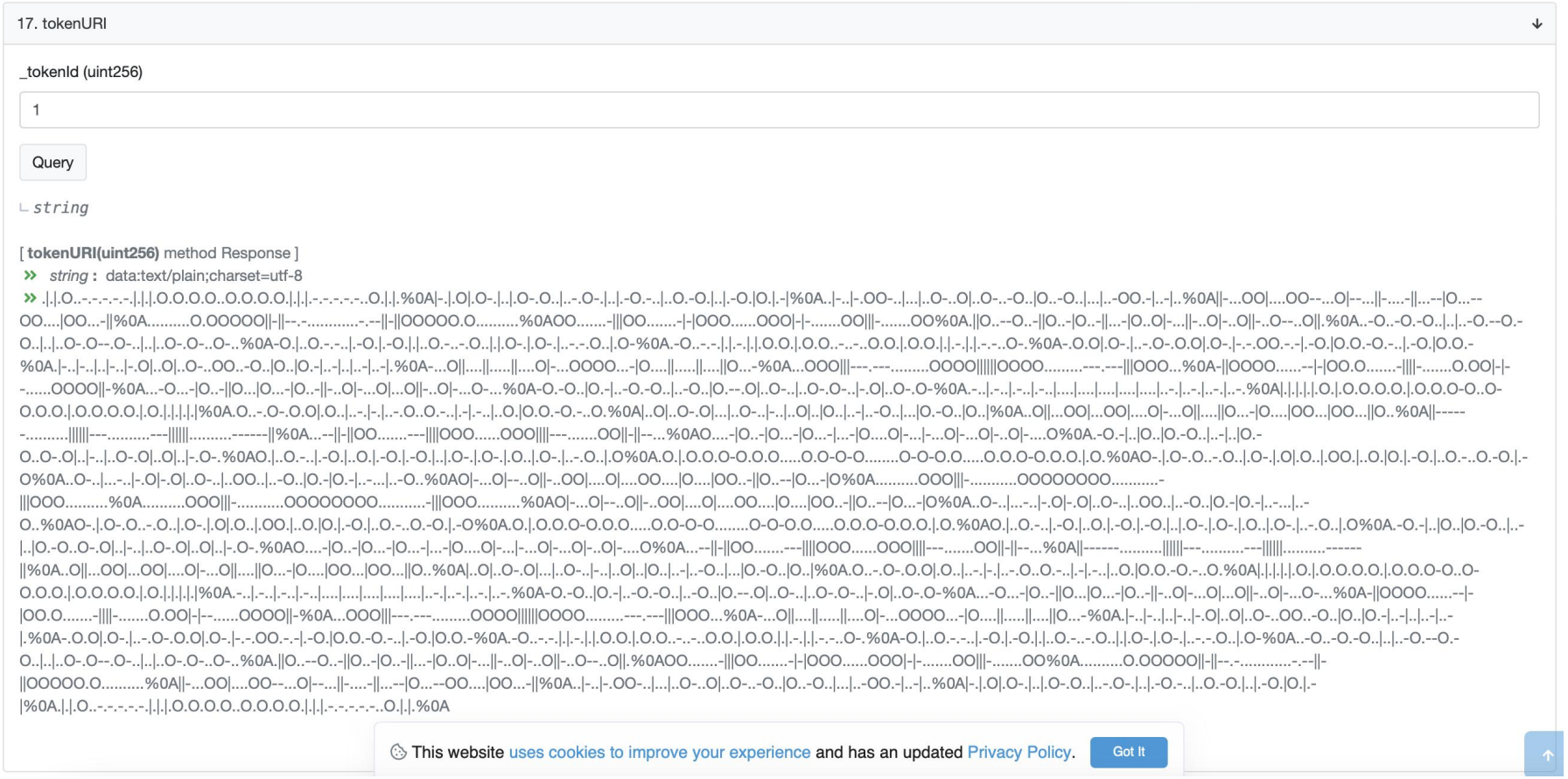
Autoglyphs
Autoglyph #487

Price
289
Last 8

153

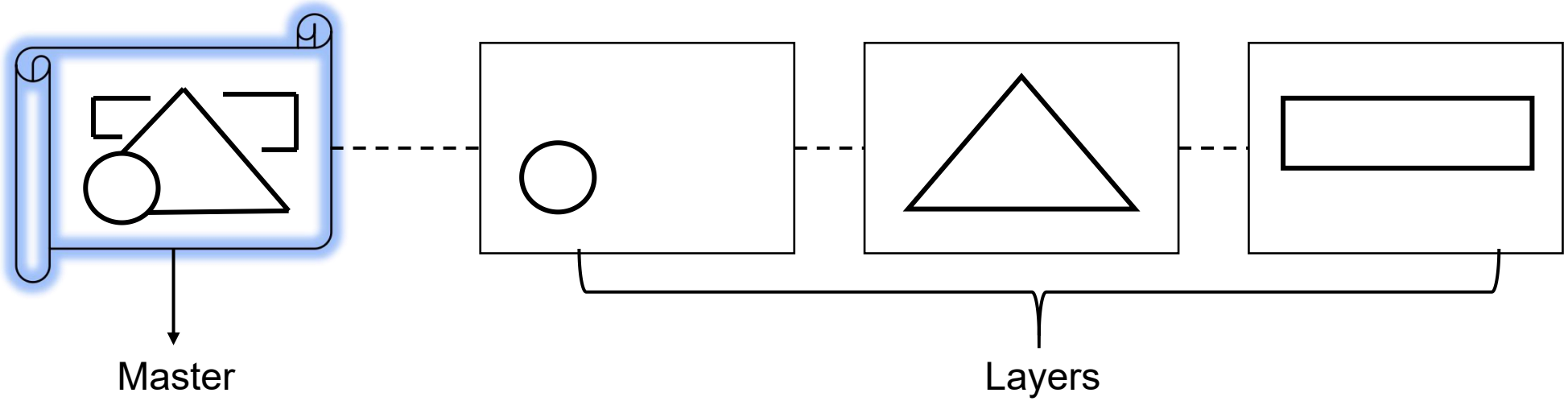
Technical Details of Autoglyphs

Contract Address: 0xd4e4078ca3495DE5B1d4dB434BEbc5a986197782



Async Art

Programable NFT Art

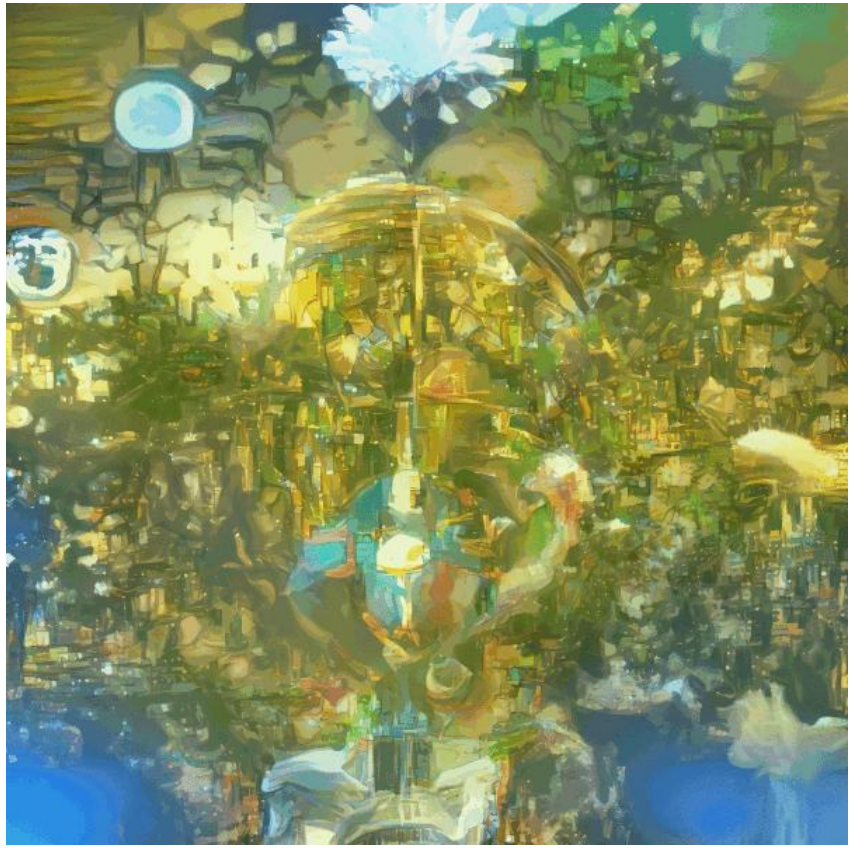


Layers: by changing the state of each layers, artists can modify the color shifting , scale, rotation, transparency and so on.

Async Art



"ETHBoy" - Trevor Jones & Alotta
Changing with ETH/BTC price



"Beautiful Worlds" or "Alien" - by Yuramiron
Changing daily (366 states in total)

Async Art



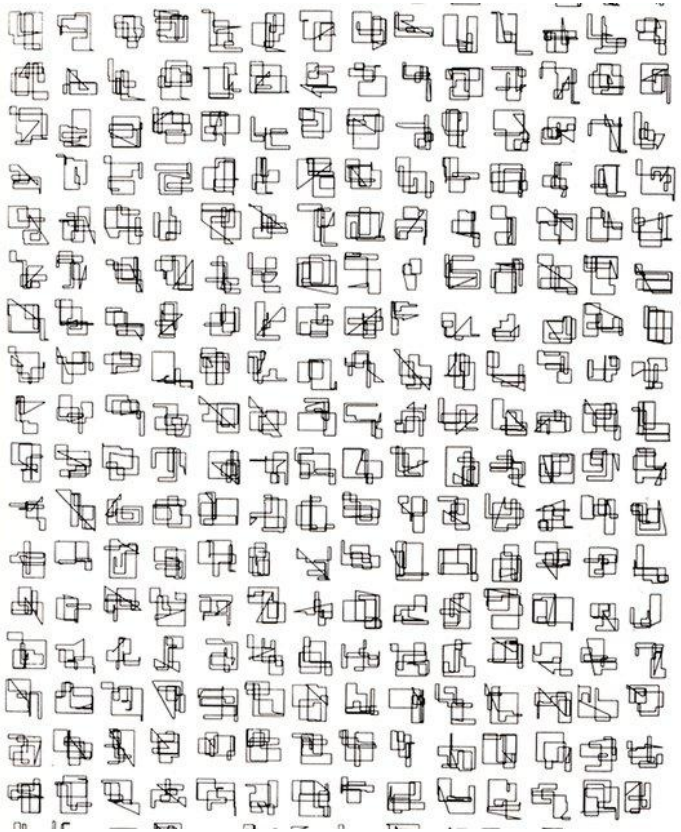
"Masked" - Jake Johns

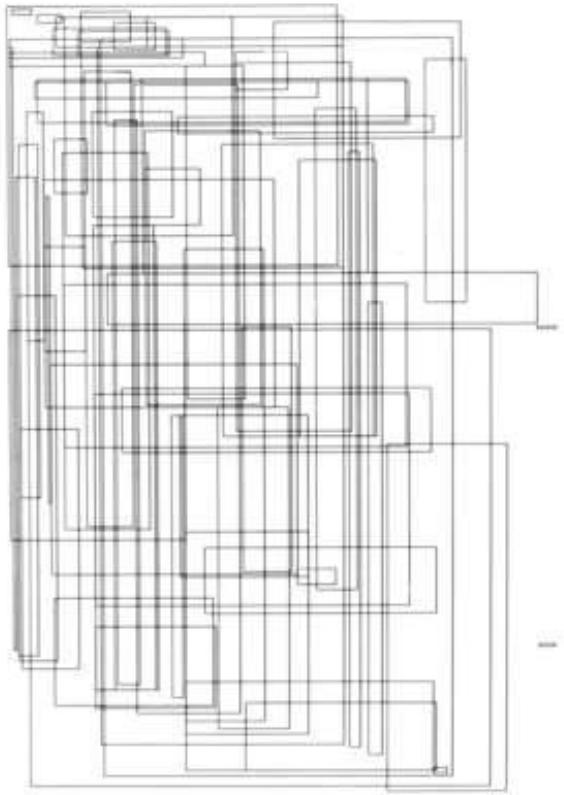
Generative Artwork

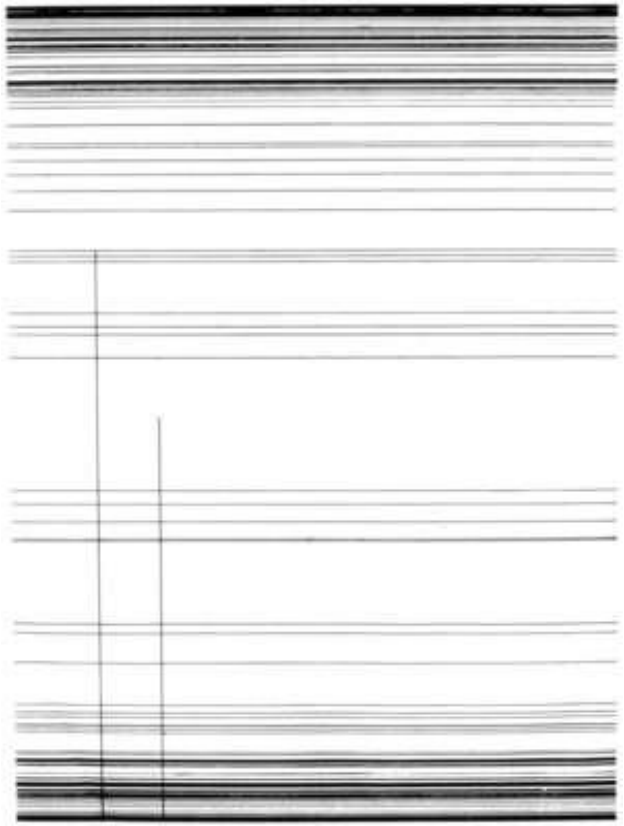
- created by computers following certain instructions (code)
- an art work created by human beings and computer together



Generative Artwork

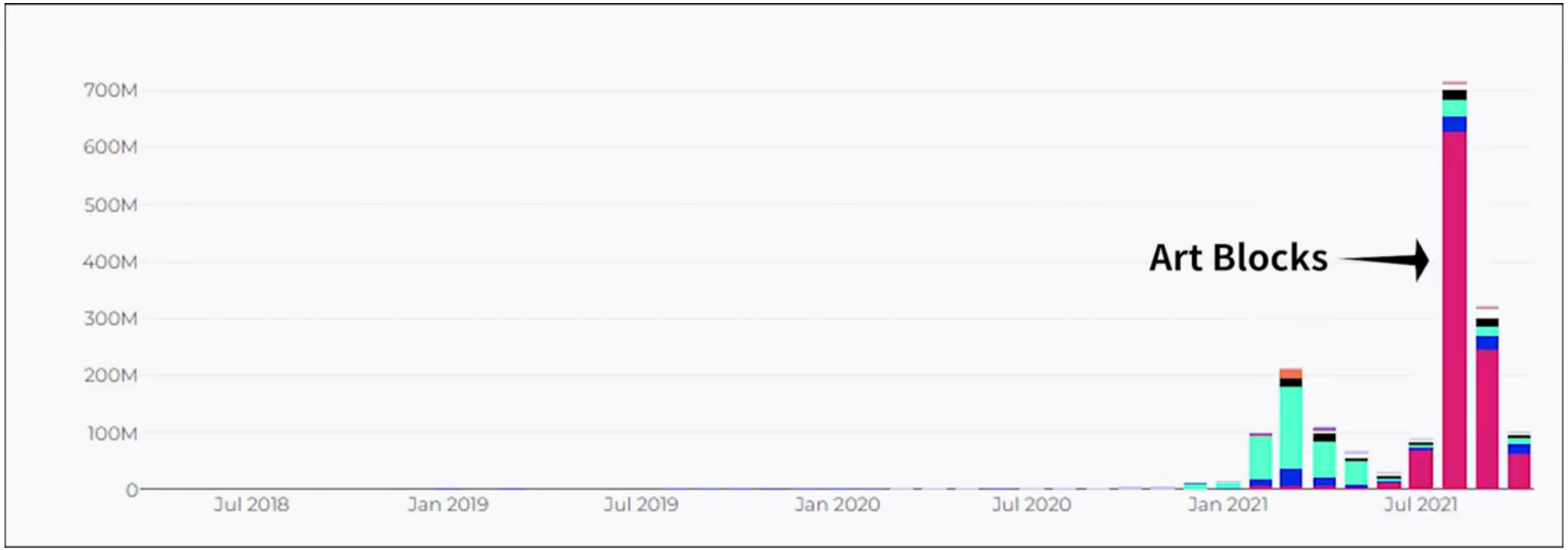






Art Blocks

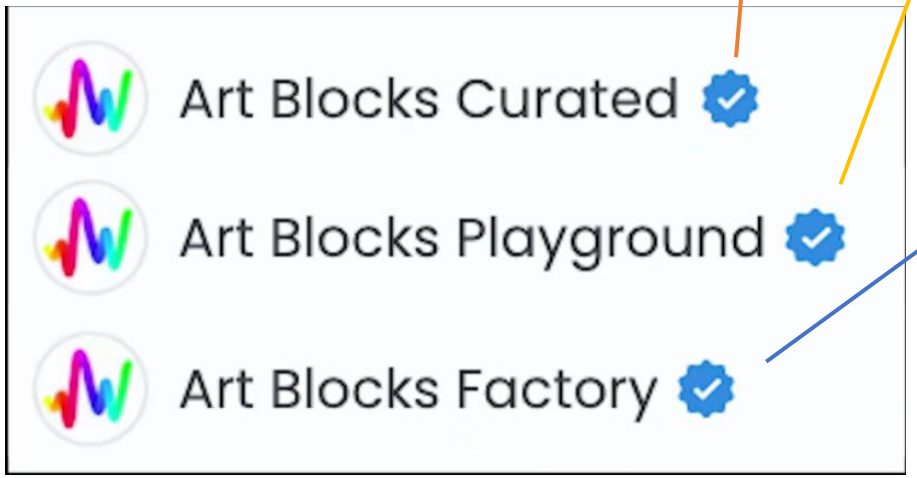
The Largest NFT trading platform for Generative Arts



CryptoArt Market data (excluding Opensea)

Art Blocks

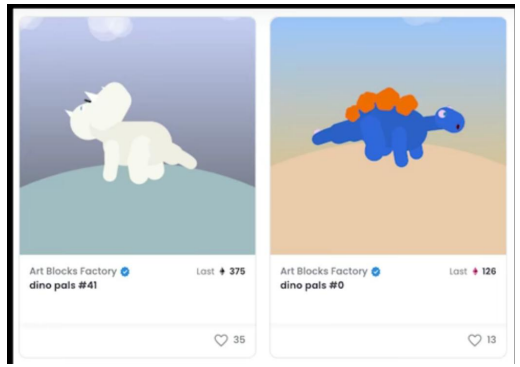
- 3 Types



- Be reviewed by a committee before publish
- High quality, and limited in numbers
- Normally expensive

- Fro artists who have published works on curated
- Normally cheaper than curated for a same artist

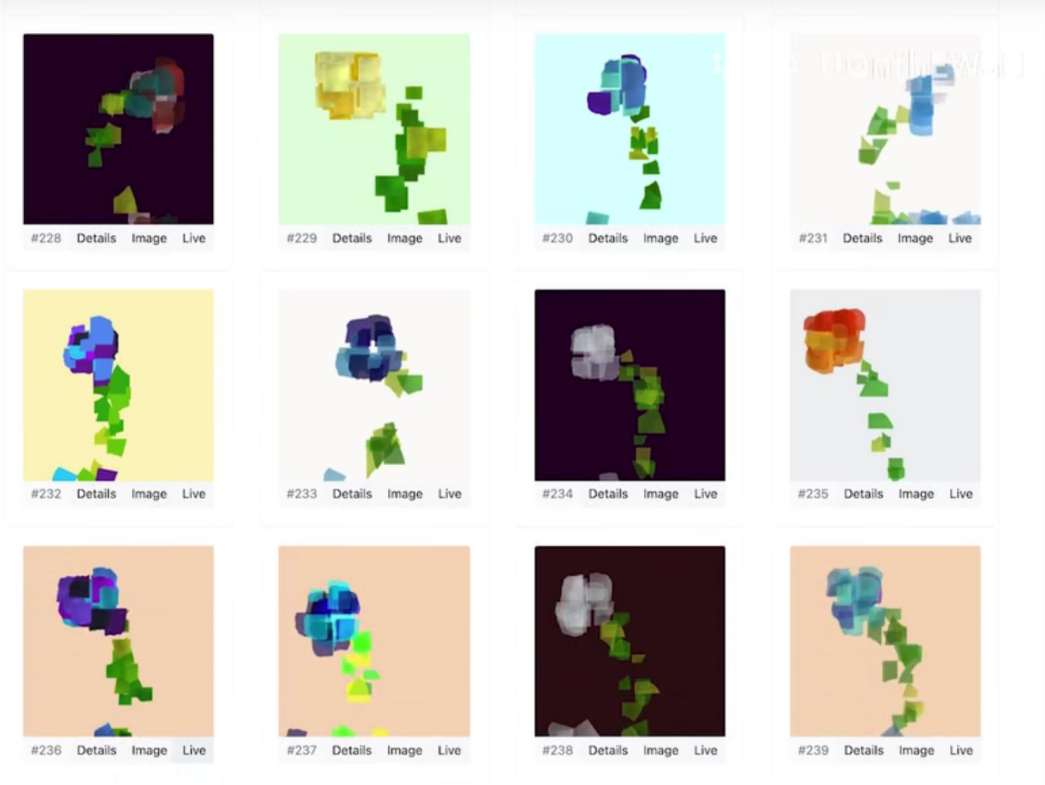
- Less restriction, but more competitive
- Still have some very excellent works
- Artist can get fame and publish works in curated

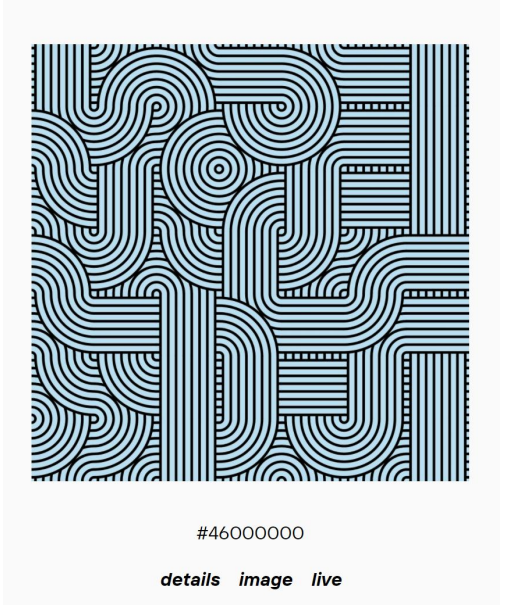


Dino Pals: up to 375 ETH

Art Blocks

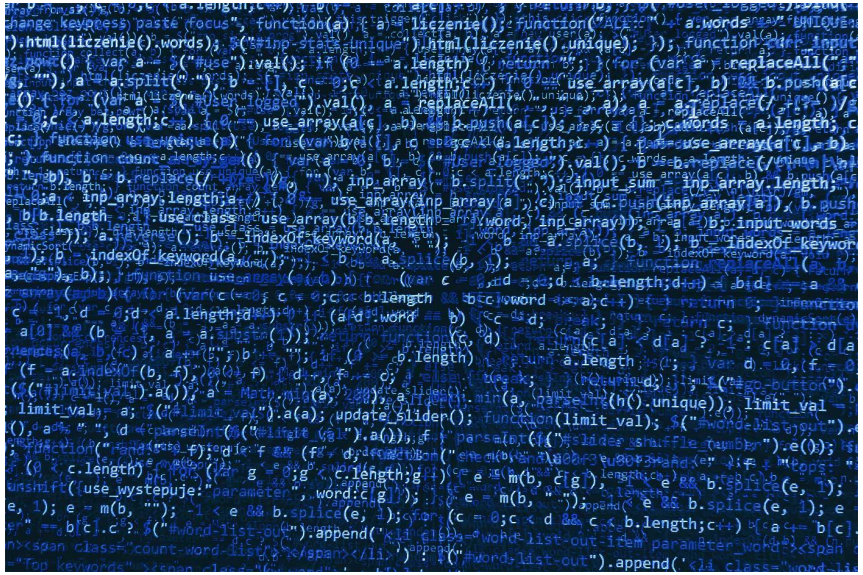
Before mint, buyers do not know what is exactly their NFT looks like.



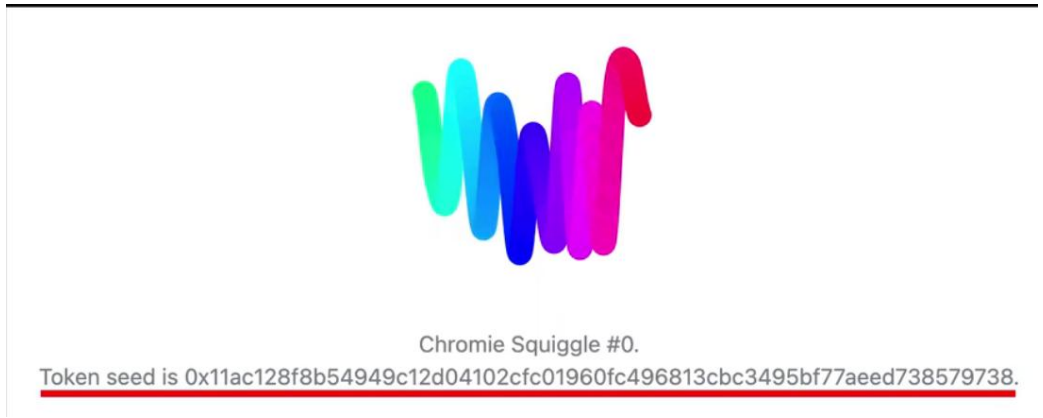


- Get an impression from previous works
- NFT in a series has a similar style

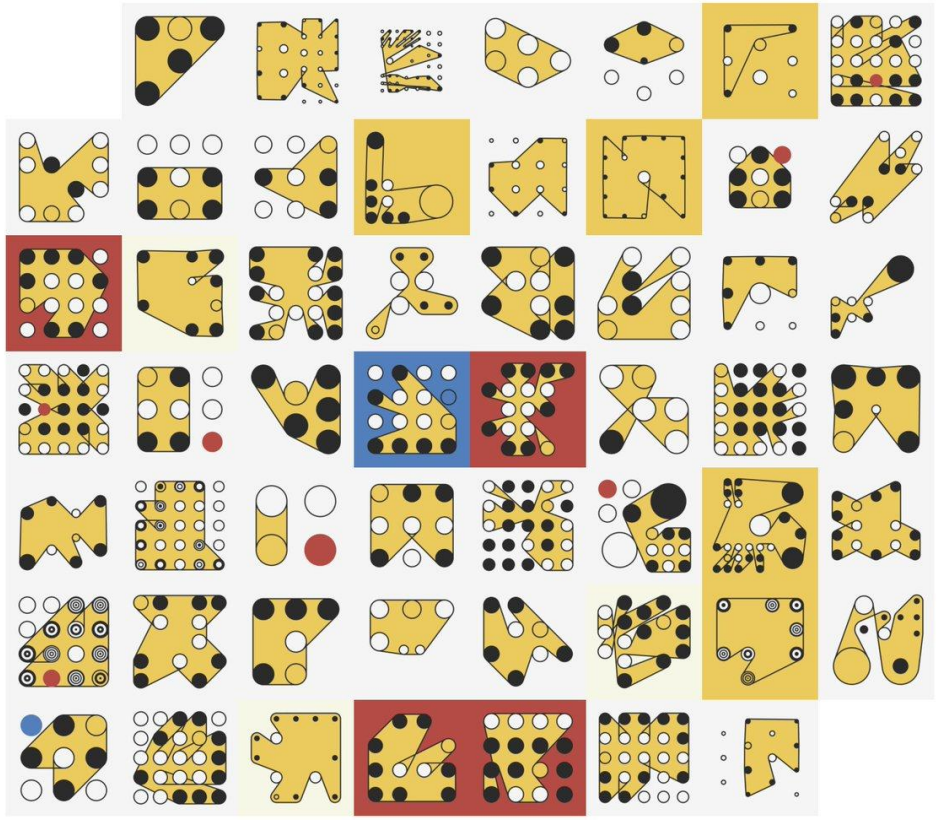
Art Blocks: Generative Artwork

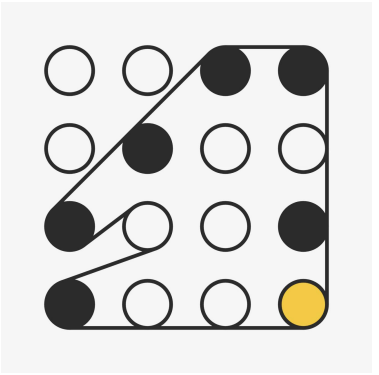




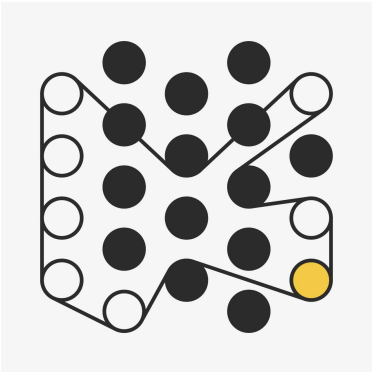


Art Blocks: Ringers series





#13000001 (seed) : 0x80592a750c34878d6172ec5a198649f1318288eec606a6c7f43953192d1fb4fc

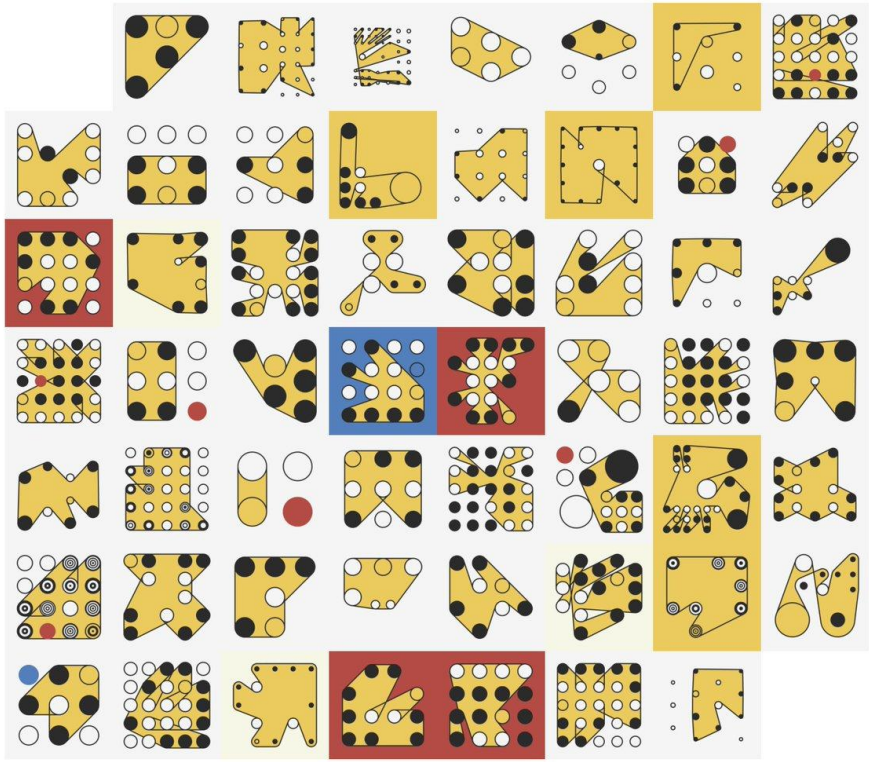


#13000001 (seed) : 0xba740640256787100ea5f8e6d1fbde8c88c402cffd7b93e72a7d6fa4678d23c7

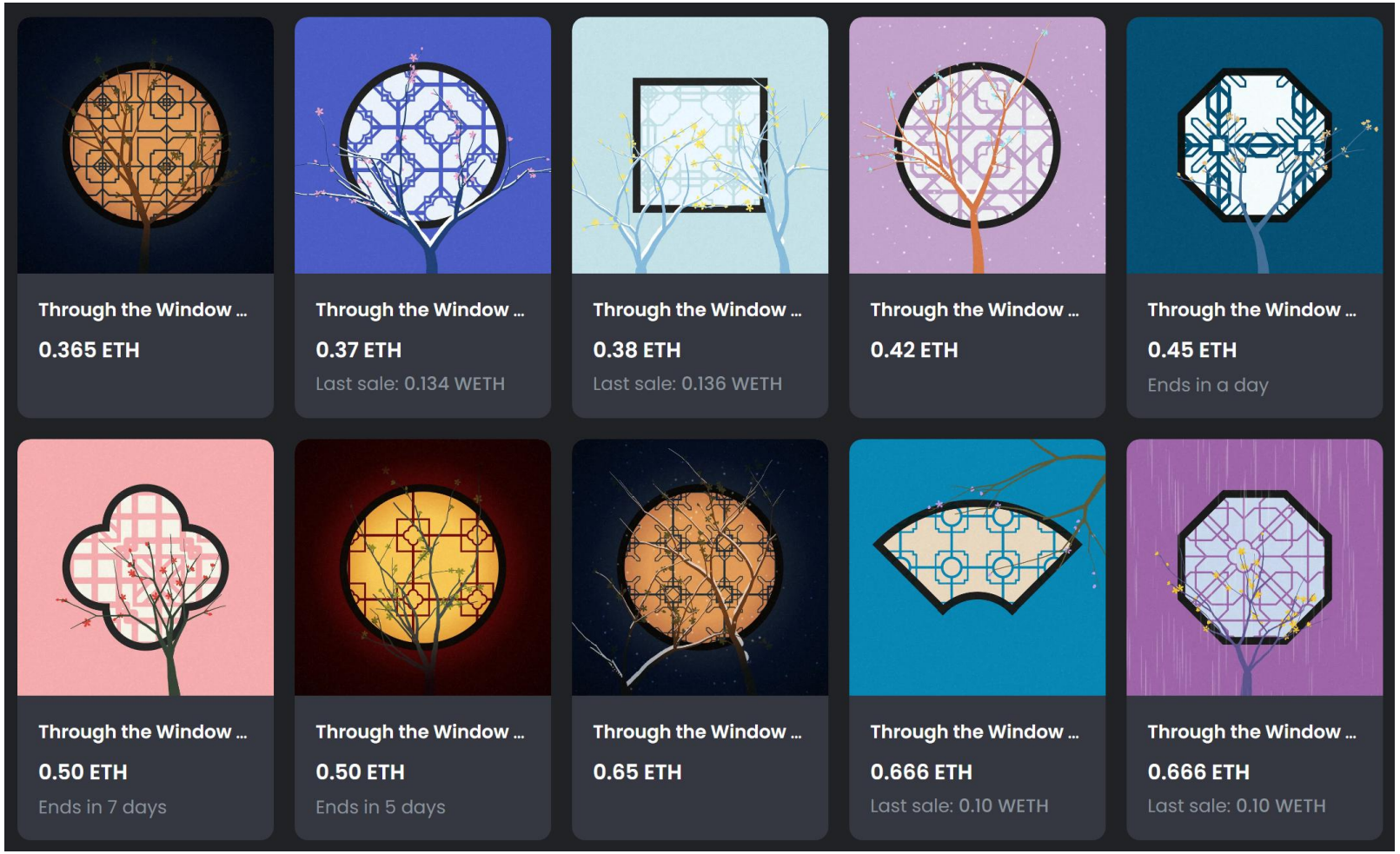
Check JavaScript code and TokenHash (seed) of Ringers series on Etherescan

```
//random seed generation
let rp=setupPs(tokenData)
seed=genSeed(tokenData);

// setting shape, color…
bC="#2b2b2b";
const ps={gridD:parseInt(mapP(rp[0] 3 6))
radius:mapP(rp[1] .5 .8)
sampleRate:mapP(rp[2] .5 .8)
wrapped:rp[3]<127
forceCentroid:rp[5]>200
bgcol:rp[22]<250?wC:bC
subtleBgcol:"#f7f7e6"
sCol:rp[22]<250?bC:wC
hlCol:rp[22]<250?N(177):bC
secCol:rp[15]>=19&&
rp[15]<=230?N(186):rp[15]>230&&rp[15]<=234?N(159):N(157)
fCCol:rp[14]>200&&(rp[6]>=127||rp[10]>220)
```

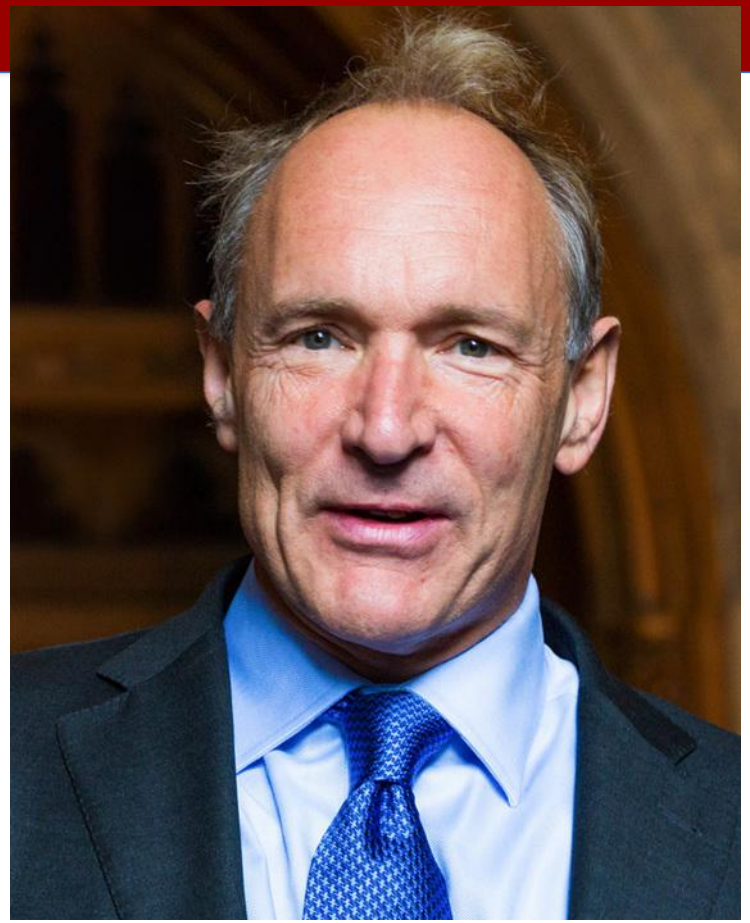


Art Blocks: Through the Window



Web 3.0 v.s. Web3

- Tim Berners-Lee's Web 3.0
- Semantic Web
- An extension of the World Wide Web through standards[1] set by the World Wide Web Consortium (W3C)



Sir Tim Berners-Lee
Inventor of the Web

Web3

- Web 1
  - 1990-2004
  - Read-Only
- Web 2
  - 2005~now
  - Read-Write
- Web 3
  - Read-Write-Own



Gavin Wood
Polkadot ($DOT)

Core ideas of Web3

decentralized

permissionless

has native payments

trustless

Web 3 Terms (Technical)

Term	Meaning
Mint	buy a completely new NFT from the creator
Burn	destroy an NFT
Aped	take on a large position relative to one's own portfolio size.
Apeing in	Buying into an NFT project, perhaps without due diligence.
Sweep the Floor	searching and buying all the lowest-priced NFTs of a particular NFT project
Airdrop	When you get an 'airdrop', you automatically receive a specific amount of a cryptocurrency or a new free NFT dropped into your wallet.
Snapshot	snapshot at a certain point in time to determine who's eligible for airdrop
Whale	Whales have the power to move markets
Schelling Point	A theory term people believe applies to NFTs: certain NFTs will become *stores of value*. As such, storing more of them makes them become even more valuable.
1/1 of X	Each one is entirely unique, but they belong to a whole collection compared to 1:1 art.

Web 3 Abbreviation

Term	Meaning	Term	Meaning
gm / gn	Good morning / evening	degen	Degenerate
WAGMI	We‘re all gonna make it	DYOR	Do Your Own Research
NGMI	Not Going To Make It	NFA	Not Financial Advice
LFG	Let’s Fxxking GO	IRL	In real life (Meatspace)
FOMO	The Fear of Missing Out	PFP	Profile picture
COPE	The opposite of FOMO	DAO	Decentralized Autonomous Organization
OG	Original gangster	MM	MetaMask
Noob/Pleb	Newbie/Plebeian	P2E	Play to Earn
Rug	Rug pulled	Rekt	Wrecked, bad investment portfolio
Szn	season, market cycle	HODL	Holding On for Dear Life
BTFD	Buy the f'ing dip	BUIDL	building useful stuff proves that it is valuable
IYKYK	If You Know, You Know	AMA	Ask me anything
FUD	Fear, Uncertainty and Doubt	ATH	All Time High

Web 3 Terms (Community)

Term	Meaning
To the moon / ToDaMoon	
When moon?	Another from CryptoTwitter refers to your NFT project's price *ascending to the moon*.
Shilling	'Shilling' means constantly promoting others to buy a certain NFT, making people feel irritable, or generating FOMO in the community. But no one likes this, don't do it.
Seems Legit	'Seems legitimate' means the project looks legitimate and not fraudulent and has potentially good prospects. It can be used directly or sarcastically about a project's chances.
Its Money Laundering	This suggests that it is easy to make profits with NFTs. Used as a joke.
Probably Nothing	
Looks Rare	Rarity is what drives value in the NFT space.
MEME	A meme is an idea, behavior, or style that spreads by means of imitation from person to person within a culture and often carries symbolic meaning representing a particular phenomenon or theme.

Web 3 Terms (Community)

Term	Meaning
JPGs	JPGs refer to NFTs
Right Click Save As	Used jokingly to describe every non-NFT person's first move when learning about NFTs.
This is the way	A way to commend socially positive behaviour.
McDonald's	A backup career plan in case the NGMI
Want fries?	Used when someone sells NFT too early or is just salty, referring to the only sentence you need to learn when going to work at McDonald's (see McDonald's above).
Salt	salty, like the tears you shed in Cope state
Love to See it	happy to see this. Sometimes it is purely literal, but sometimes it is used oppositely.
Valhalla	An enormous hall where dead Vikings party alongside others with valuable NFTs.
Diamond hands	someone who has a high-risk tolerance for high volatility stocks or assets that they own.
Paper/Weak Hands	Opposite to diamond hands
PND	pump and dump

Workshop:

Deploy an ERC-20 token on the testnet

Testnet in blockchain ecosystem

A testnet is the test environment in any blockchain network where the developers could test their application, smart contracts, and changes free from potential damage of the real assets or the security breach of the main blockchain.

Preparation

Step1: Prepare your [MetaMask] wallet. (follow the steps mentioned by last workshop)

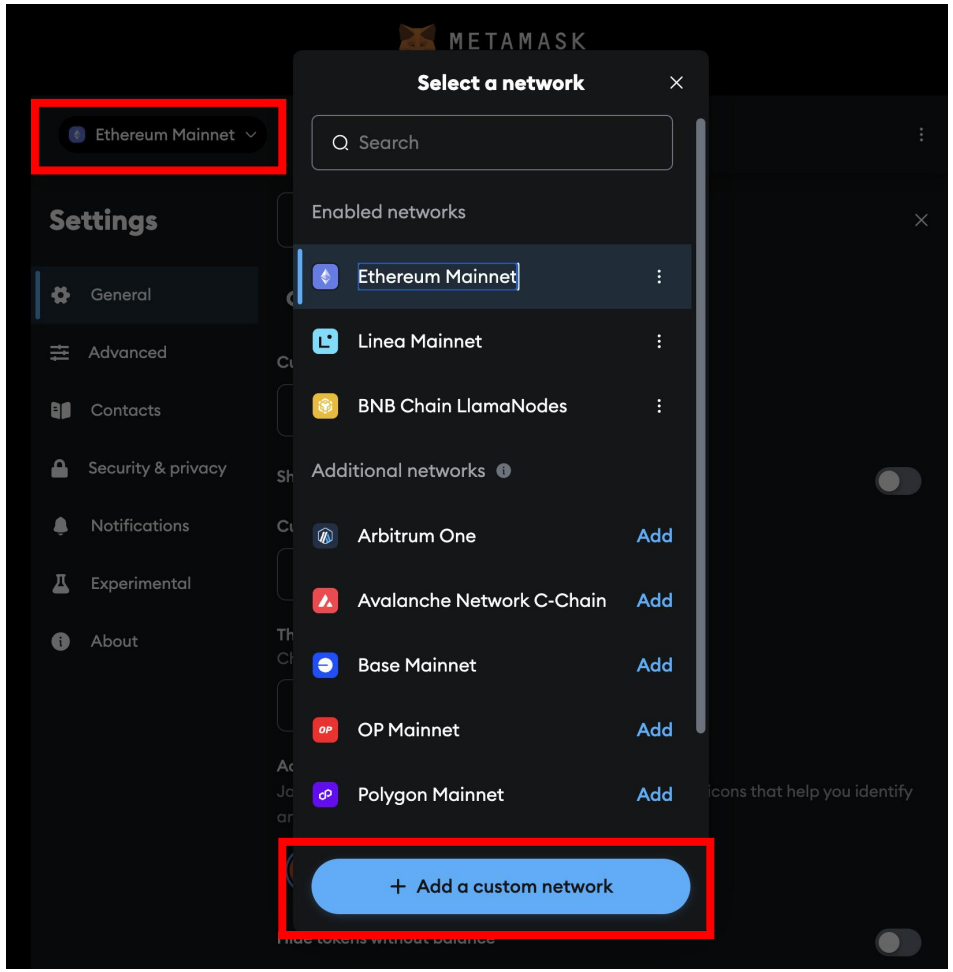
Add the MetaMask into the Google Chorme extensions.

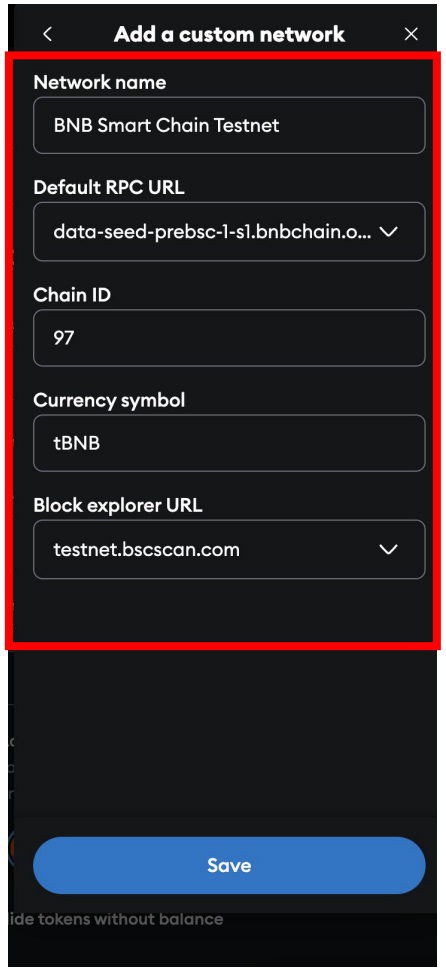


Preparation

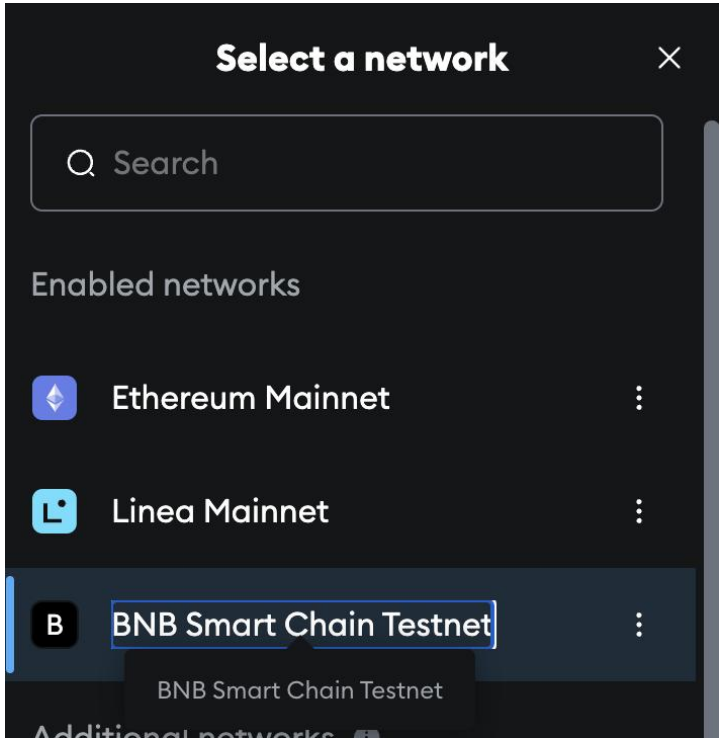
Step2: Select the testnet and connect to the testnet.

Recommended: BNB Smart Chain Testnet; Ethereum Sepolia





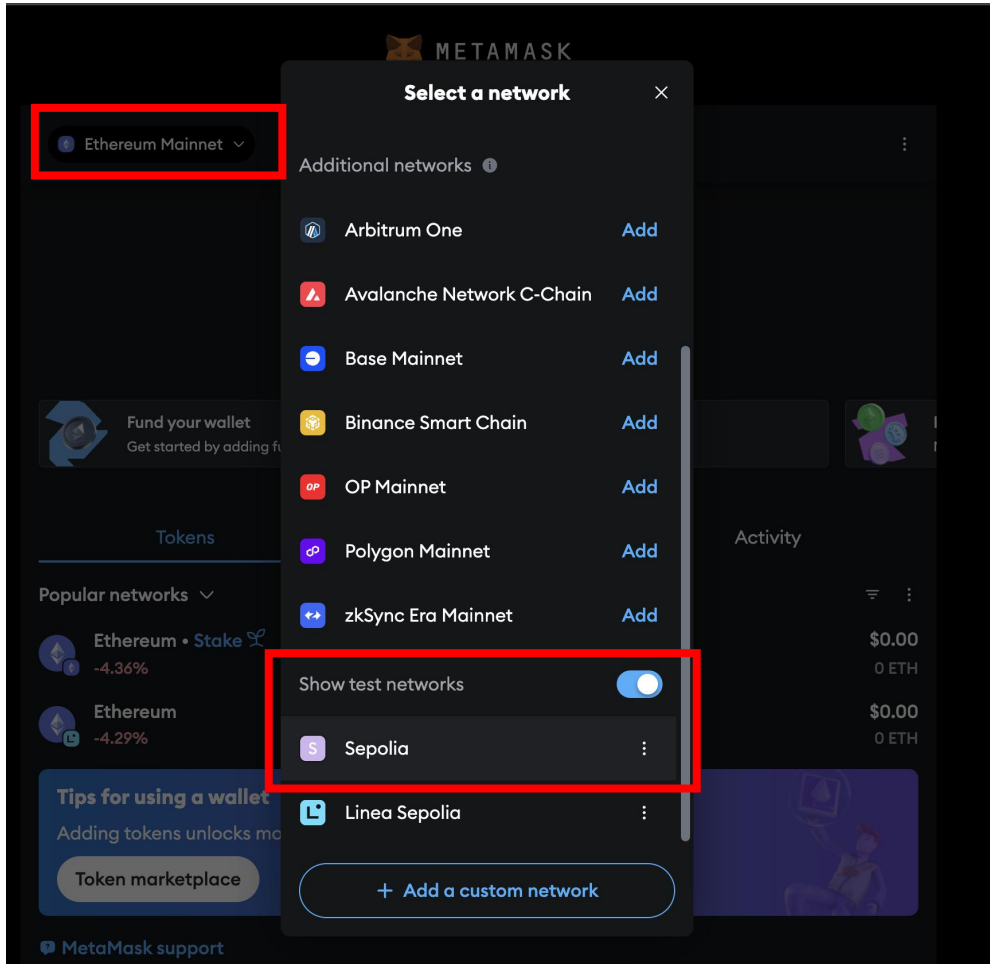
Network Name: BNB Smart Chain Testnet
RPC URL: https://bsctestapi.terminet.io/rpc
Chain ID: 97
Currency Symbol: tBNB
Block Explorer URL: https://testnet.bscscan.com

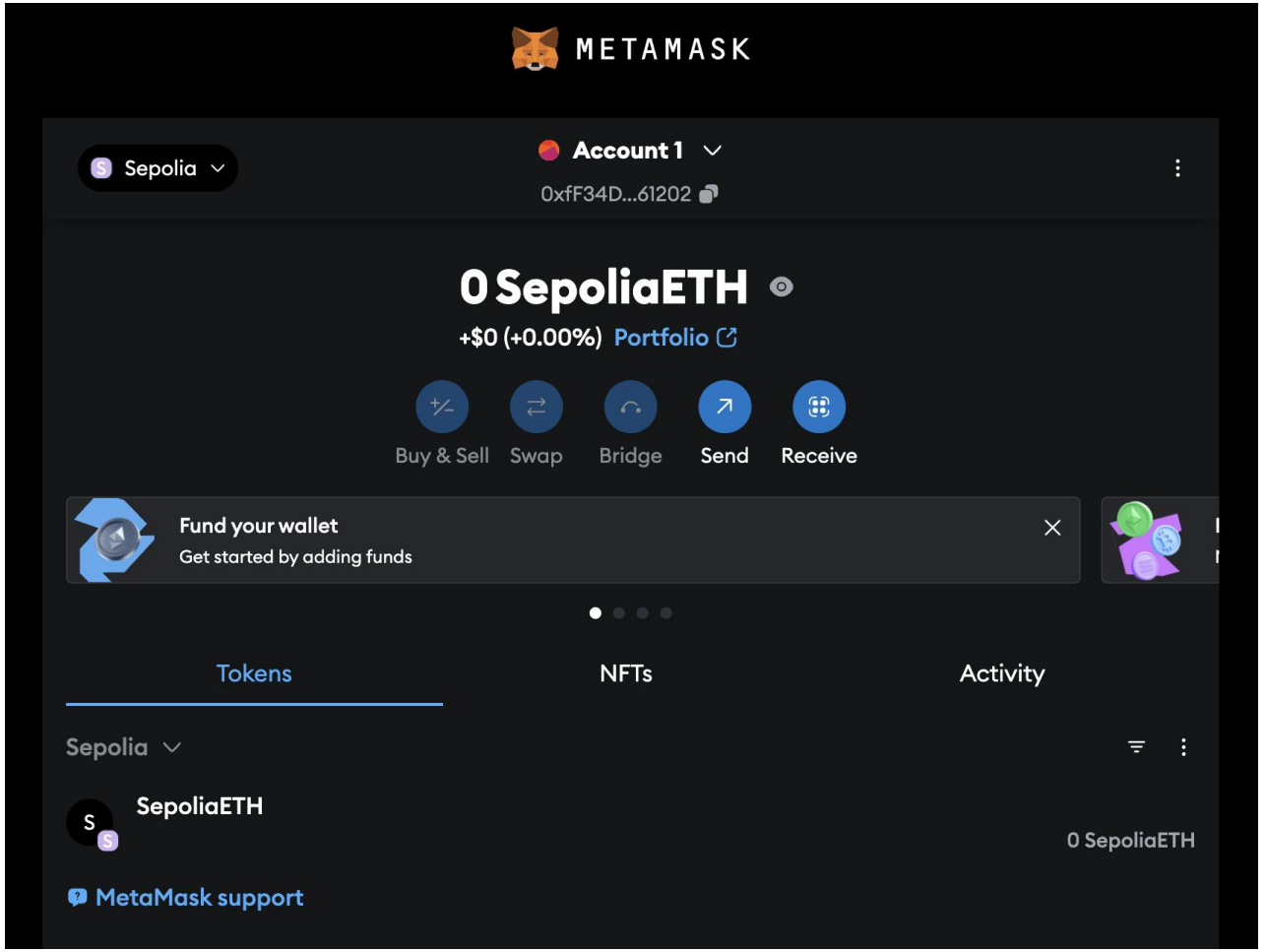


Preparation

Step2: Select the testnet and connect to the testnet.

Recommended: BNB Smart Chain Testnet; Ethereum Sepolia

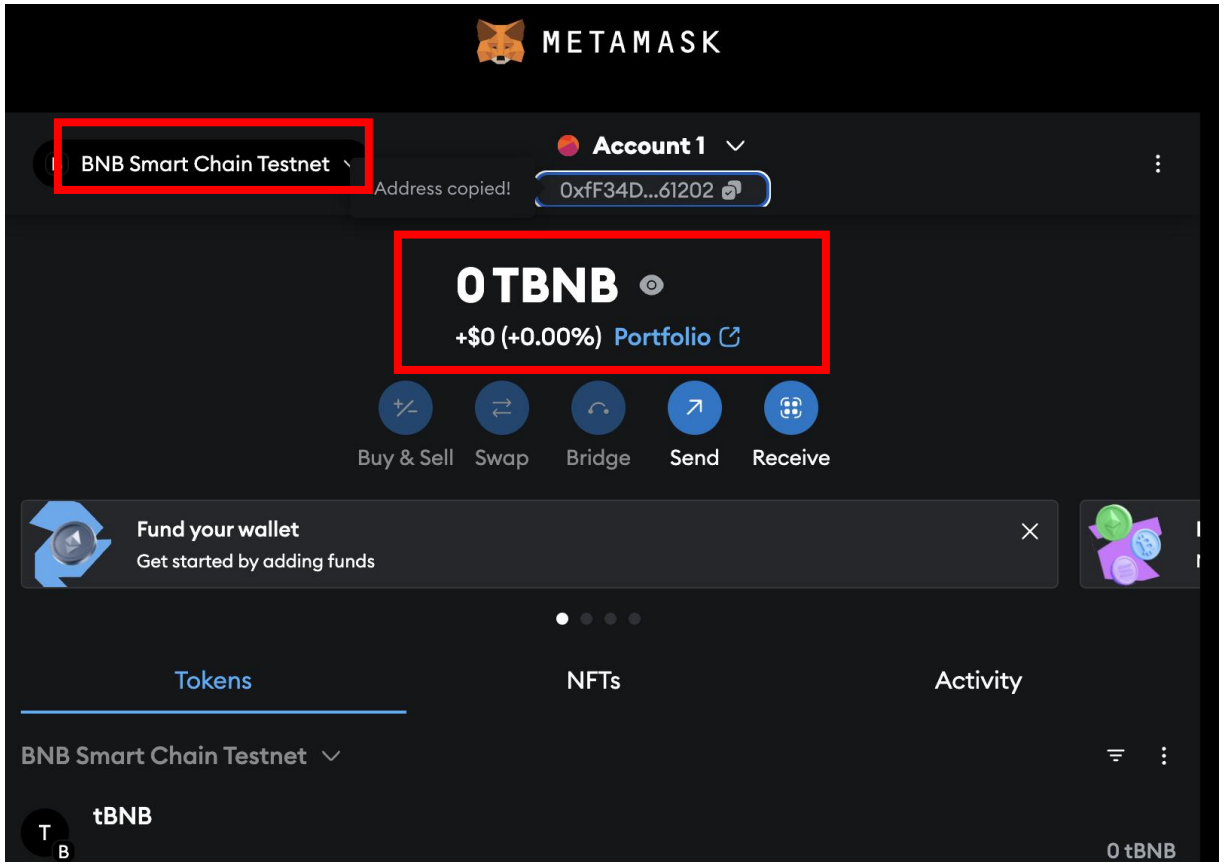


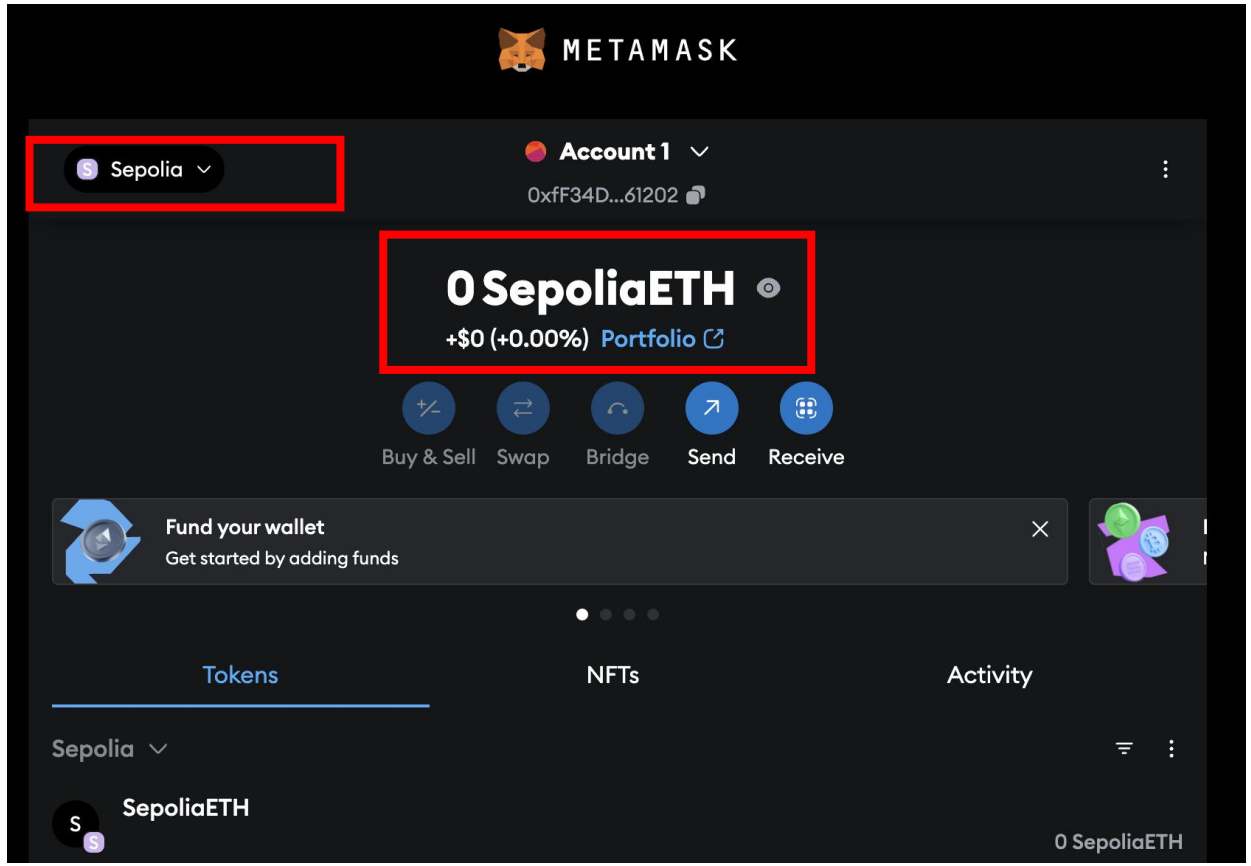


Preparation

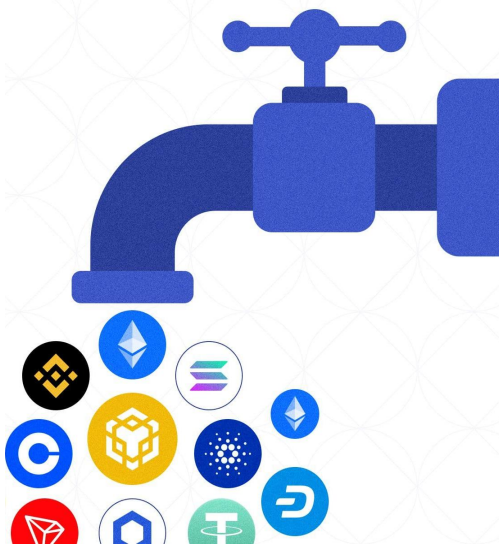
Step3: Obtain test tokens to cover the gas fees required during deployment.

Example: tBNB for BNB Smart Chain Testnet; SepoliaETH for Ethereum Sepolia





Preparation



Step3: Obtain test tokens to cover the gas fees required during deployment.

Example: tBNB for BNB Smart Chain Testnet; SepoliaETH for Ethereum Sepolia

Important: Where can you get the tesk token?

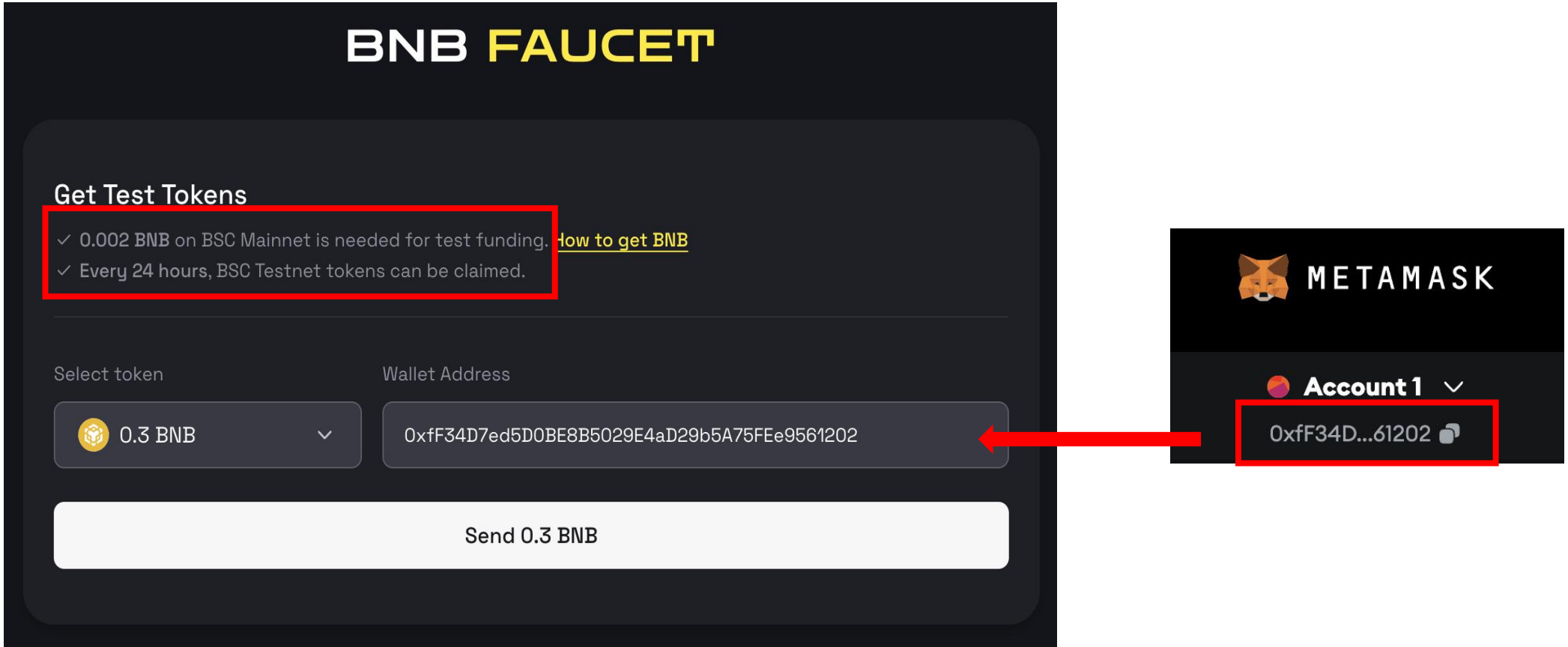
- **Search: "XXXXXX + faucet"**
- **Testnet faucets are the sources of testnet coins used for testing purposes. They're set up by the organisation running the testnet to distribute testnet coins to users.**
- **Example: tBNB faucet: https://www.bnbchain.org/en/testnet-faucet**

  SepoliaETH faucet: https://cloud.google.com/application/web3/faucet/ethereum/sepolia

Preparation

Token Acquisition Limitations

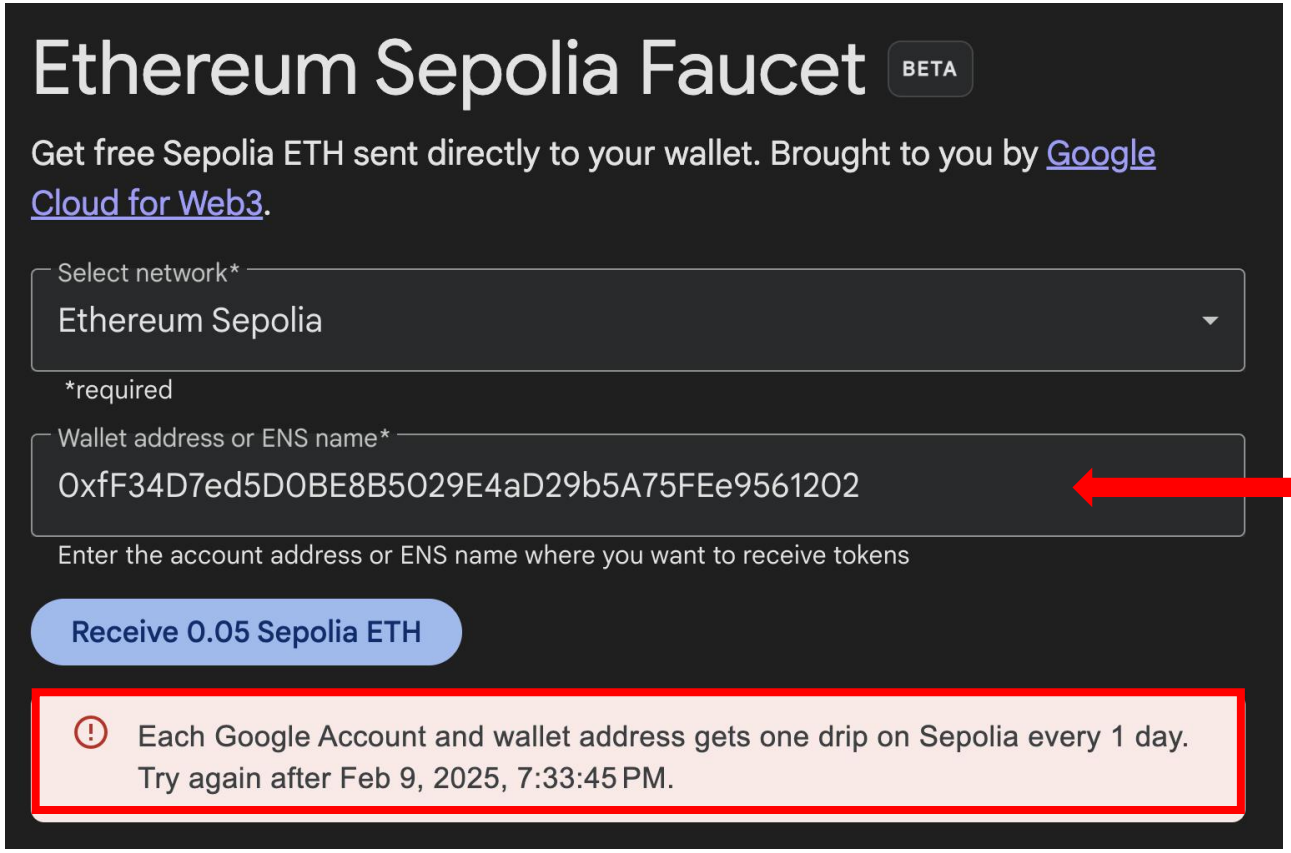
- **Mainnet Token Requirement:** Your account might need a certain amount of mainnet tokens. If you don't plan to invest funds, you'll need to find faucets without this requirement.
- **Daily Limits:** There may be restrictions on the number of tokens you can claim each day.

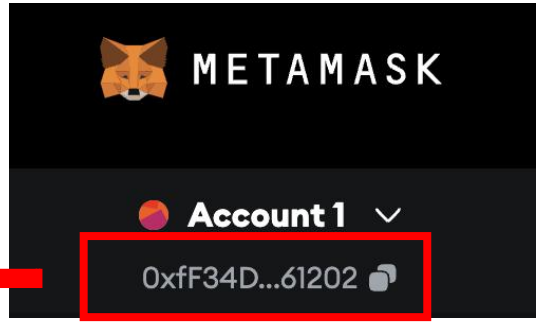


Preparation

Token Acquisition Limitations

- **Mainnet Token Requirement:** Your account might need a certain amount of mainnet tokens. If you don't plan to invest funds, you'll need to find faucets without this requirement.
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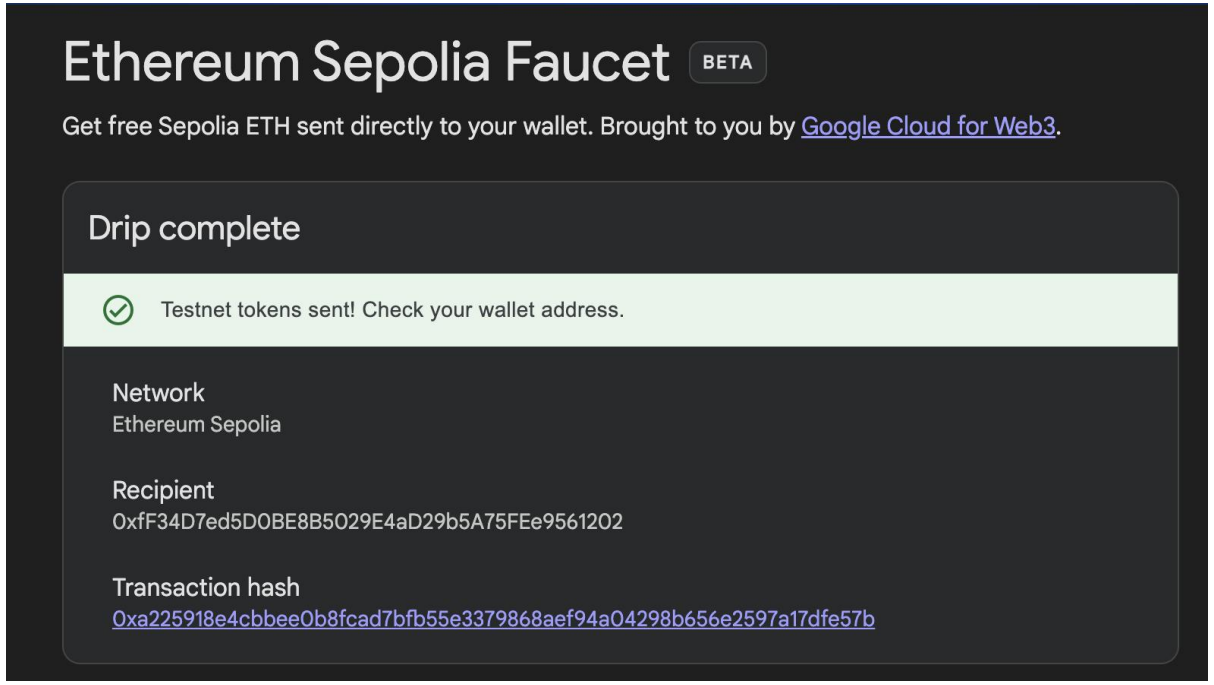


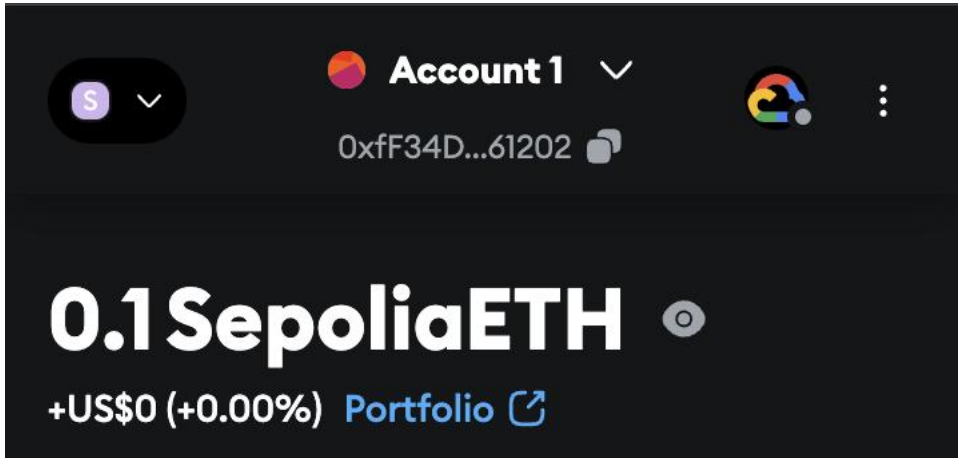


Preparation

Important Task!!!

- **Start accumulating your test tokens from today to ensure you have enough gas fees for your final project implementation!**





ETH Gas Price History: https://sepolia.beaconcha.in/gasnow

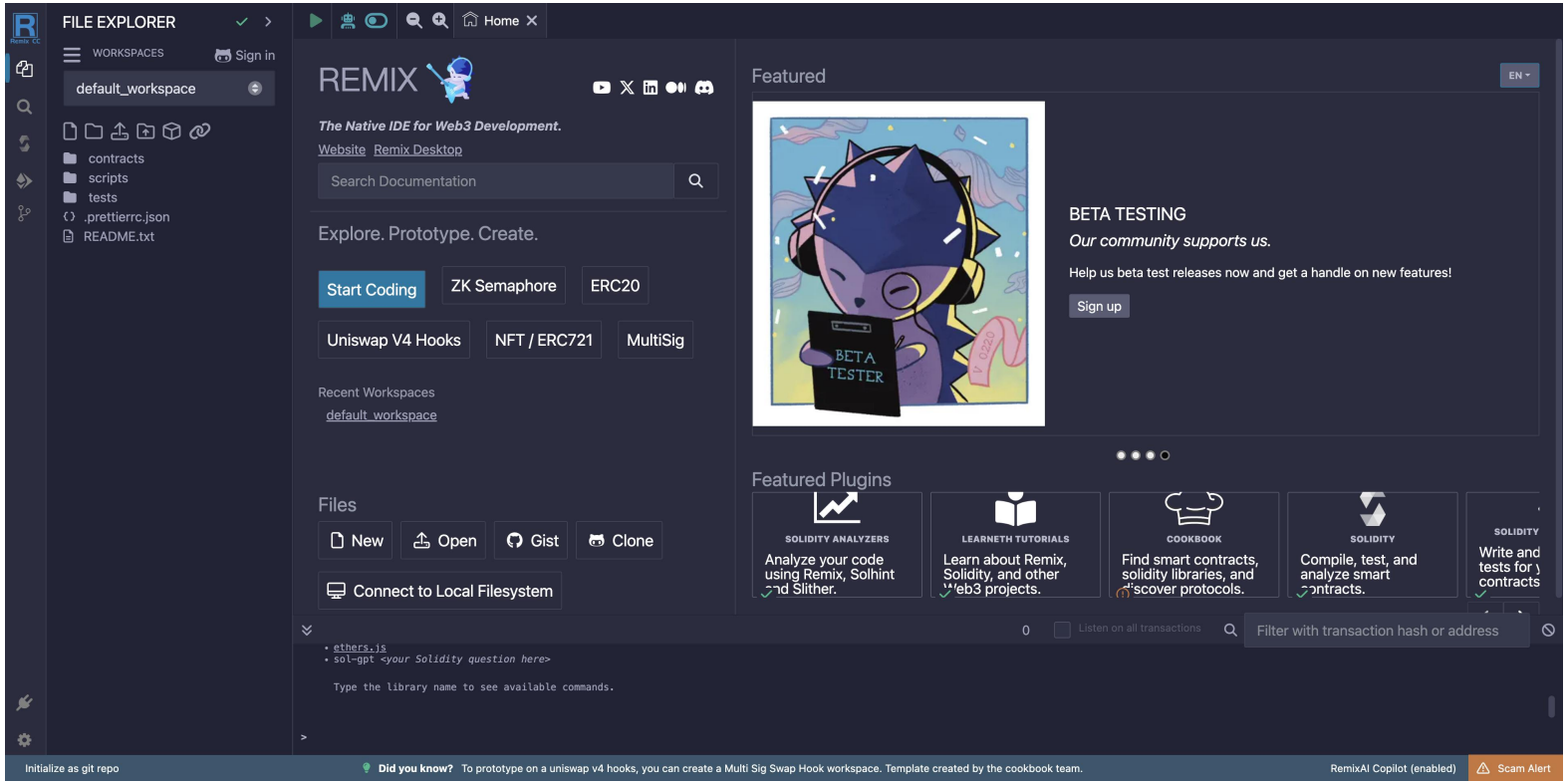
Implementation Step

Step1: Open Remix IDE

In the default workspace, create a new file named TestToken.sol under the contracts folder.

Official: https://remix-project.org/?lang=zh

Chinese Mirror Version: https://remix.learnblockchain.cn/



Implementation Step

Step2: Write the ERC20 Contract

Creat the “TestToken.sol” file, and use the ERC20 contract template provided by OpenZeppelin (open source smart contract library) to create a new ERC20 token (this is a simple ERC20 token contract example):

```
//SPDX-License-Identifier: MIT

pragma solidity ^0.8.0;

import "https://github.com/OpenZeppelin/openzeppelin-contracts/blob/master/contracts/token/ERC20/ERC20.sol";

contract TestToken is ERC20 {

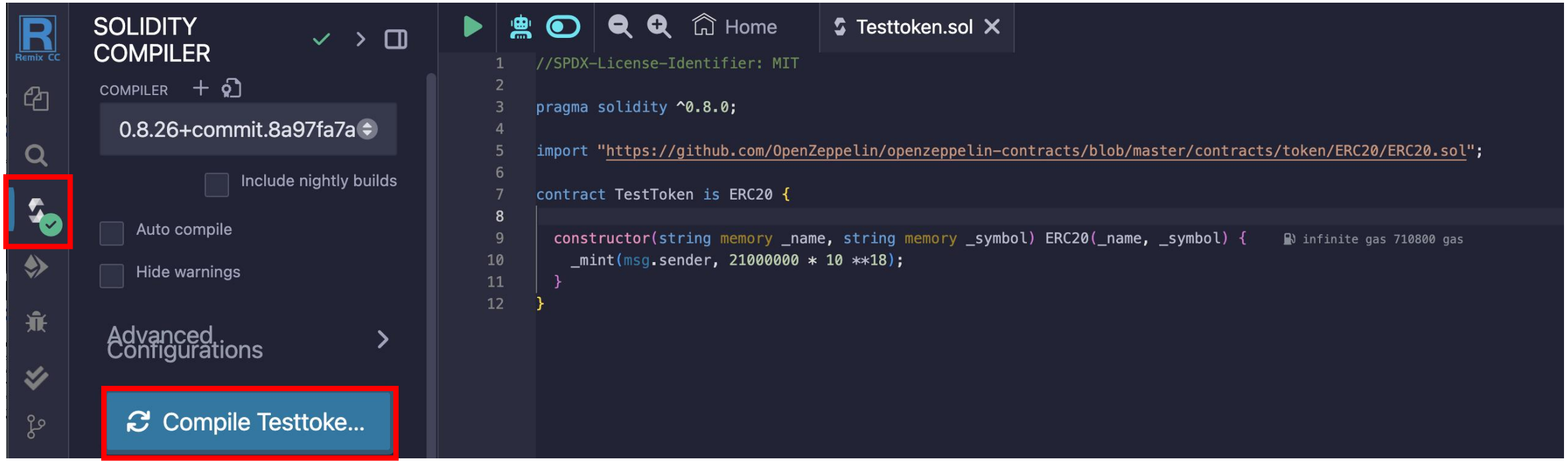
  constructor(string memory _name, string memory _symbol) ERC20(_name, _symbol) {
    _mint(msg.sender, 21000000 * 10 **18);
  }
}
```

Implementation Step

Step3: Compile the Contract

In the Remix IDE, click on the "Solidity Compiler" icon on the left sidebar.

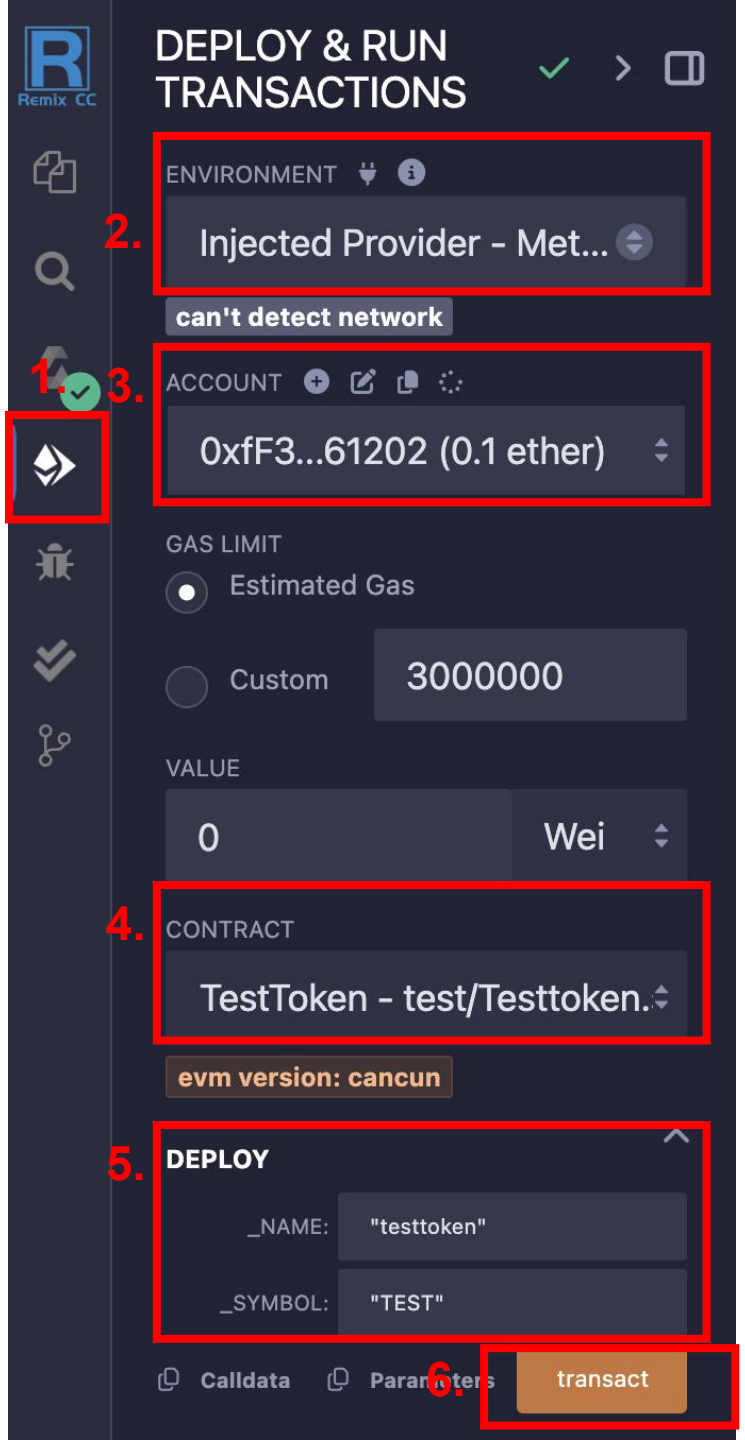
Select a suitable compiler version (e.g., 0.8.0) and click "Compile TestToken.sol".



Implementation Step

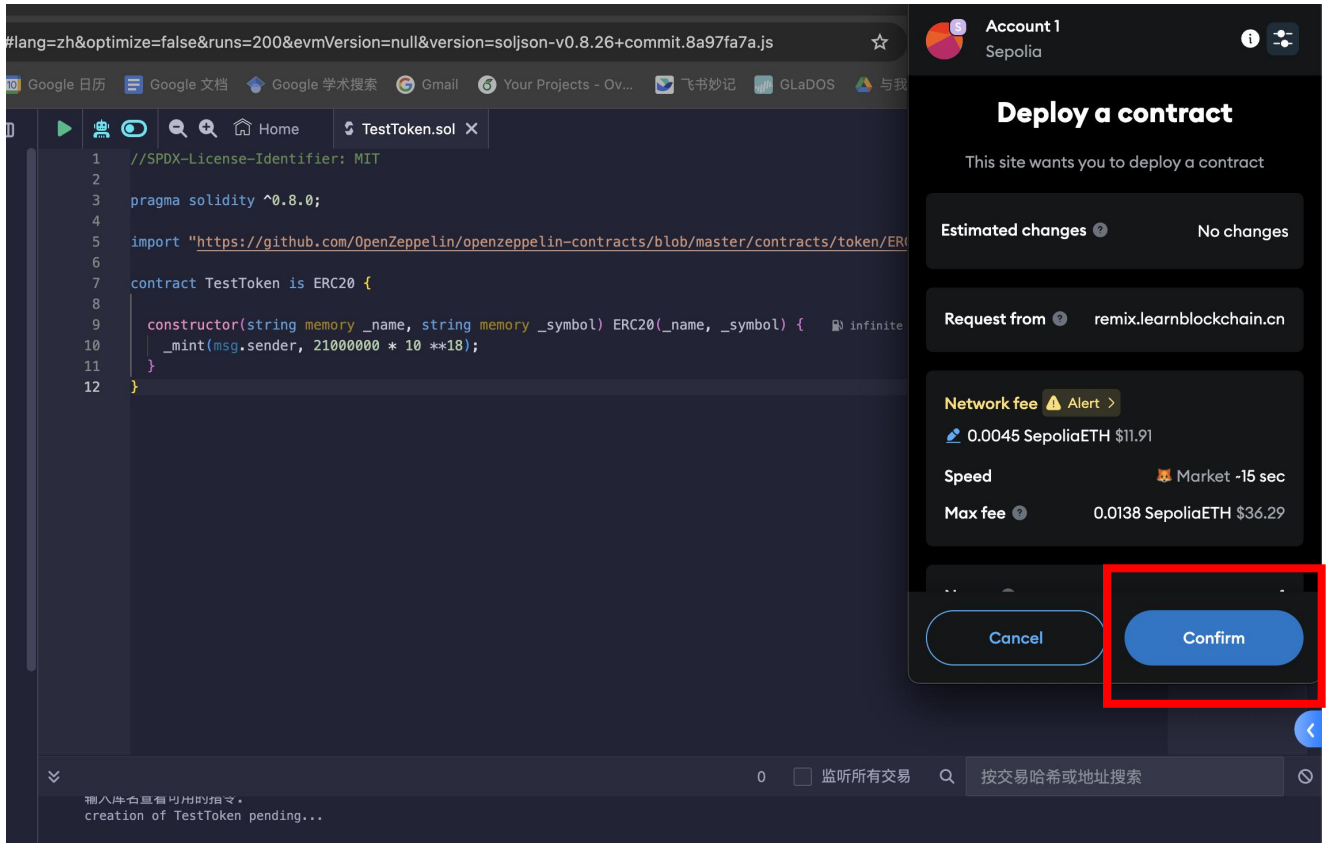
Step4: Deploy the Contract

1. **Click on the "Deploy & Run Transactions" icon on the left sidebar.**
2. **In the "Environment" dropdown menu, select "Injected Web3" to use MetaMask.**
3. **Ensure MetaMask is connected to the Sepolia Test Network.**
4. **In the "Contract" dropdown menu, select "TestToken - [Contract Name]".**
5. **Set the "-NAME" and "-SYMBOL".**
6. **Click the "transact" button and confirm the transaction in MetaMask.**



Implementation Step

Step4: Deploy the Contract




```
creation of TestToken pending...

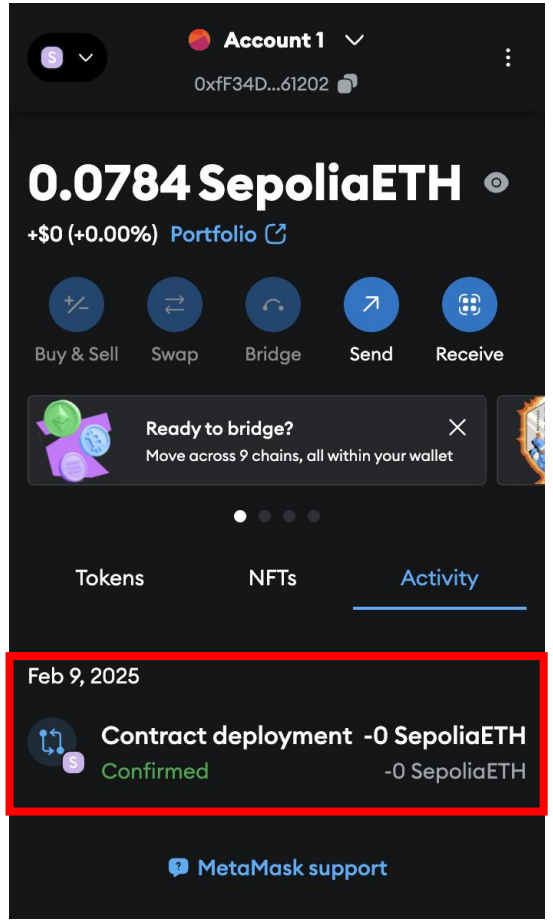
view on etherscan

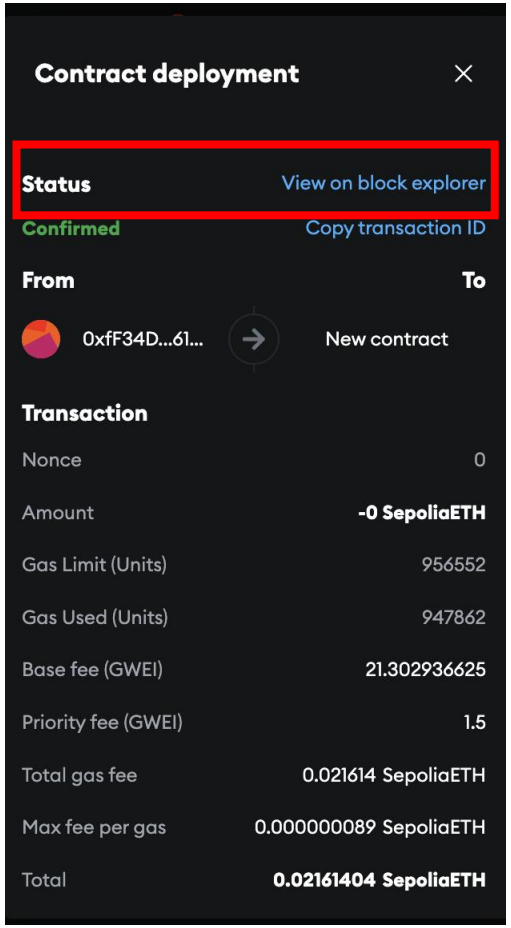
[block:7677421 txIndex:33] from: 0xff3...61202 to: TestToken.(constructor) value: 0 wei data: 0x608...00000 logs: 1
hash: 0xef5...304e4
```

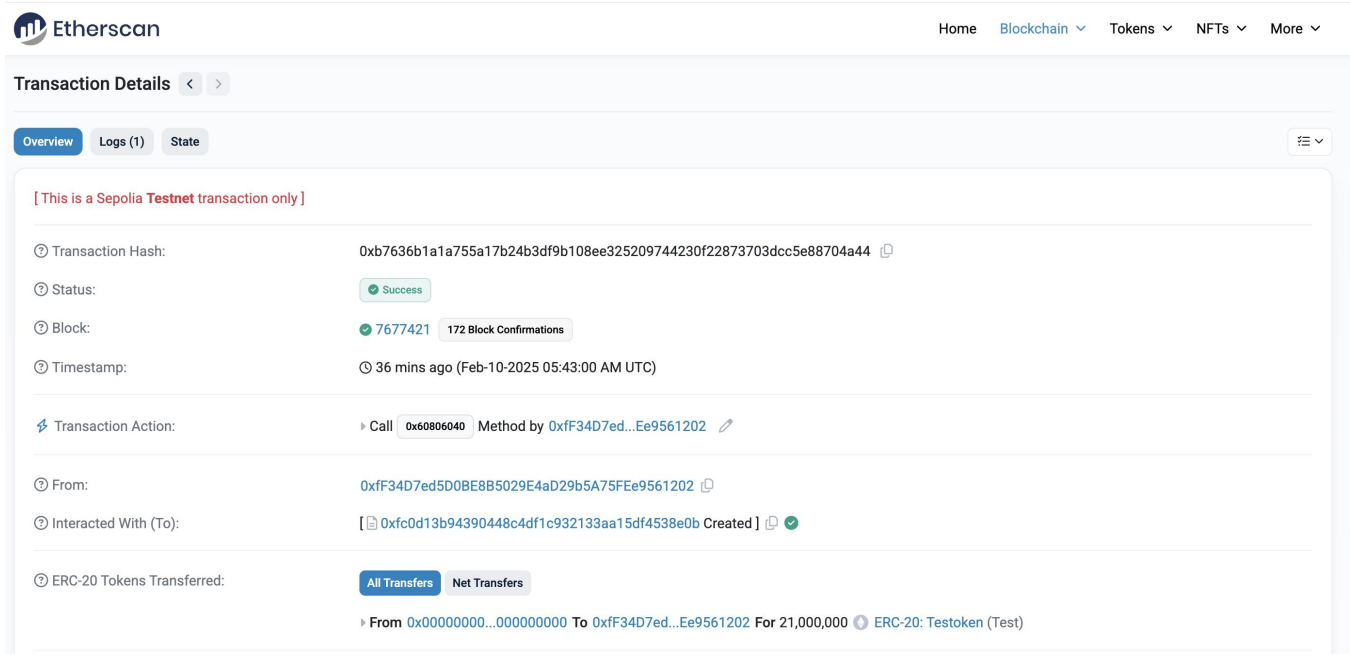
Implementation Step

Step 5: Verify Contract Deployment

Visit the block explorer to view contract information and transaction history.



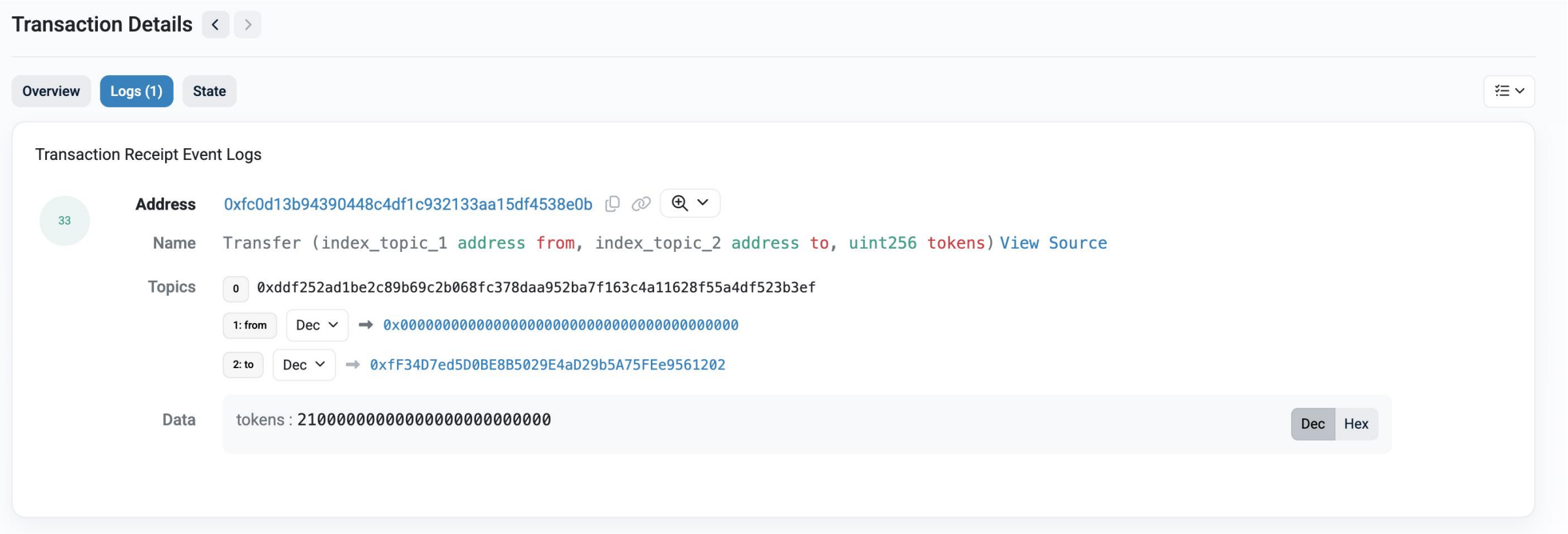




https://sepolia.etherscan.io/tx/0xb7636b1a1a755a17b24b3df9b108ee325209744230f22873703dcc5e88704a44

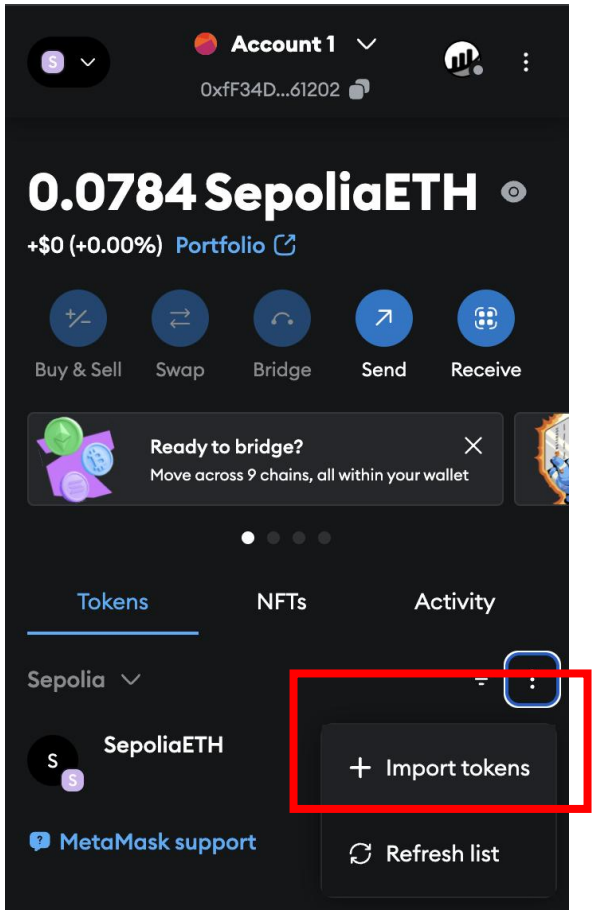
Implementation Step

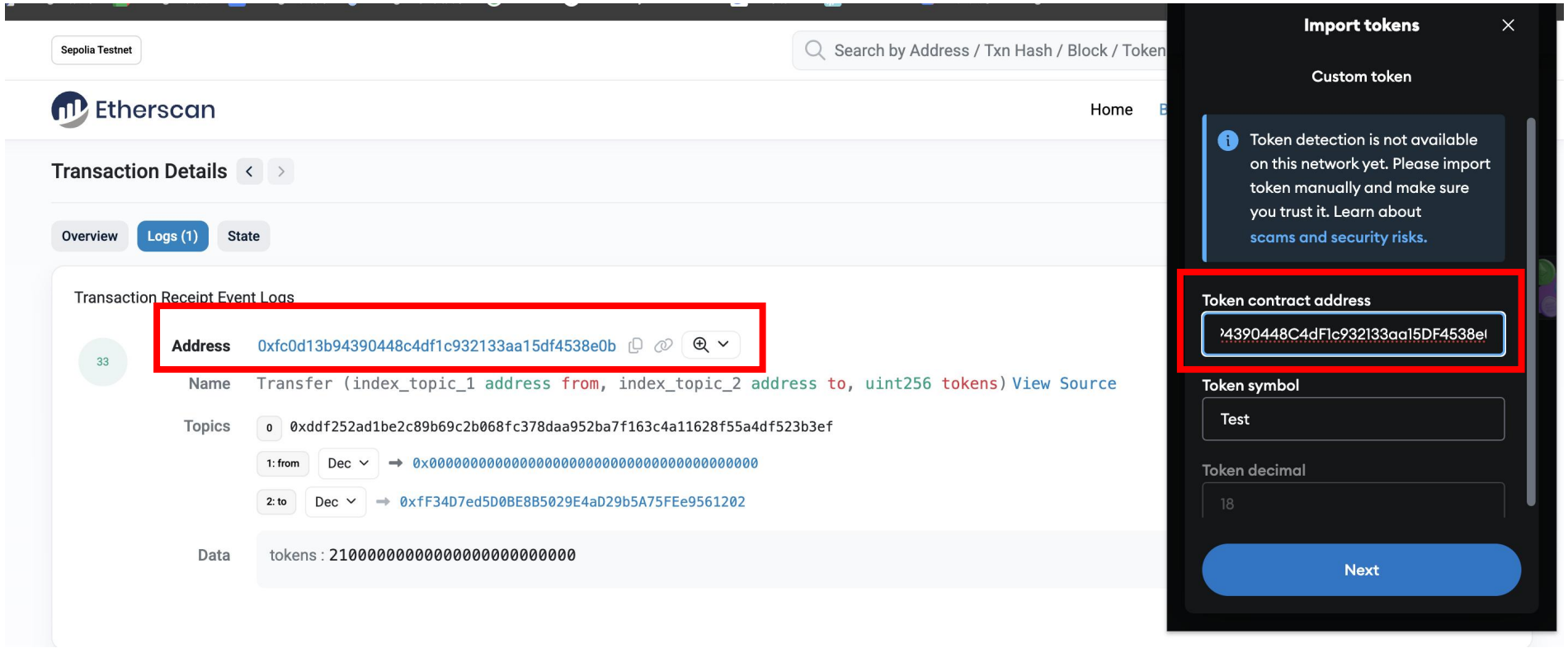
Step 5: Verify Contract Deployment



Implementation Step

Step 5: Verify Contract Deployment





Implementation Step

Step 5: Verify Contract Deployment

